

Karl Zilles

The Cortex of the Rat

A Stereotaxic Atlas



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A Stereotaxic Atlas

With 130 Figures

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Introduction

The rat brain is the most widely used animal model in neurobiology. The frequency of its use underlines the need for stereotaxic atlases of the rat brain. The first attempt to fulfill this requirement was made by Krieg (1946c), but his atlas contains only very rough data based on 30 schematic drawings of coronal sections. The drawings are exaggeratedly schematic and significantly distorted. This, together with the fact that Krieg delineated far too few structures in the di- and rhombencephalon, makes the accurate placement of electrodes difficult. Accurate electrode placement is essential for advanced techniques in modern neuroanatomical, neurophysiological, and neurochemical studies. Nevertheless, Krieg's work represents the first delineation of cortical areas in the rat. These delineations were more elaborately produced and partly corroborated by experimental work by Krieg (1946a, b, 1947). Despite its shortcomings, this fundamentally valuable series of reports greatly influenced experimental approaches in the four decades following its publication. The work of König and Klippel (1963) was another landmark in the development of stereotaxic atlases of the rat brain. This atlas is one of the most widely used works of reference on stereotaxic coordinates of the rat brain. Some difficulties, however, arise from the drastic changes there have been in the accepted parcellation of the thalamus, septum, and amygdala since the publication of this atlas. In addition, stereotaxic coordinates for structures of the pons, medulla oblongata, and cerebellum are lacking. A further deficiency is the absence of cortical areal boundaries. Yet another source of considerable electrode placement error is that the work was based on the rather idiosyncratic use of young (150 g) female rats. Older and heavier rats are used for most current research (cf. Paxinos and Watson 1982). Some additional subcortical structures are delineated in the atlas by Pellegrino and Cushman (1979). Stereotaxic coordinates for newborn and growing rats have been established by

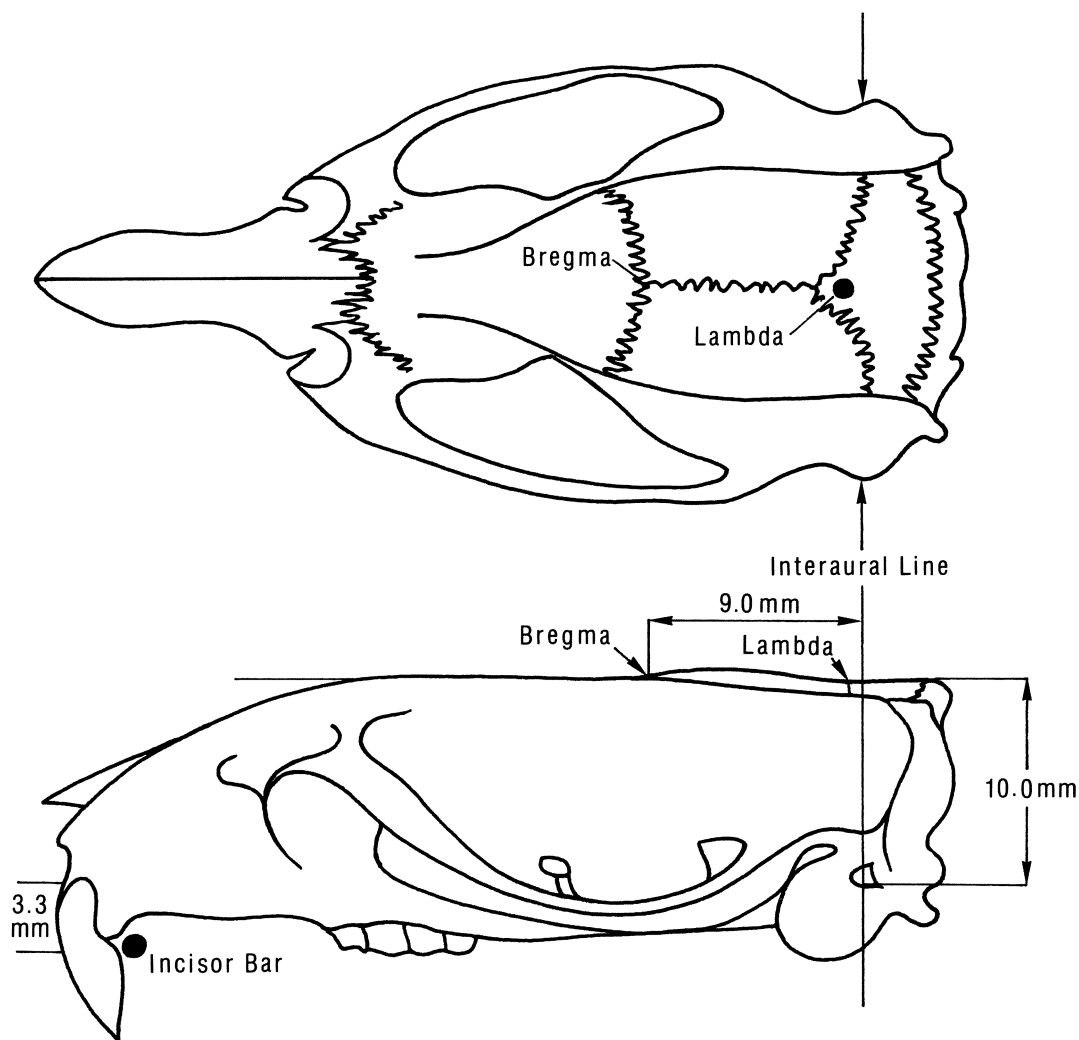
[1]

Sherwood and Timiras (1970) and by Heller et al. (1979). These works also present data on subcortical structures only. Finally, the architectonic work of Wünscher et al. (1965) deals exclusively with the pons and medulla of the rat.

Paxinos and Watson (1982) made a significant advance in that they used the flat-skull position, which can be reproduced with adequate precision. Bregma, lambda, and the midpoint of the interaural line can be used as reference points and are described in the figures. The coordinates are calculated for male Wistar rats weighing 250–350 g, that is to say, for the weight and strain most widely used in experimental work. Fresh brains frozen in the skull were cut and sections presented at 0.5-mm intervals in the coronal, sagittal, and horizontal planes. With this technique the usual distortions caused by histological shrinkage have been avoided. Cytoarchitectural, hodological, and histochemical data were the basis for the delineation of subcortical structures. The presently accepted organizational schemes were maintained throughout these delineations. Cranio-metric and stereotaxic data for correction of numerous distances, as in the case of other rat strains or younger animals, are also given. This atlas by Paxinos and Watson (1982) is the most comprehensive and advanced work of reference for stereotaxic studies on subcortical structures of the rat brain.

[2]

The primary reason for the present atlas, therefore, is not to present even more data on stereotaxic coordinates of subcortical structures, but to give stereotaxic data on *cortical* areas of the rat brain. This aspect is not considered by Paxinos and Watson (1982) or in any of the earlier atlases. It seems that cortical delineations are in great demand because advanced techniques in neuroanatomy, neurophysiology, and neurochemistry of the cortex provide data whose interpretation could be ambiguous without precise knowledge of the cortical areas involved. Very often it is totally inadequate to make a crude subdivision of cortical tissue into neo- and allocortex or even into frontal and occipital cortex. Numerous architectural, electrophysiological, histochemical, metabolic, psychological, and other studies have unequivocally established the heterogeneity of the different cortical areas. These results should be accessible for topographic analysis by scientists who are inexperienced in the difficult and often controversial problems of cortical parcellation. In some cases, the experimental methods used or staffing and instrumentation restrictions of the particular laboratory may make the precise identification of cortical areas impossible. This atlas presents stereotaxic coordinates for cortical areas of the rat brain in close analogy to the Paxinos and Watson (1982) atlas, which is gaining acceptance as the standard work of reference



[3]

Fig. 1. Dorsal and lateral views of a rat skull with the positions of bregma, lambda, and the interaural line. Lambda is about 0.3 mm anterior to the coronal plane passing through the interaural line. (After Paxinos and Watson 1982)

on stereotaxis. The original histological sections of this atlas have been used in the present work for the delineation of cortical areas. Subcortical structures have been delineated in the very few instances in which they are helpful or necessary for orientation. This arrangement has several advantages. First, it makes republication of low-power micrographs of the original histological sections on which the plates are based unnecessary, because these can be found in the Paxinos and Watson (1982) atlas. This allows space for the reproduction of low- and medium-power photomicrographs of histological sections specifically prepared for the demonstration of cortical structures of the rat and their variability in structure and position. Second, a fairly complete description of the subcortical and cortical structures

of a representative brain can be established, because all the information on subcortical structures found in the Paxinos and Watson (1982) atlas can be transferred directly to this atlas of cortical structures if necessary.

The main features of this atlas are:

1. Line drawings of 30 coronal (distance: 0.5 mm), 12 horizontal (distance: 0.5 mm), and 2 sagittal sections with stereotaxic coordinates based on the flat-skull position. The boundaries of 44 isocortical and allocortical areas are delineated and cortical maps are graphically reconstructed by orthogonal projection.
2. The frontal and occipital poles are shown from varying angles of inspection. These figures were based on computer-aided reconstruction and rotation of the line drawings.
3. Structures are delineated on the basis of Nissl-, myelin-, and acetylcholinesterase-stained sections from 12 additional rat brains.
4. The delineation of cortical structures is based on a quantitative approach using television image analyzers.
5. Low- and medium-power photomicrographs illustrate the overall appearance and laminar structure of the different cortical areas.
6. Cortical maps of different rat brains demonstrate the intra-specific variability in regional patterns.

Nomenclature

A review of the nomenclature for cortical areas in the rat reveals a confusing variety of terms and terminological systems. Brodmann's (1909) nomenclature is designed mainly for primates. Although Brodmann personally used this numerical system to describe many nonprimate species, including rodents, the system seems to be inadequate for such use, because problems of comparative anatomy and functional equivalence between primates and rodents are unresolved. One shortcoming is evident insofar as Brodmann (1909) incorrectly identified the auditory cortex in rodents. He presented a peculiar areal pattern in the temporal region. Krieg (1946, 1947) revised and expanded Brodmann's nomenclature for use with rats. A second, more biologically relevant problem arises from the implied comparability between the cortical areas of primates and rodents when the same areal terms are used to describe both. In most cases this problem of homology of cortical areas is unresolved. Many other systems of nomenclature have been designed for special studies, but these are not generally applicable outside the specialized fields for which they were designed.

[5]

A purely functional nomenclature is not appropriate, because clearly analyzed functions have been found for only some of the anatomically defined areas. A nomenclature system which is neutral, and therefore open to future supplementation without alteration to the entire system, has been used in an earlier review (Zilles and Wree 1985) of the areal and laminar pattern of the rat cortex. This alphanumeric system uses abbreviations to denote topographic regions (Fr = frontal cortex; Par = parietal cortex; Oc = occipital cortex; Cg = cingulate cortex, etc.) or to denote traditionally used terms (CA = Cornu ammonis; RSA = retrosplenial agranular cortex). Numbers are used to supplement these abbreviations wherever necessary. In the isocortex, for example, primary and supplementary somatosensory cortices are denoted as Par1 and Par2, respectively. The secondary visual

cortex requires further parcellation. This is indicated by additional alpha indices (Oc2M=medial secondary visual cortex; Oc2MM=medial part of Oc2M; Oc2ML=lateral part of Oc2M). The cortical maps in the present volume correspond to maps previously published in papers by Zilles et al. (1984) and Zilles and Wree (1985). These papers present detailed information on delineation and identification problems. The area described as 29D by Zilles et al. (1984) is defined in this atlas as the medial part of the secondary visual cortex and is therefore described as Oc2MM. Axonal transport studies with HRP (personal unpublished studies) suggest this revised identification.

Coronal, Horizontal, and Sagittal Sections in Stereotaxic Coordinates

The line drawings (Figs. 2–45) which include stereotaxic coordinates, are based on the original histological sections of adult male Wistar rats published by Paxinos and Watson (1982). The maps in Figs. 46–49 are also prepared from these drawings by orthogonal projection.

The brains of these rats were either frozen within the skulls (sagittal and horizontal series) or removed from the calvarium and then frozen (coronal series). Brains were frozen with dry ice or CO₂ and were sectioned at 40-μm intervals. They were then stained with either cresyl violet or AChE (Koelle and Friedenwald 1949; Lewis 1961). A detailed description of these procedures is given in the Paxinos and Watson (1982) atlas. The boundaries of cortical areas in the present study were determined in the original Nissl- and AChE-stained sections with a quantitative morphological method using a computer-controlled image analyzer (Zilles et al. 1980, 1982, 1984; Zilles and Wree 1985). The equipment used to carry out the measurements on Nissl sections consists of a Micro-Videomat 2 image analyzer (Zeiss, Oberkochen, West Germany) linked to a Wang 2200 MVP computer. A gray level index (GLI) was determined by an automatic scanning procedure. In the present investigation the GLI is the amount of the areal proportion of Nissl-positive particles in a measuring field of defined size (30 × 30 μm or 50 × 50 μm). A previous study has shown that the GLI correlates with the nerve cell-packing density (Wree et al. 1982), even though glial cell nuclei are included by this measurement. Since the glial cell-packing density makes a constant contribution to the GLI values (Wree et al. 1982), it does not affect the GLI-to-neuron packing density ratio. The results of these measurements are presented in computer plots, in which the regions of lower or higher GLI values are plotted with lower or higher densities (Figs. 55–67). The GLI values ranging from 0 to 100% are classified in five different ranges.

Additional information on cellular morphology which cannot be shown by the GLI measurement has been included. This was based on a detailed inspection of the original histological sections. This qualitative analysis of morphological features, combined with the quantitative analysis using the GLI, provides a complex aspect of laminar pattern based on a reliable quantitative method. Both analyses served as the basis for the delineation of areal and laminar borders.

The investigation of Nissl-stained sections was further supplemented, in many cases, by measurements of the gray value distribution in alternate AChE-stained sections. These gray value measurements were generated by an IBAS 2 (Kontron, Munich, West Germany) image analyzer. In these analyses, the quantitatively described pattern of AChE activity distribution was used as an additional criterion for the delineation of cortical areas and layers.

Detailed information concerning stereotaxic procedure, use of the flat-skull position, and variability of craniometric data is provided in the atlas by Paxinos and Watson (1982). Figure 1 summarizes the most important landmarks for the stereotaxic coordinates. In summary, the procedure is:

[8] Anesthesia is initiated with halothane vapor, followed by an IP injection of barbiturate. For adult rats (350 g body weight) a solution consisting of 1 ml Nembutal (equivalent to 60 mg Na phenobarbitate) in 3 ml sterile Ringer's solution is used, 0.25 ml of this solution being injected per 100 g body weight (37.5 mg Na phenobarbitate/kg body weight). If necessary, a further 0.1 ml/100 g body weight can be injected 10 min after the initial injection. This provides deep anesthesia for 60–70 min.

The flat-skull position is reached by lowering the incisor bar of the stereotaxic apparatus to an average of 3.3 mm below horizontal zero. Horizontal zero is defined by the interaural line, which extends between the tips of the ear bars. The lambda point is located approximately 0.3 mm anterior to the interaural line. This is just caudal to the actual intersection of the lambdoid and sagittal sutures. The bregma point is defined by the intersection of the sagittal and coronal sutures.

In the *coronal drawings* (Figs. 2–31) the anteroposterior distance in millimeters from the vertical plane passing through the interaural line is displayed at the bottom left of each drawing and the anteroposterior distance from the bregma at the bottom right. The dorsoventral distance in millimeters from the horizontal plane passing through the interaural line is shown by the numbers in the left margin, and the dorsoventral distance from the horizontal plane passing through bregma and lambda on

the surface of the skull is shown by the numbers in the right margin. The mediolateral distance from the interhemispheric cleft and the mediosagittal plane of the brain is given by the numbers in the top and bottom margins.

In the *horizontal drawings* (Figs. 32–43) the dorsoventral distance in millimeters from the horizontal plane passing through the interaural line is displayed at the bottom left and the dorsoventral distance from the horizontal plane passing through bregma and lambda on the surface of the skull, at top right. The anteroposterior distance in millimeters from the coronal plane passing through the interaural line is given in the bottom margin, and the anteroposterior distance from the coronal plane passing through bregma is shown in the top margin. Positions ventral to the interaural horizontal plane (position = 0) or posterior to either the interaural line or bregma are indicated by minus signs. The mediolateral distance in millimeters is shown in the right and left margins.

The *sagittal drawings* (Figs. 44 and 45) show the distance in millimeters from the midline at bottom left. The distance from the horizontal plane passing through the interaural line and the distance from the horizontal plane passing through bregma and lambda on the surface of the skull are given in the left and right margins, respectively. The distance in millimeters from the coronal plane passing through bregma is given in the top margin. The numbers in the bottom margin refer to the distance from the coronal plane passing through the interaural line.

[10]

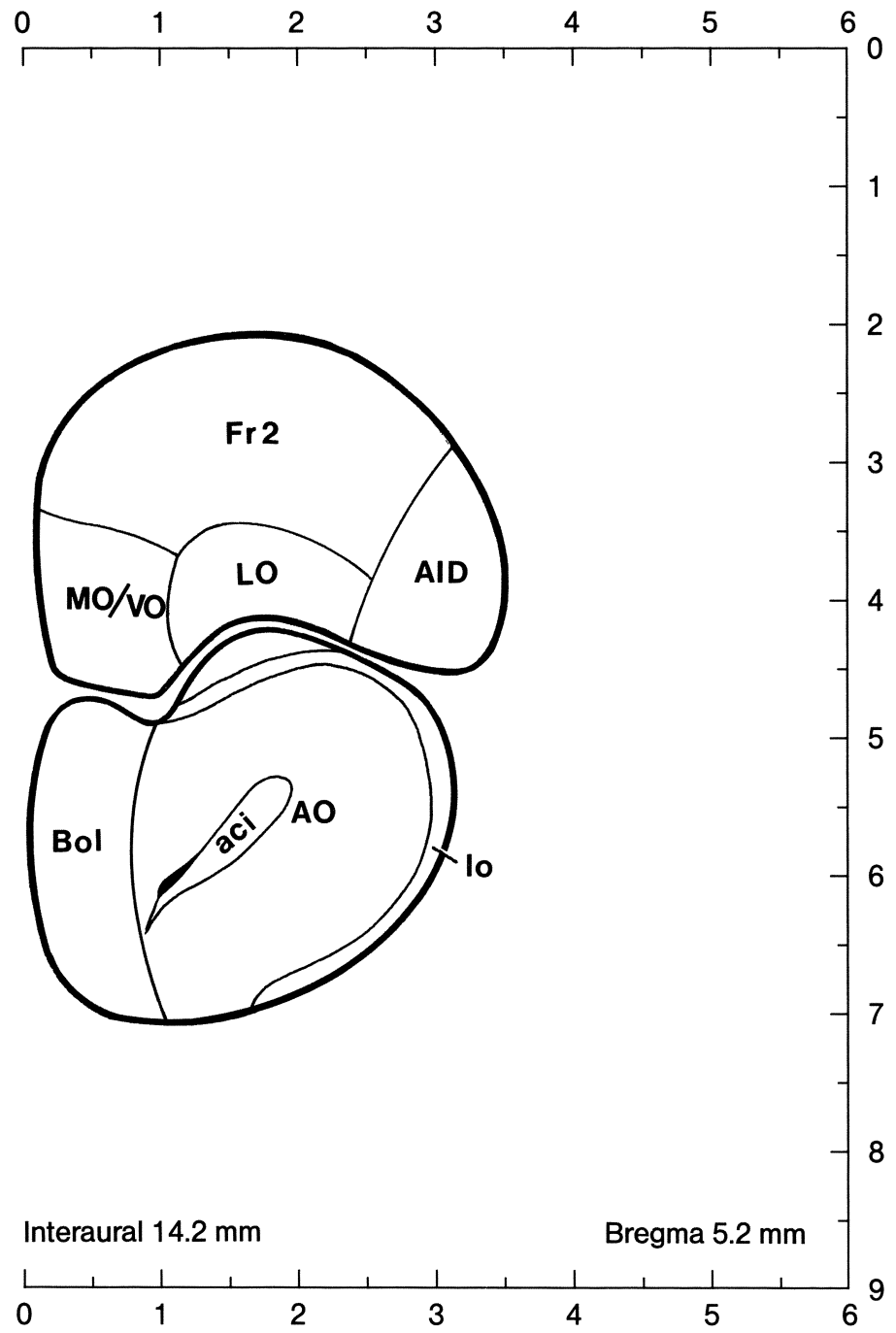
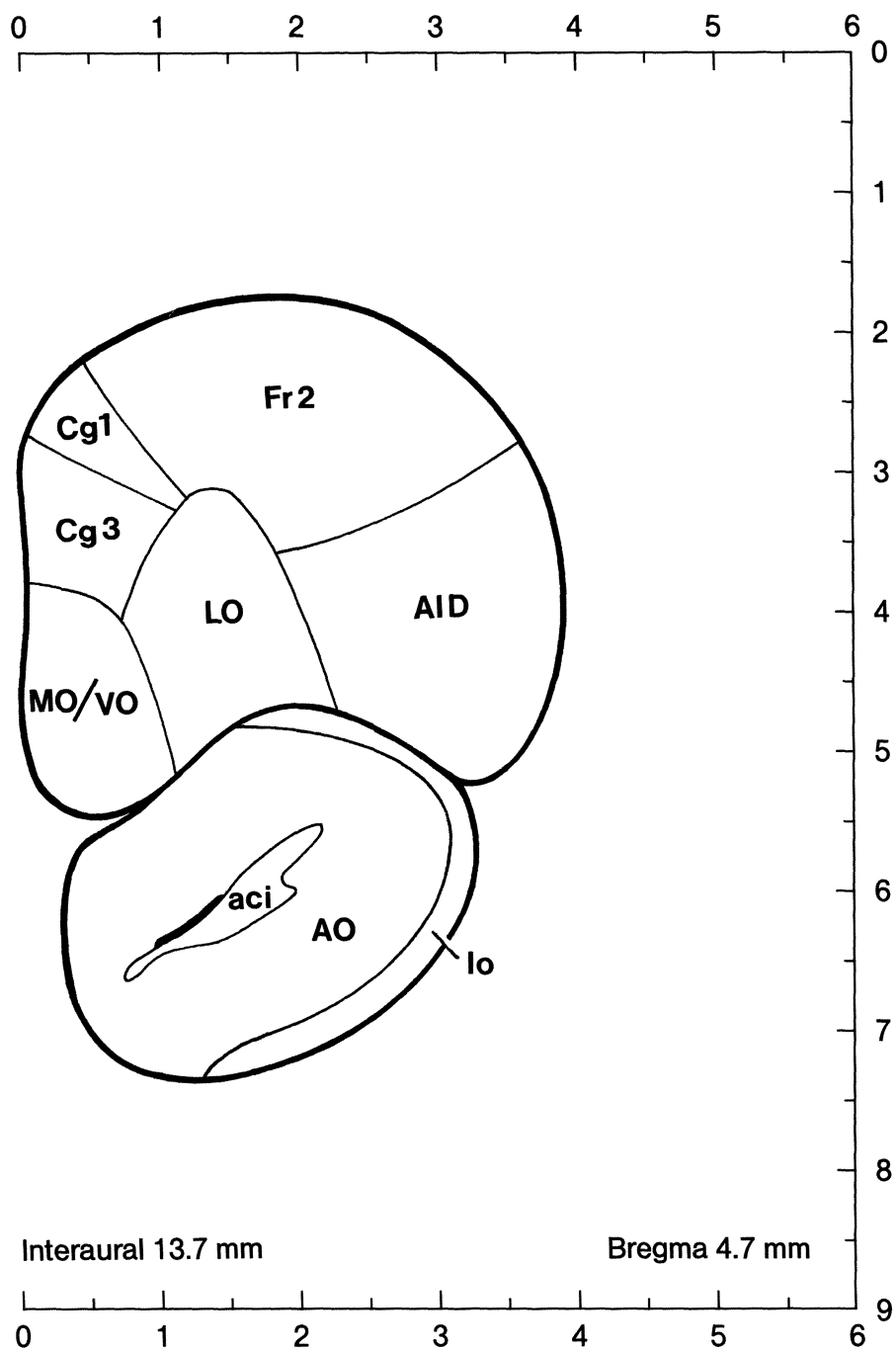


Fig. 2



[11]

Fig. 3

[12]

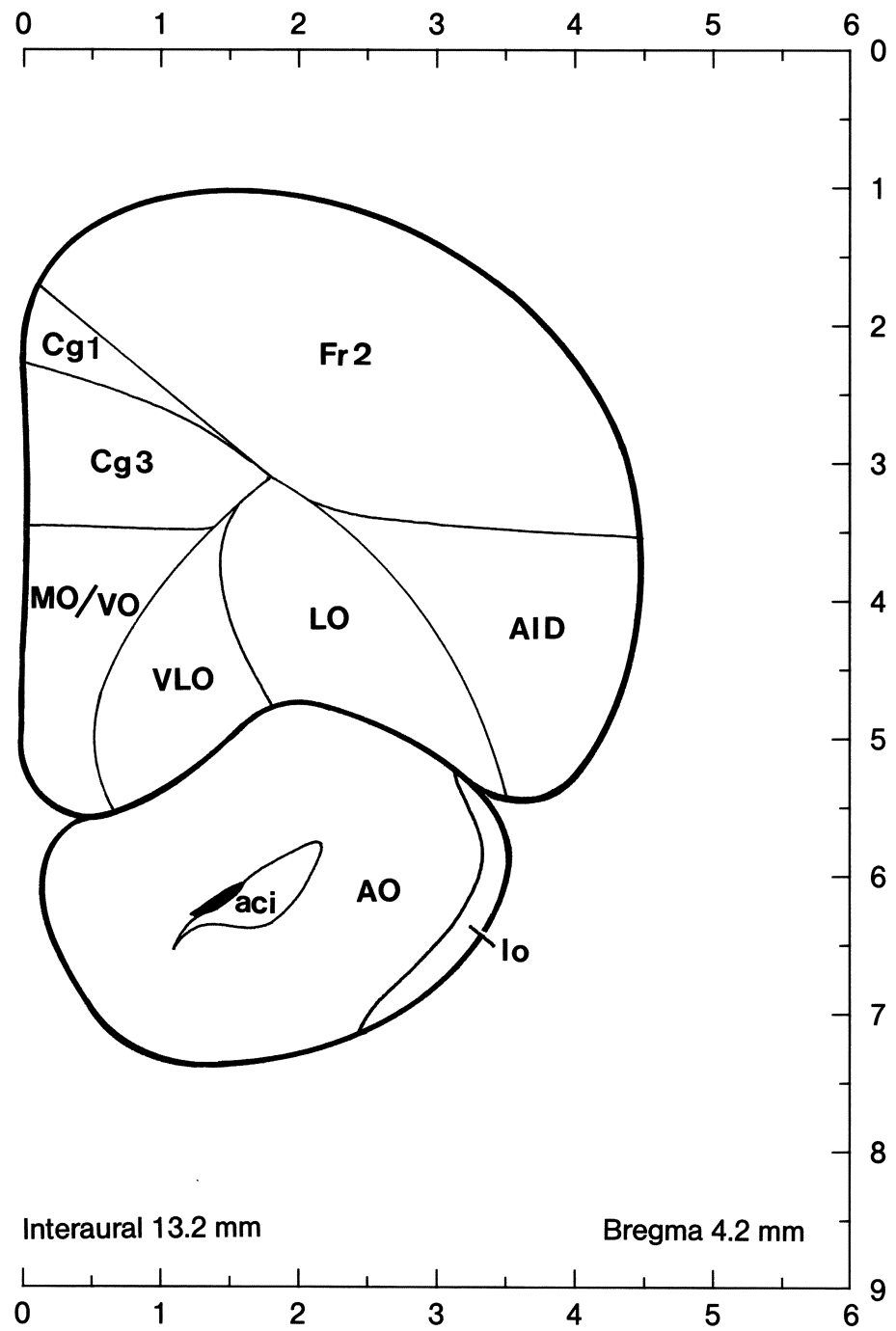
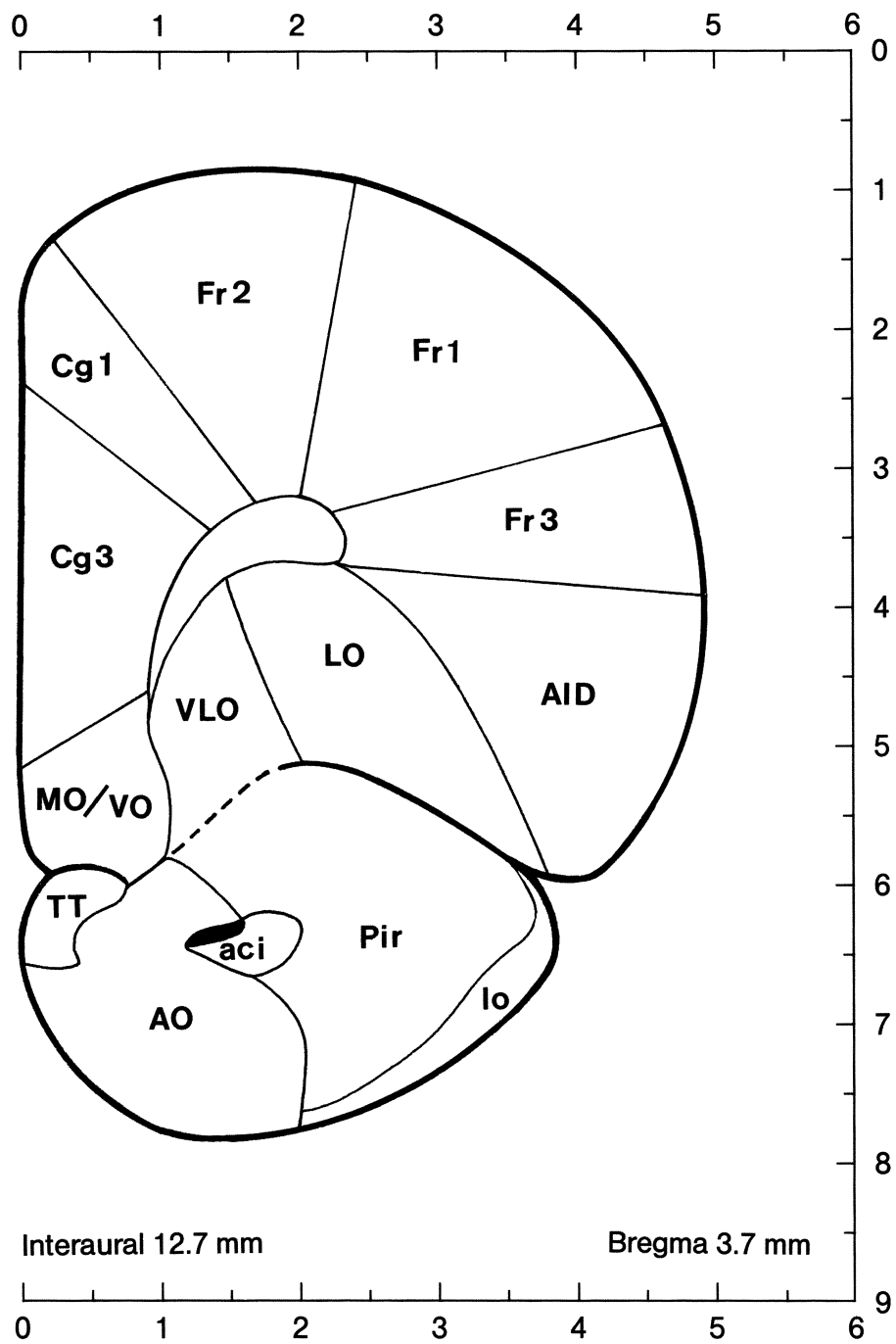


Fig. 4



[13]

Fig. 5

[14]

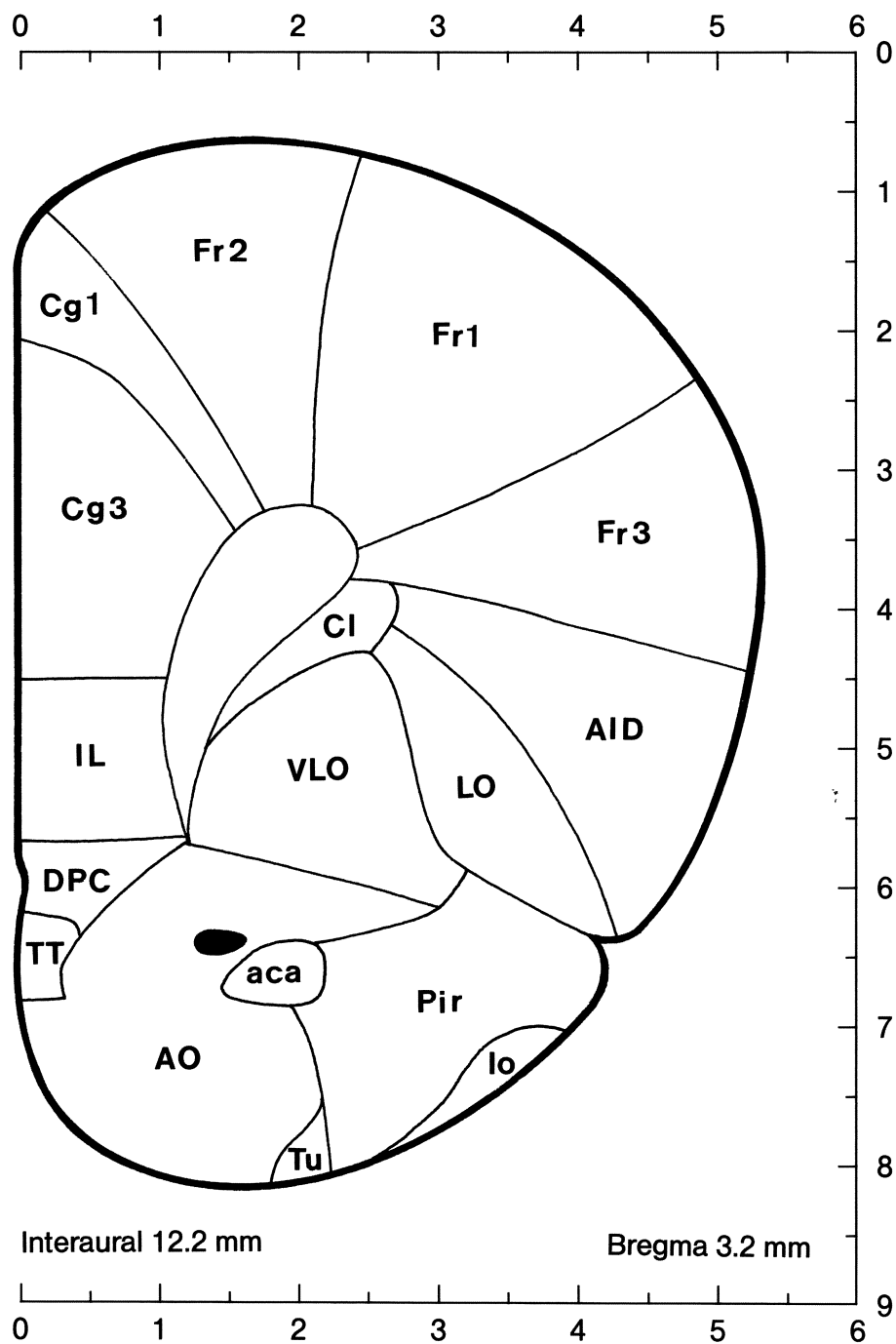
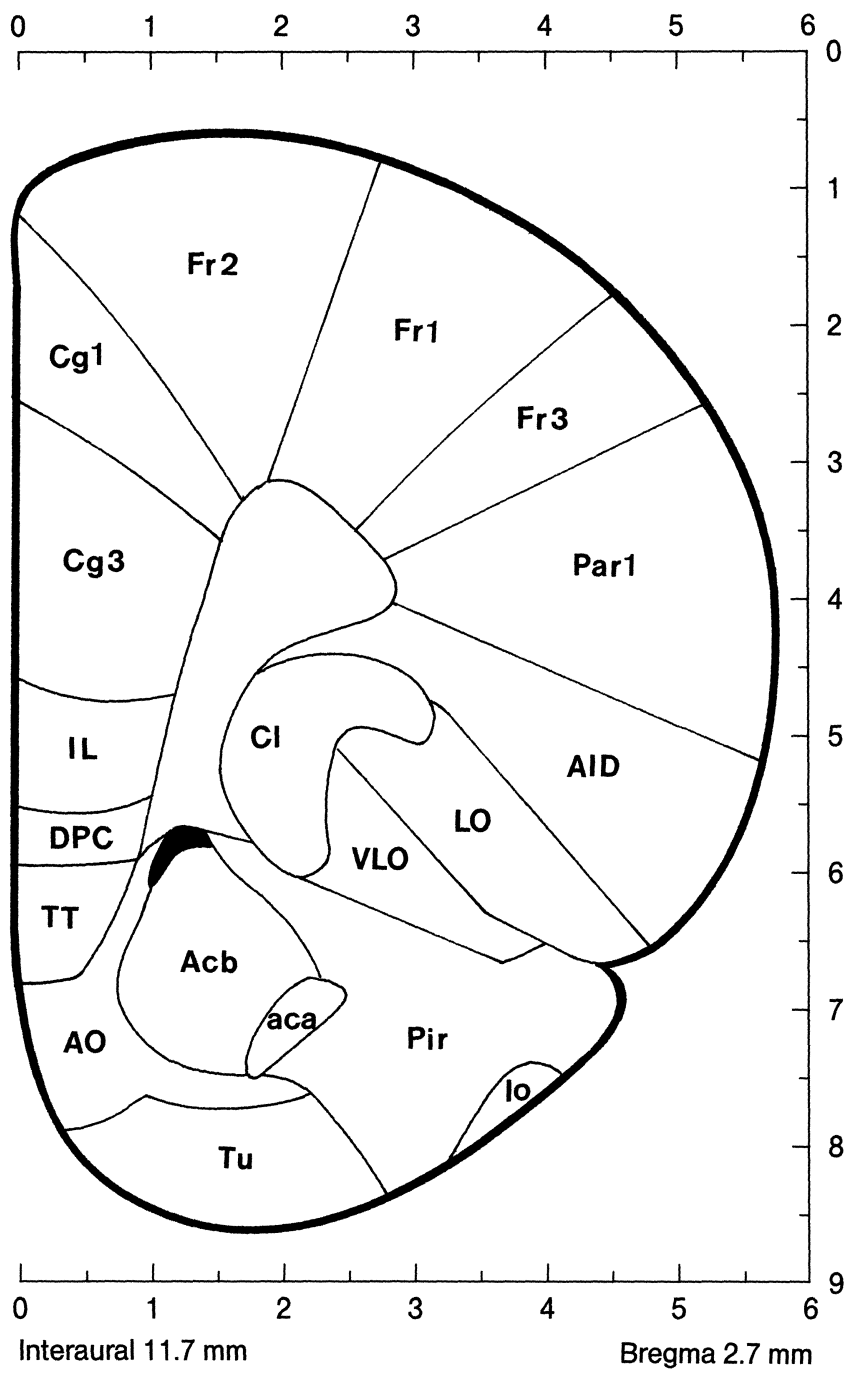
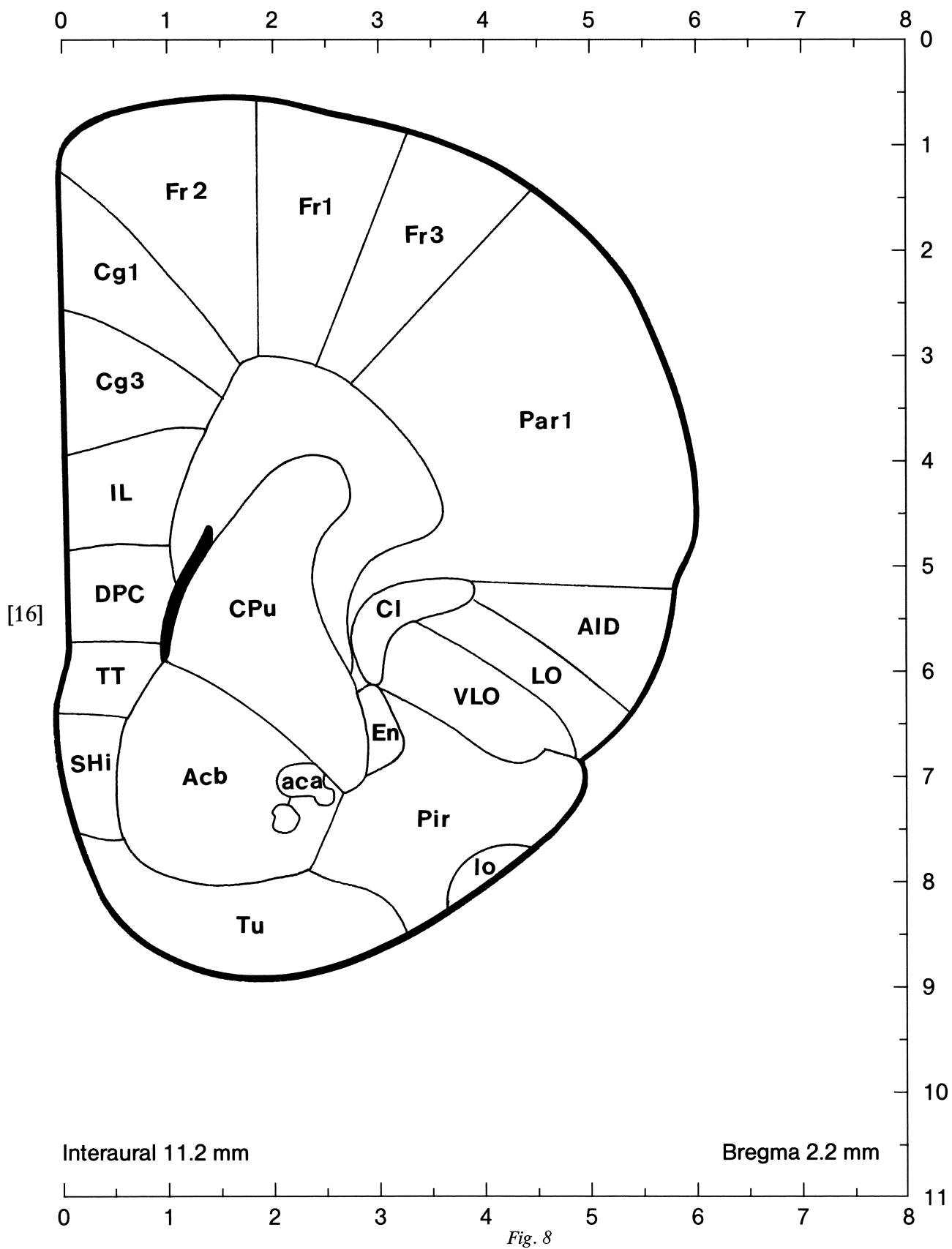


Fig. 6



[15]

Fig. 7



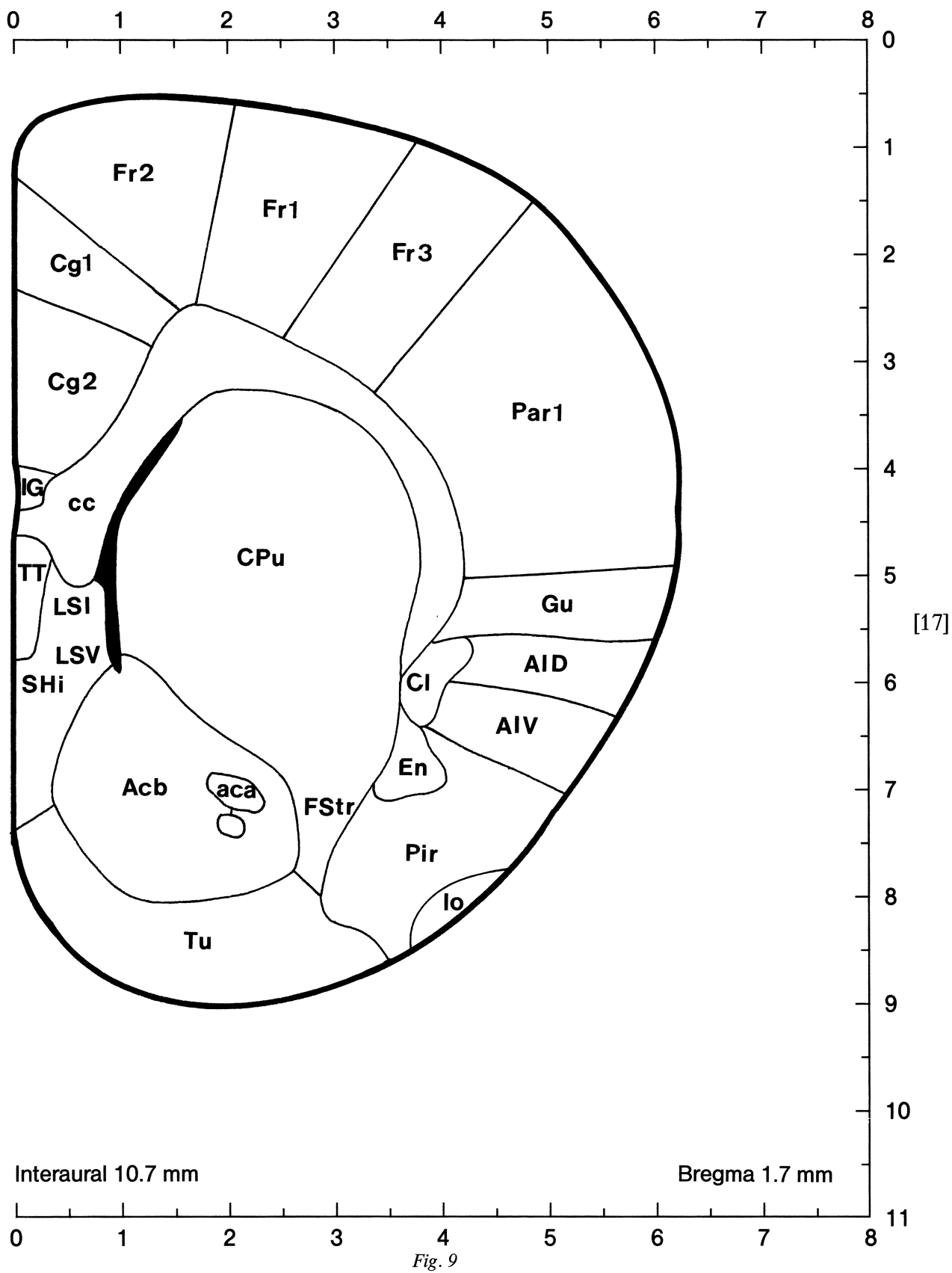


Fig. 9

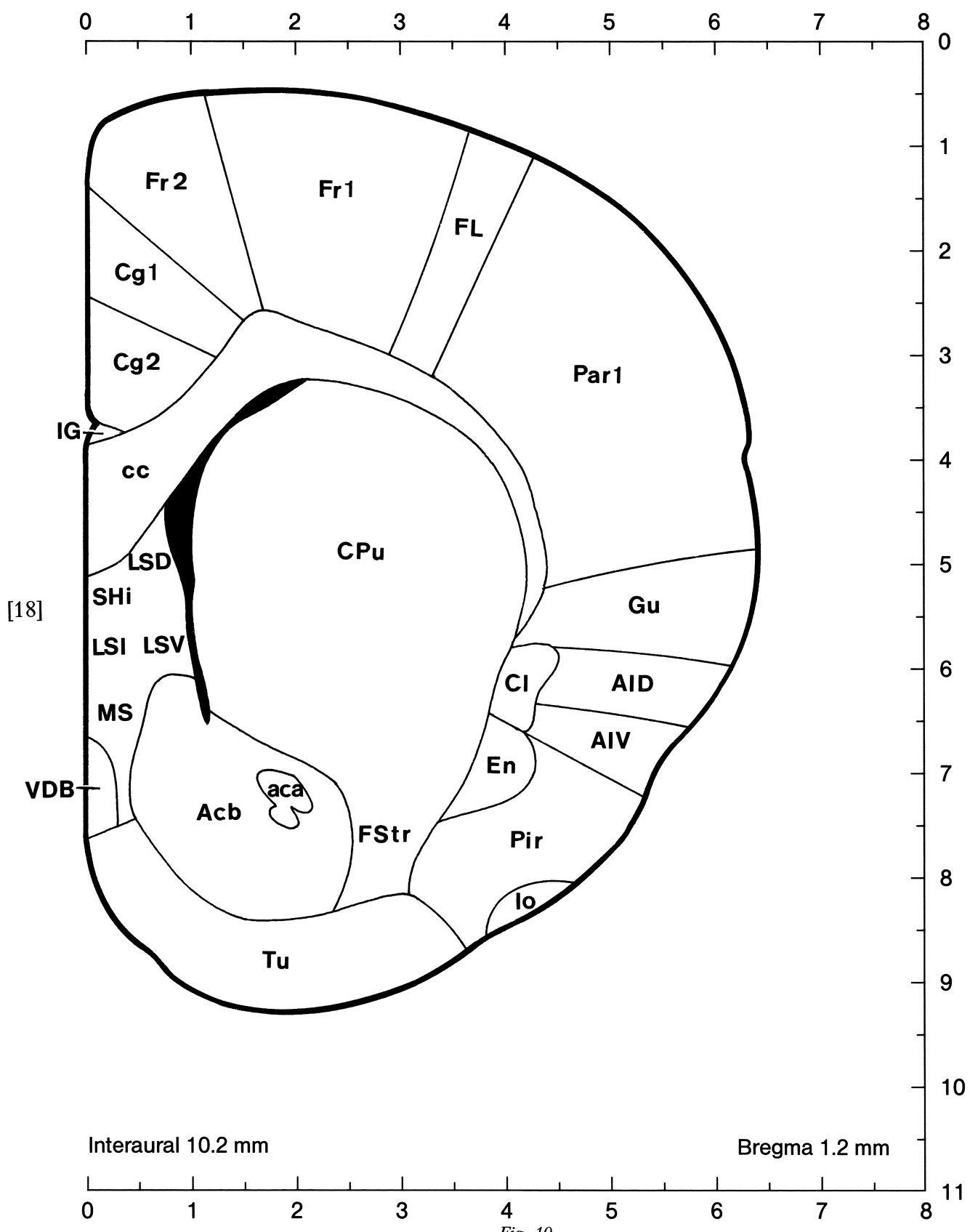


Fig. 10

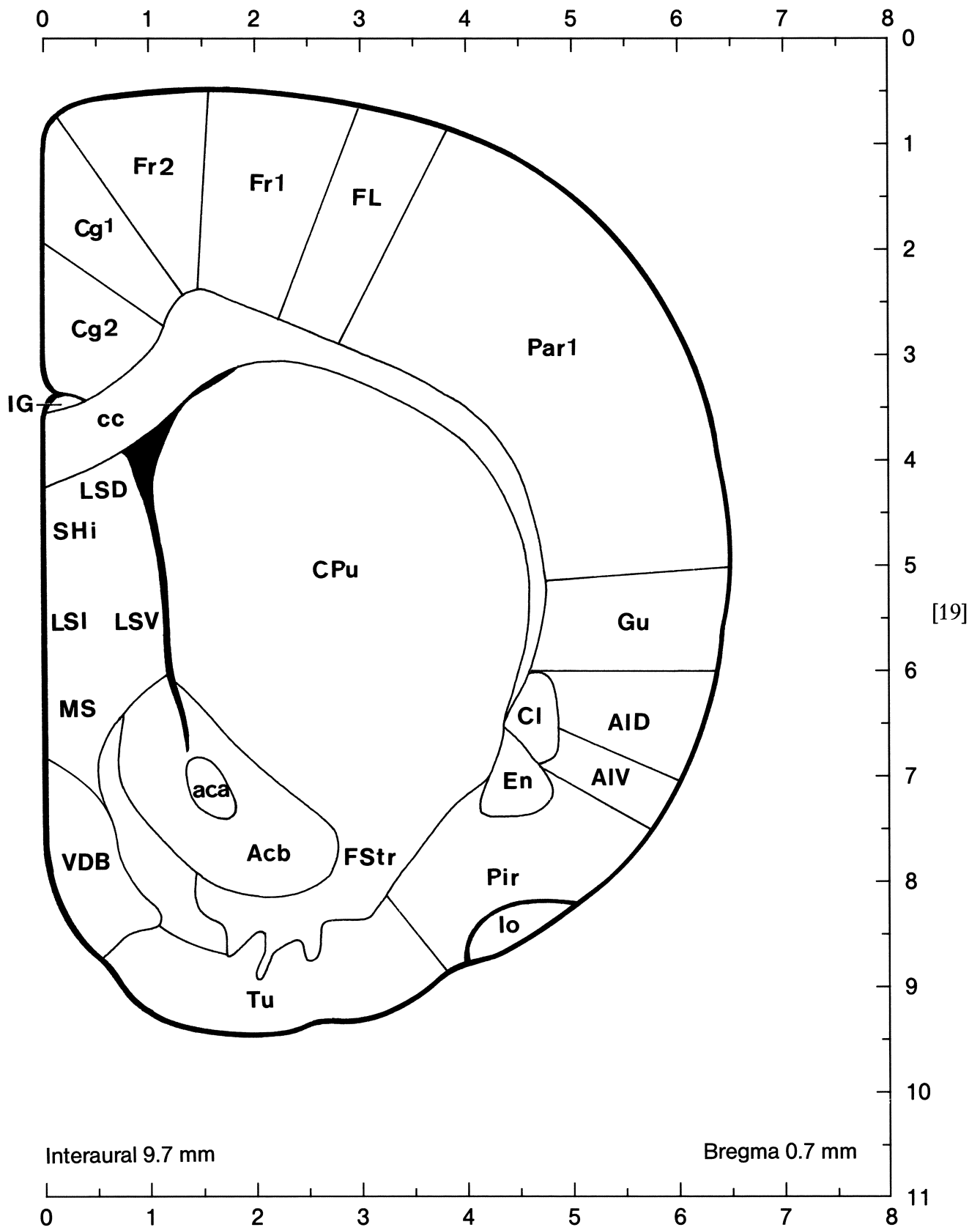


Fig. 11

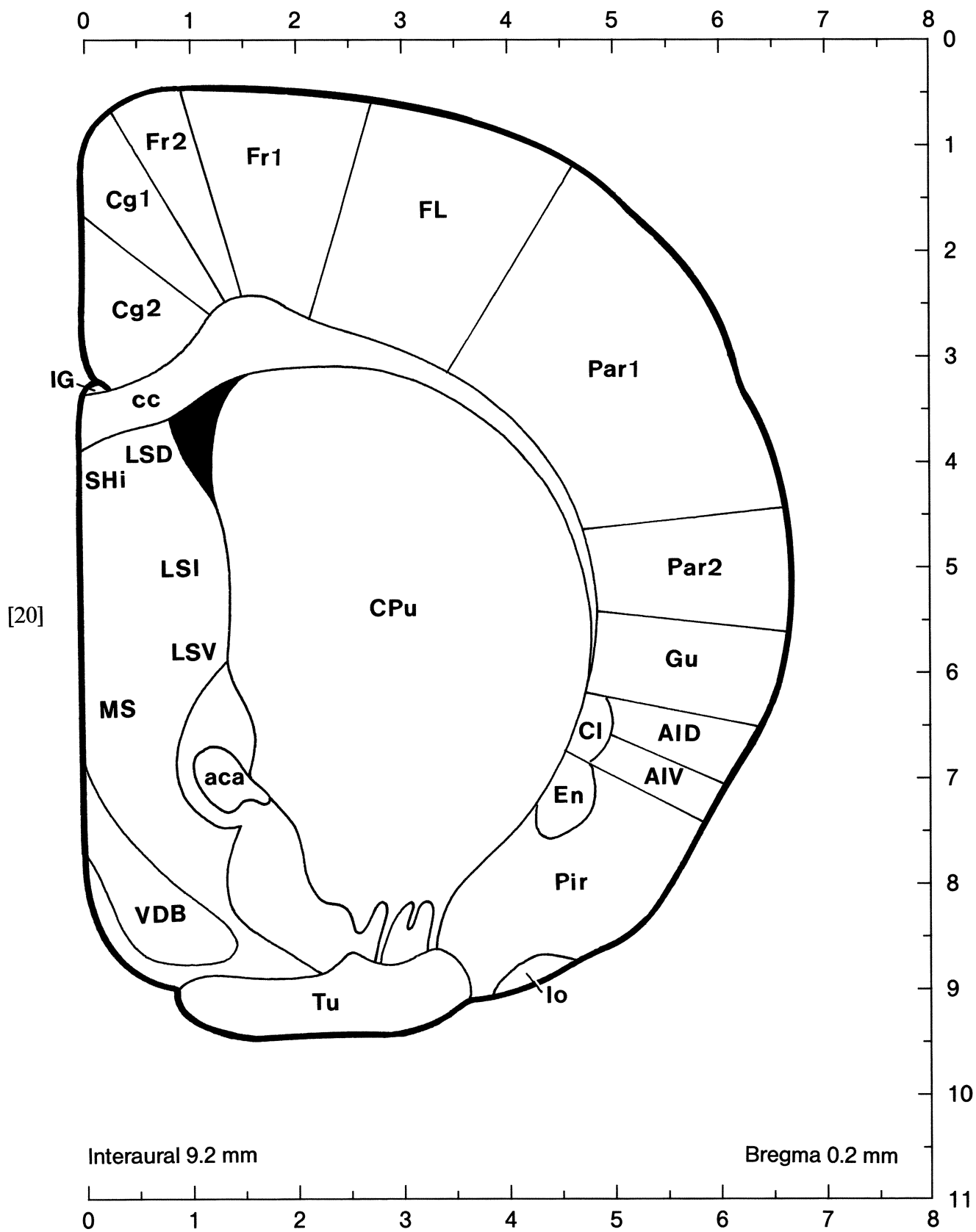


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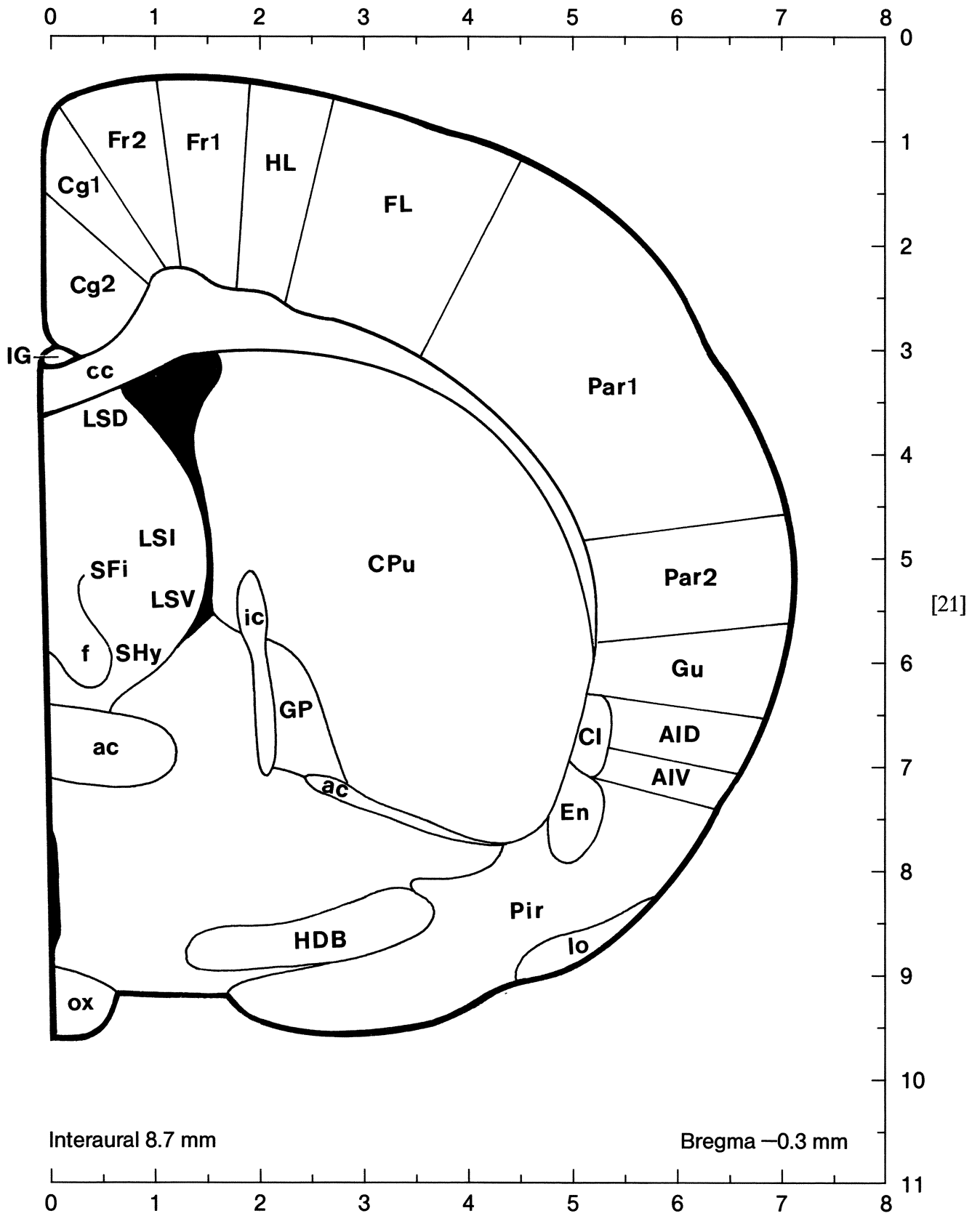


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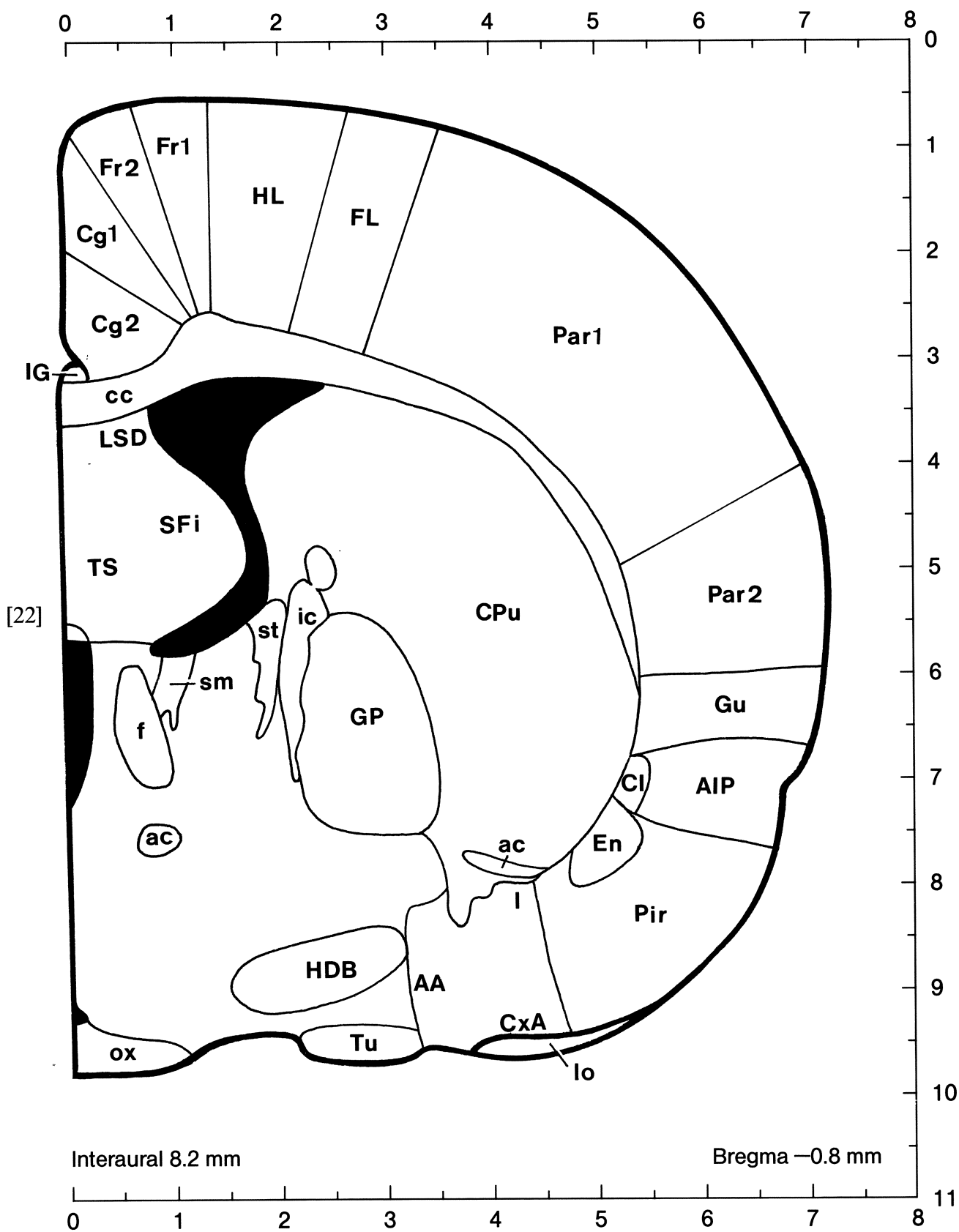


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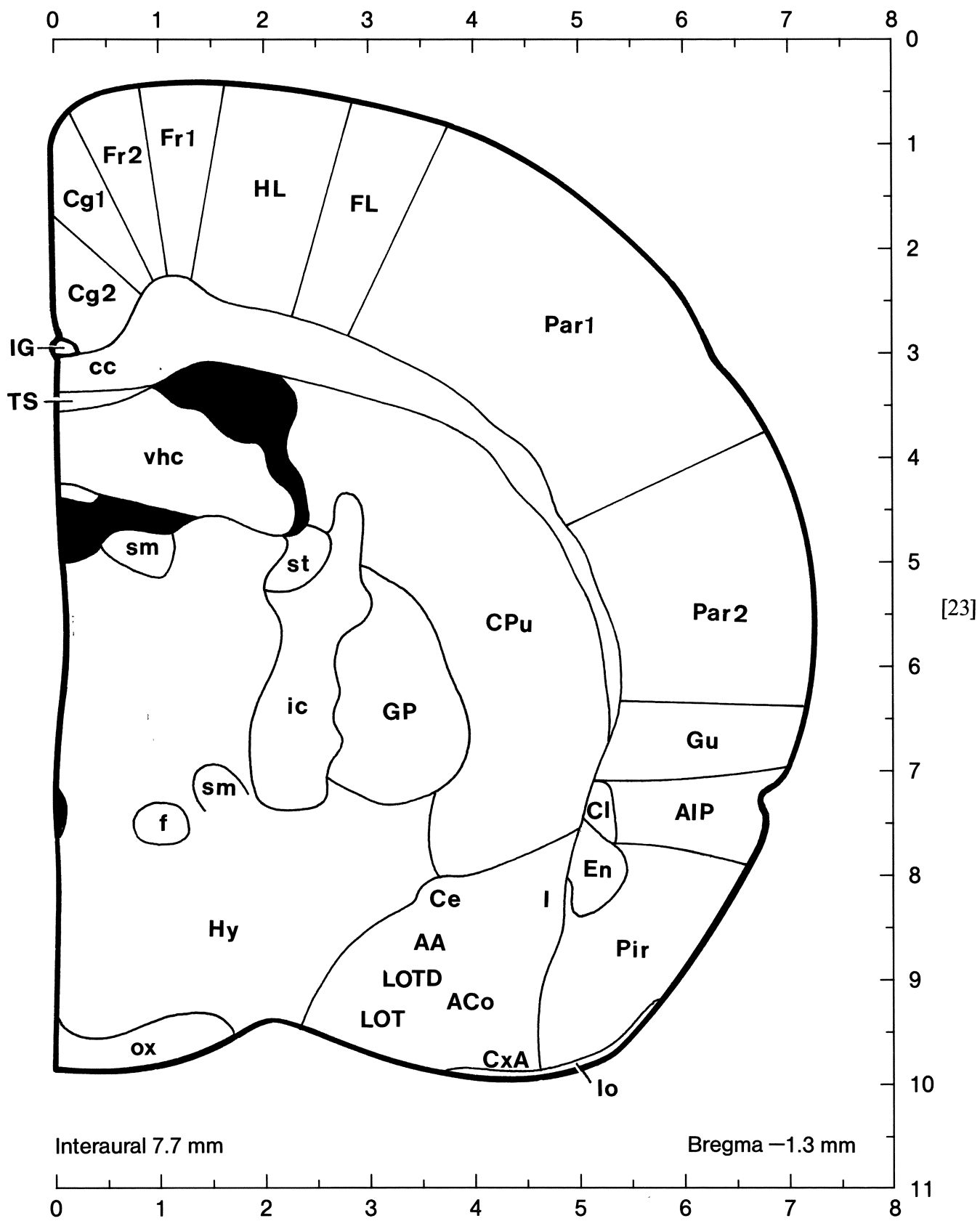


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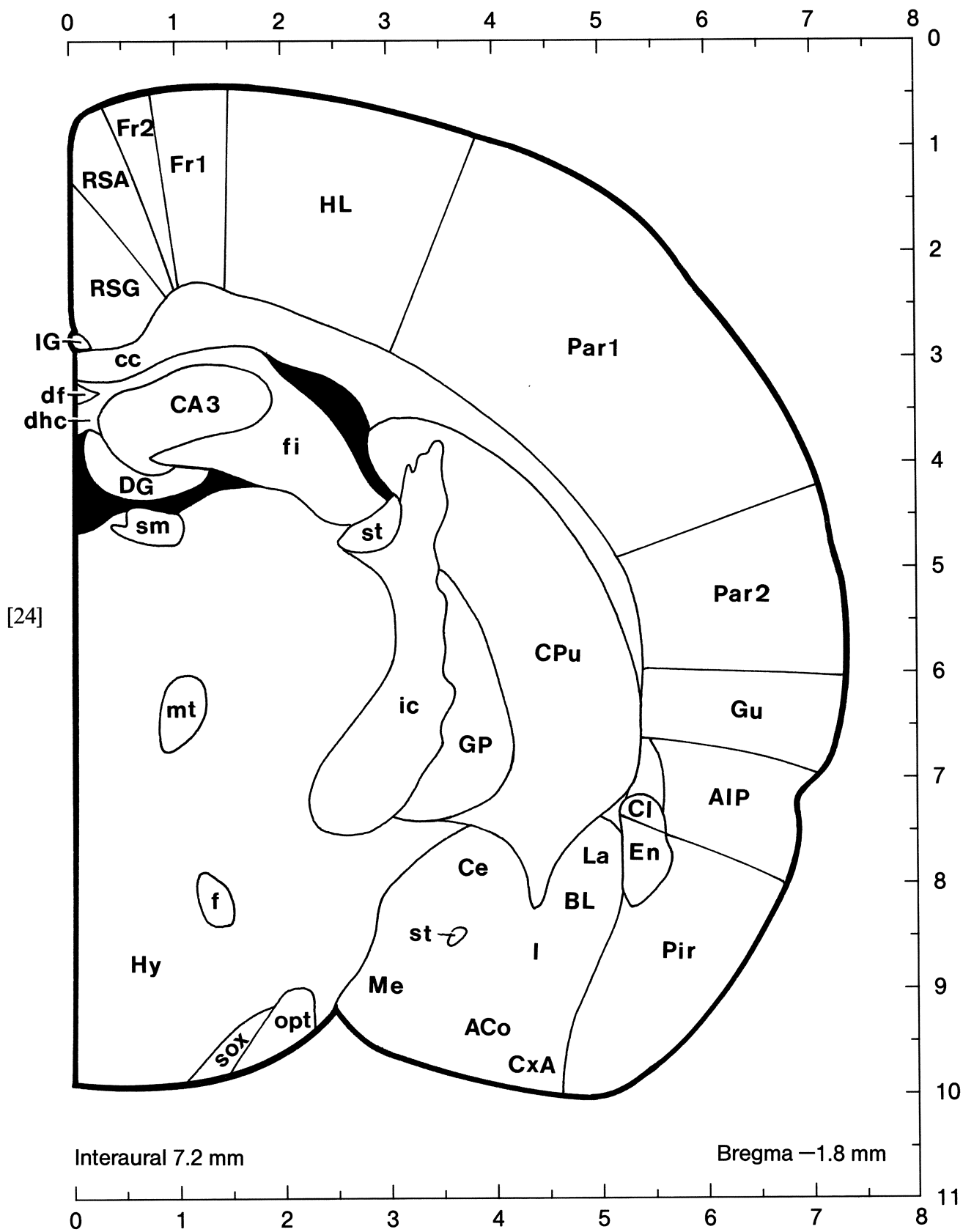


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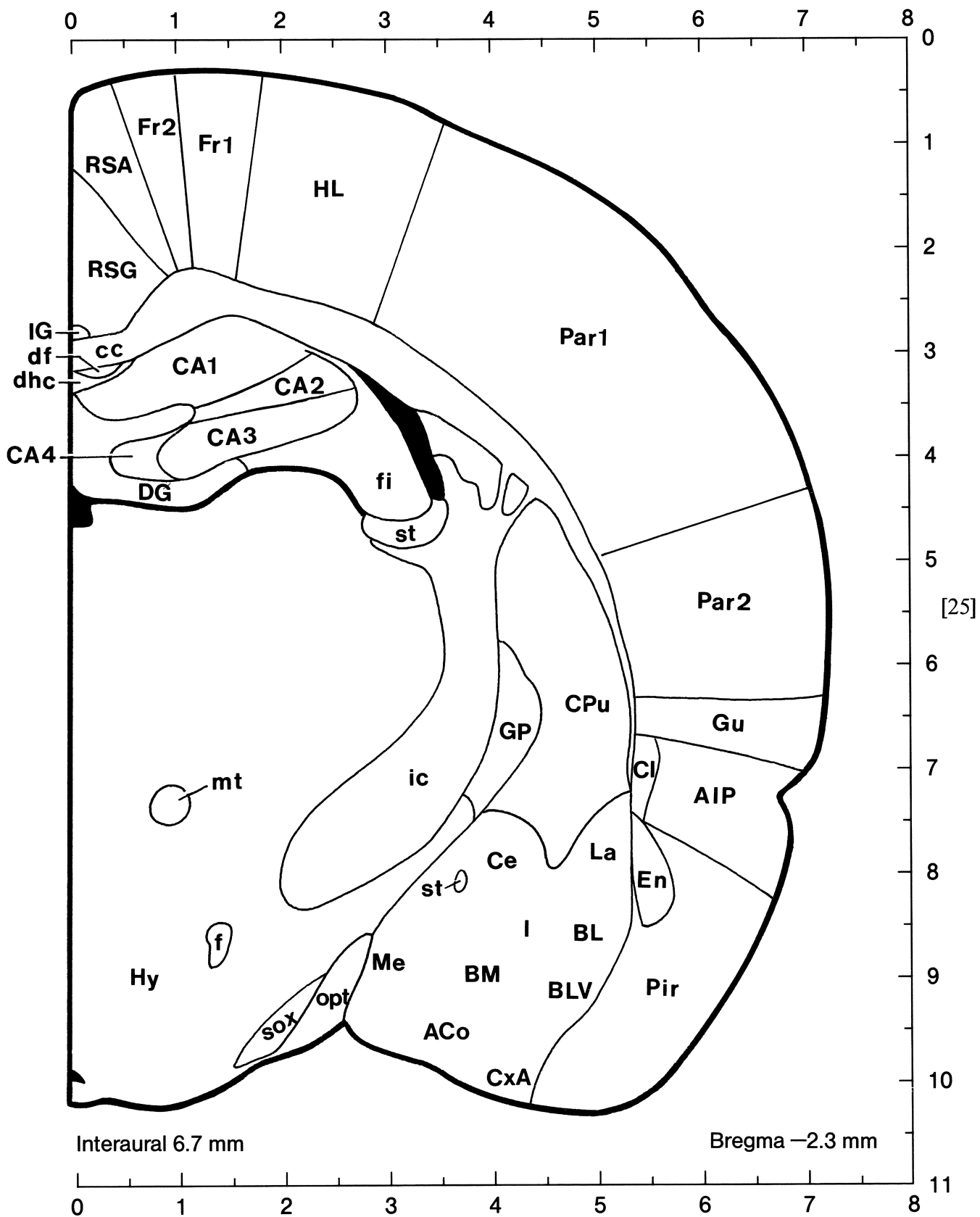


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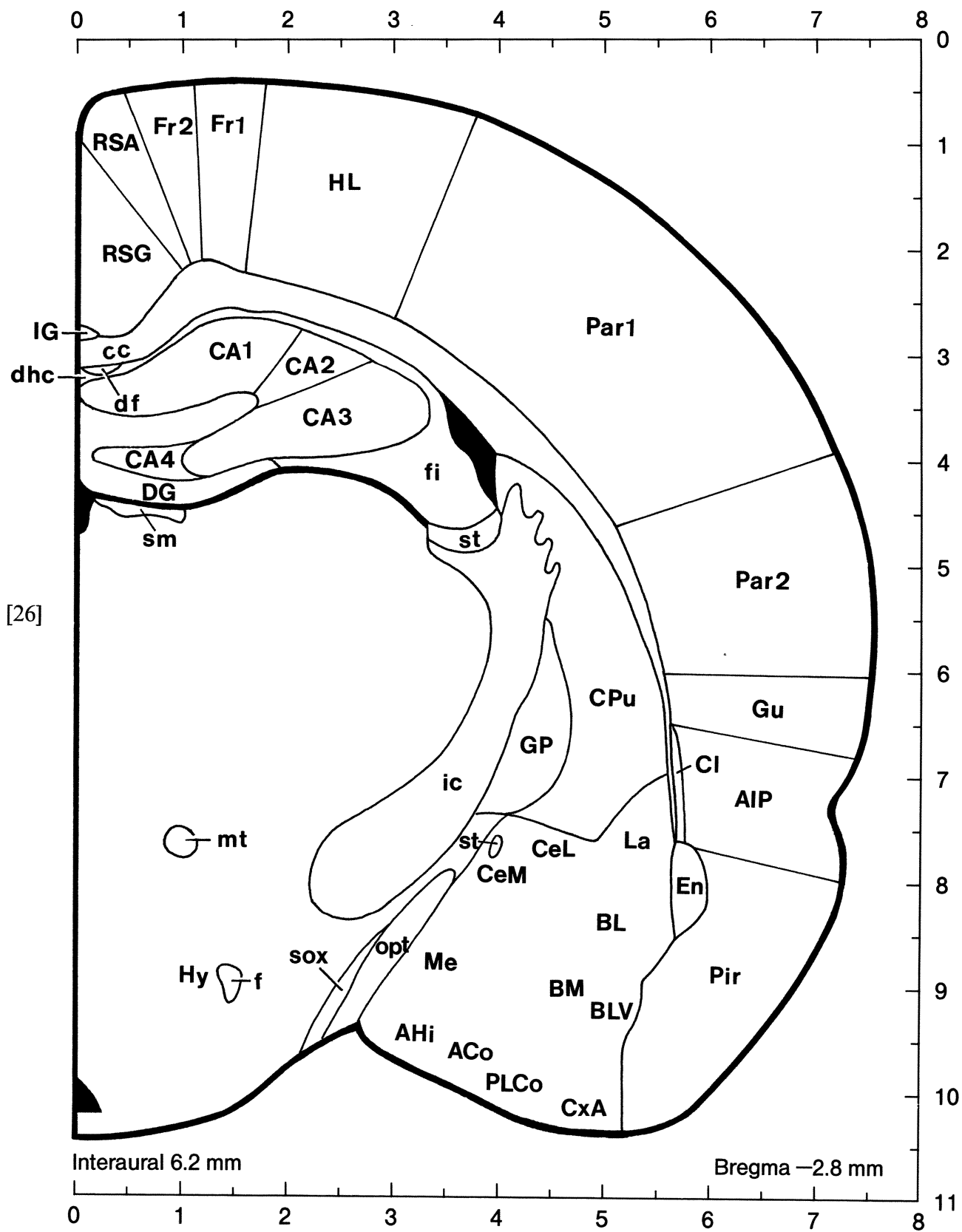


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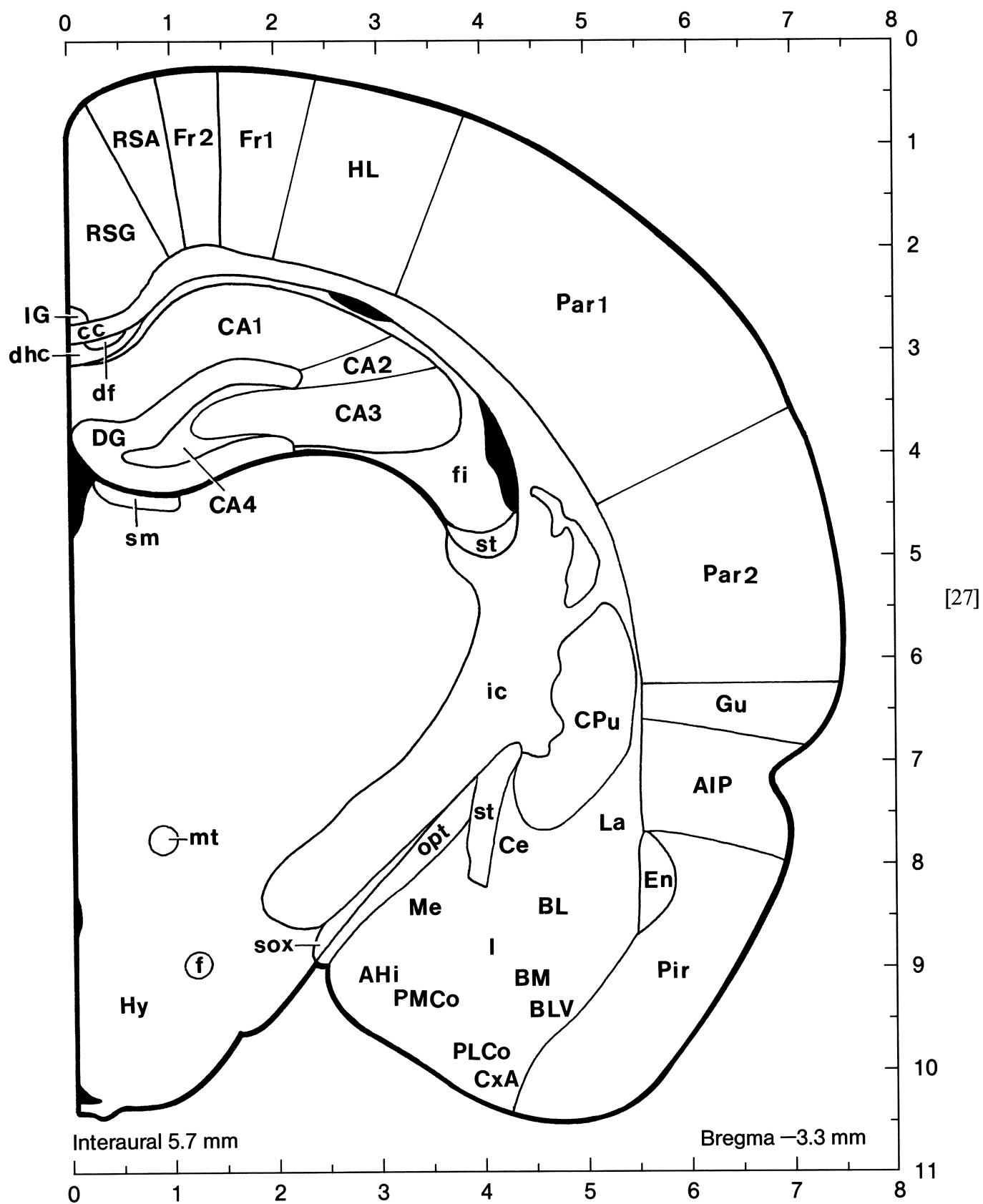


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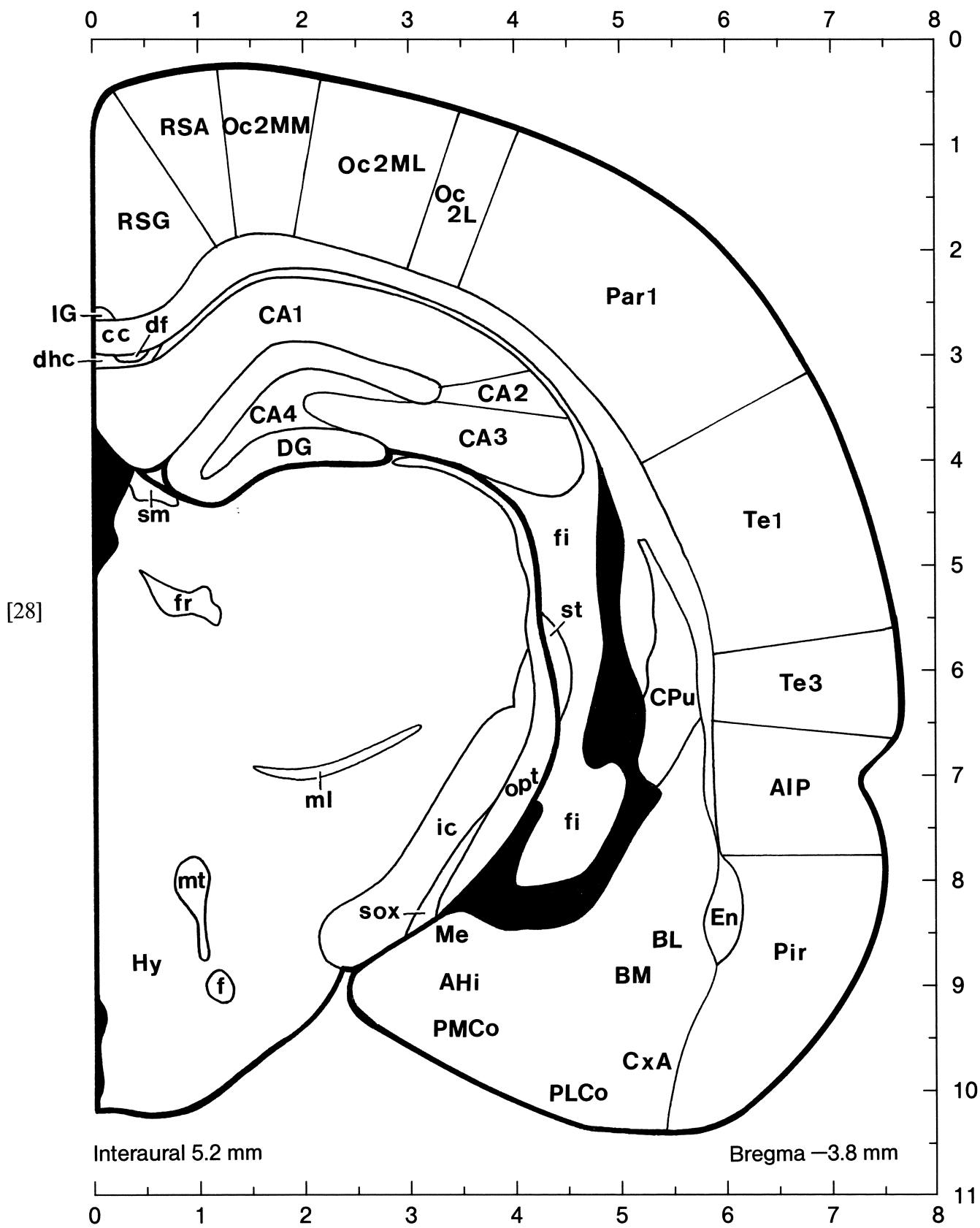


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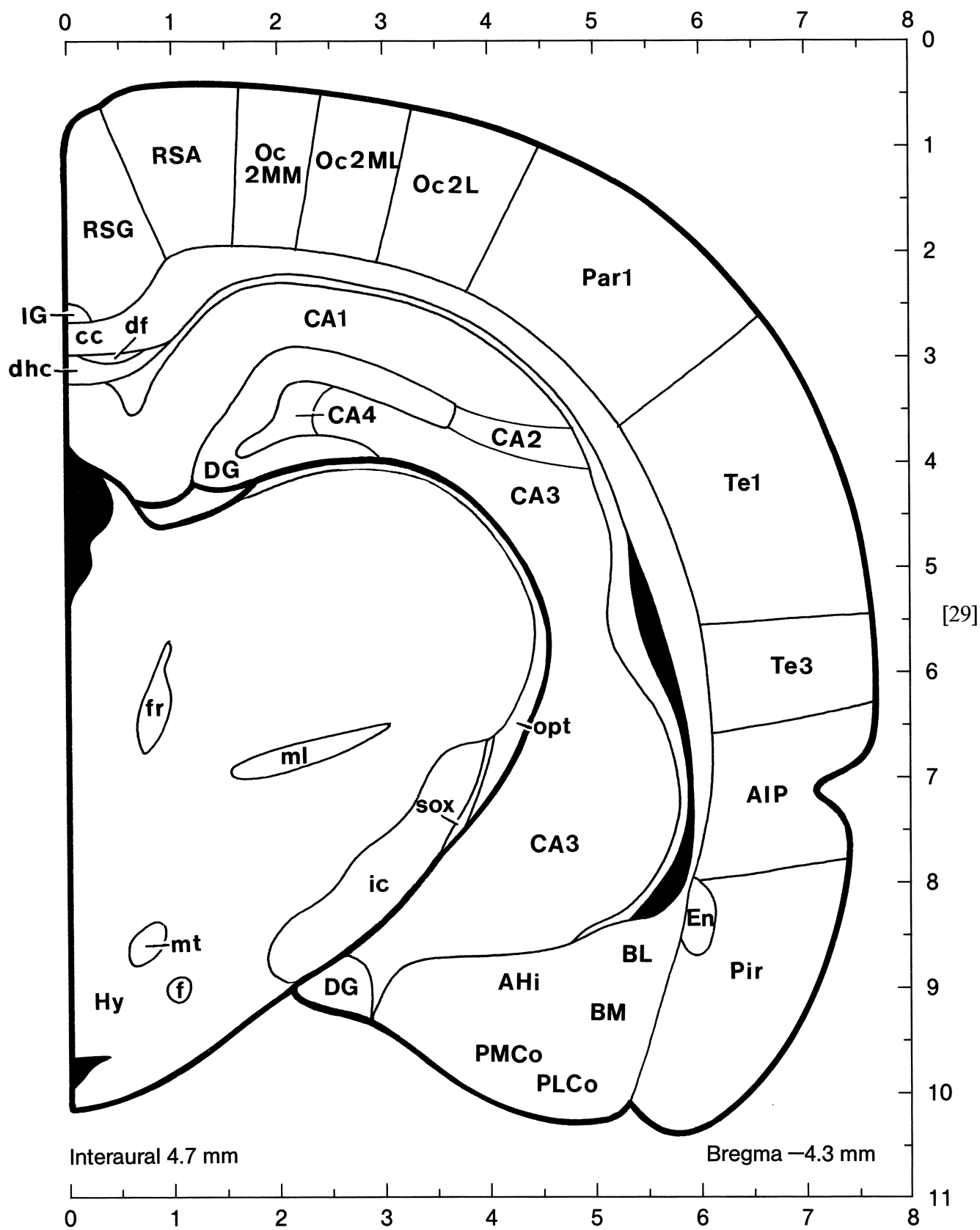


Fig. 21

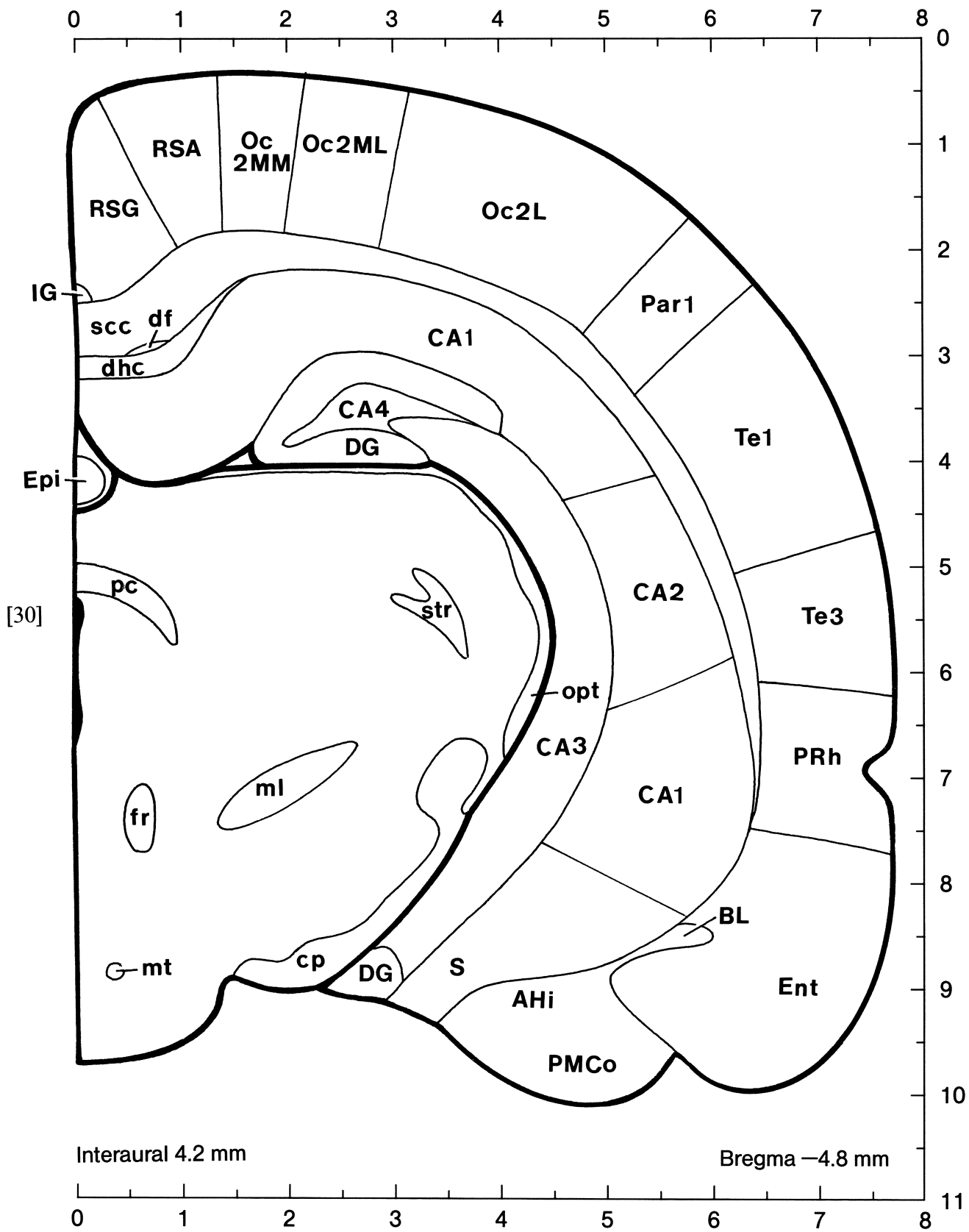
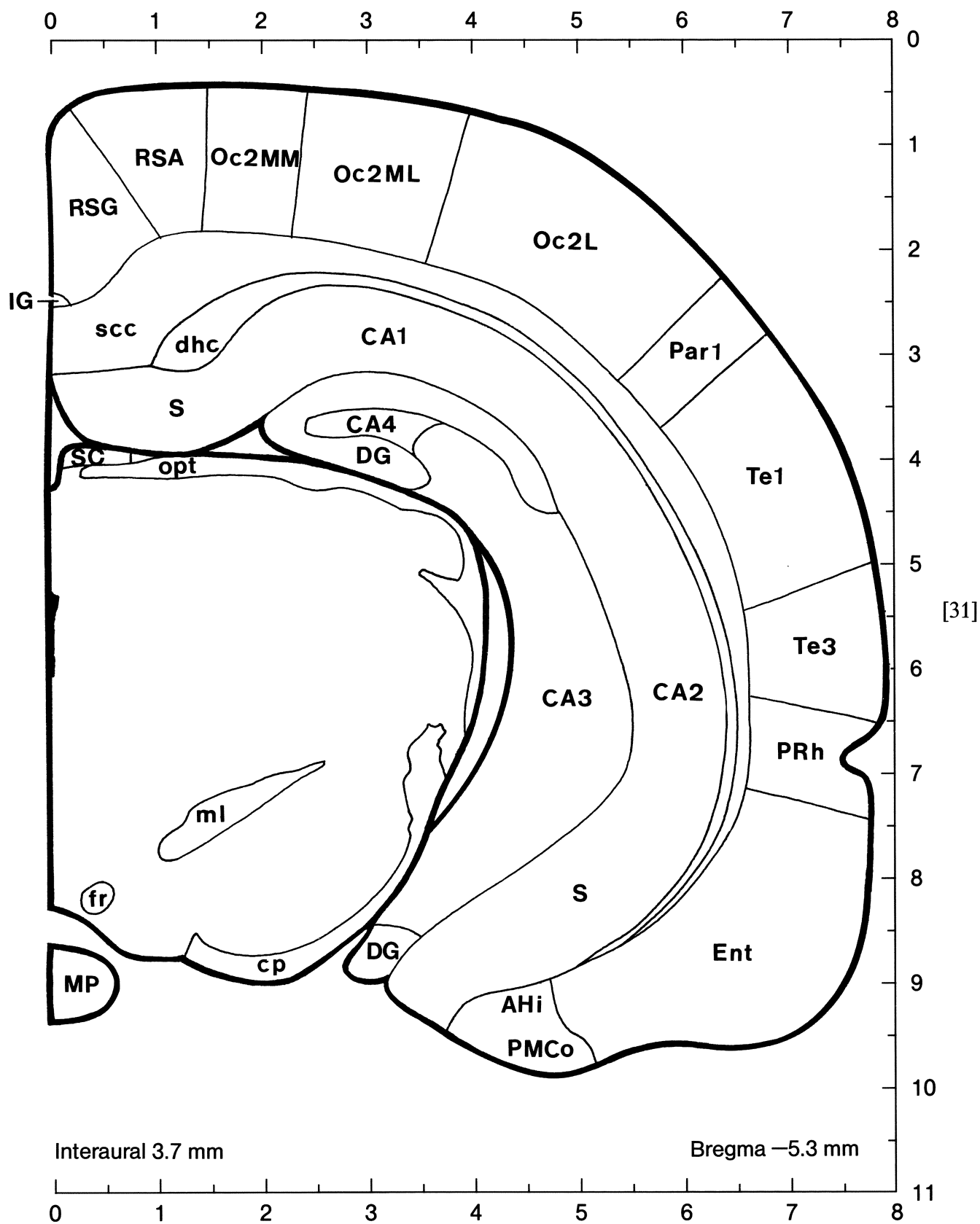


Fig. 22



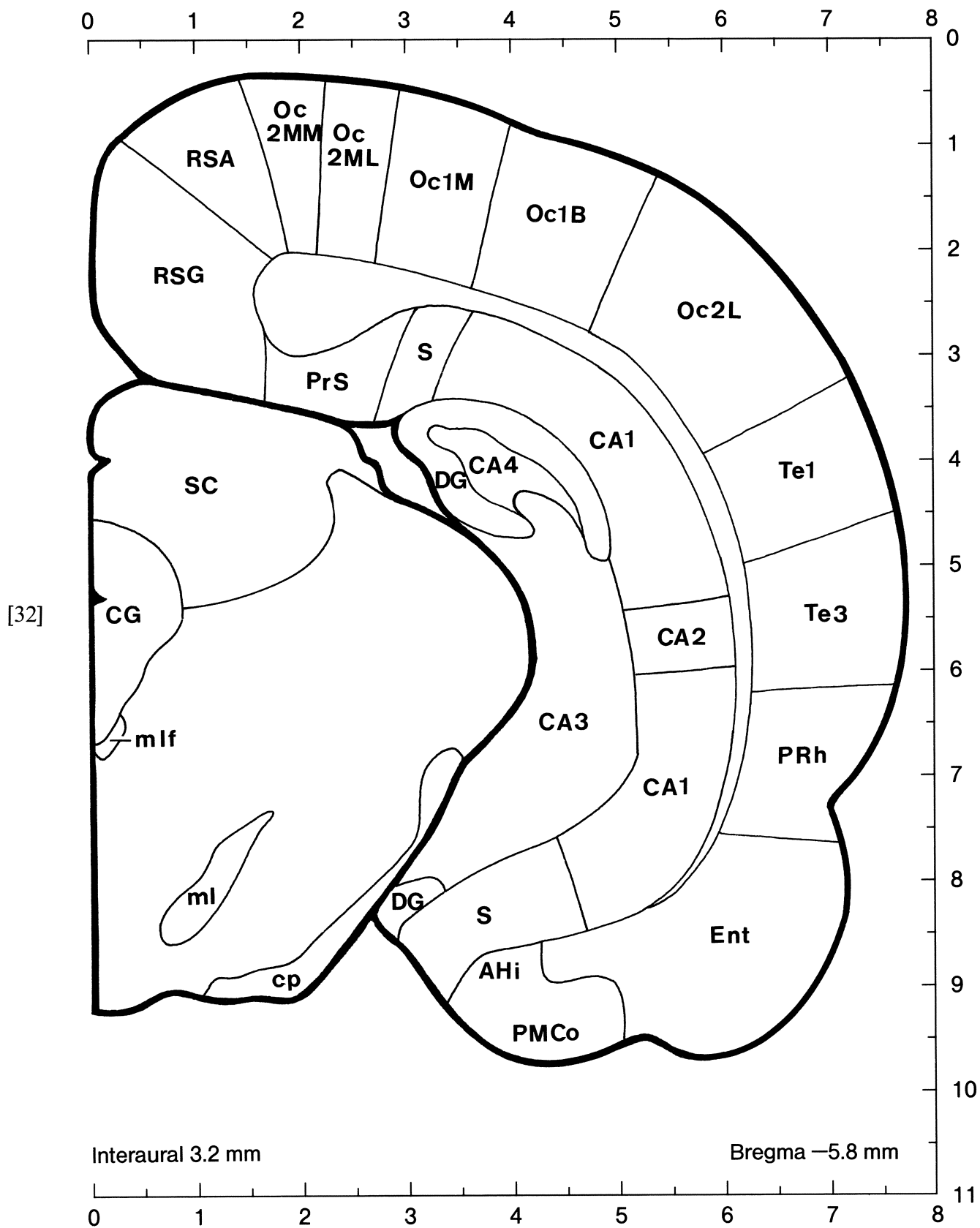


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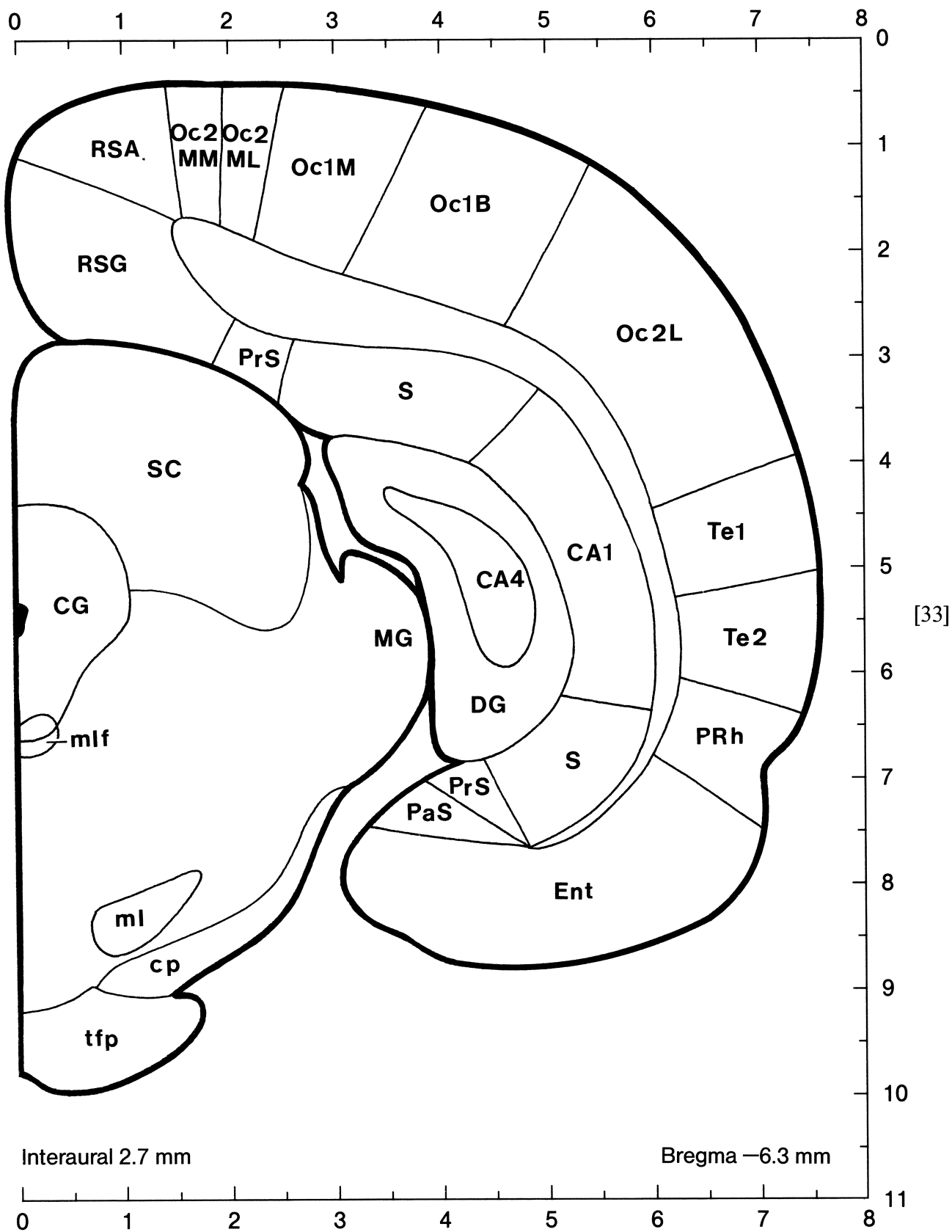


Fig. 25

[34]

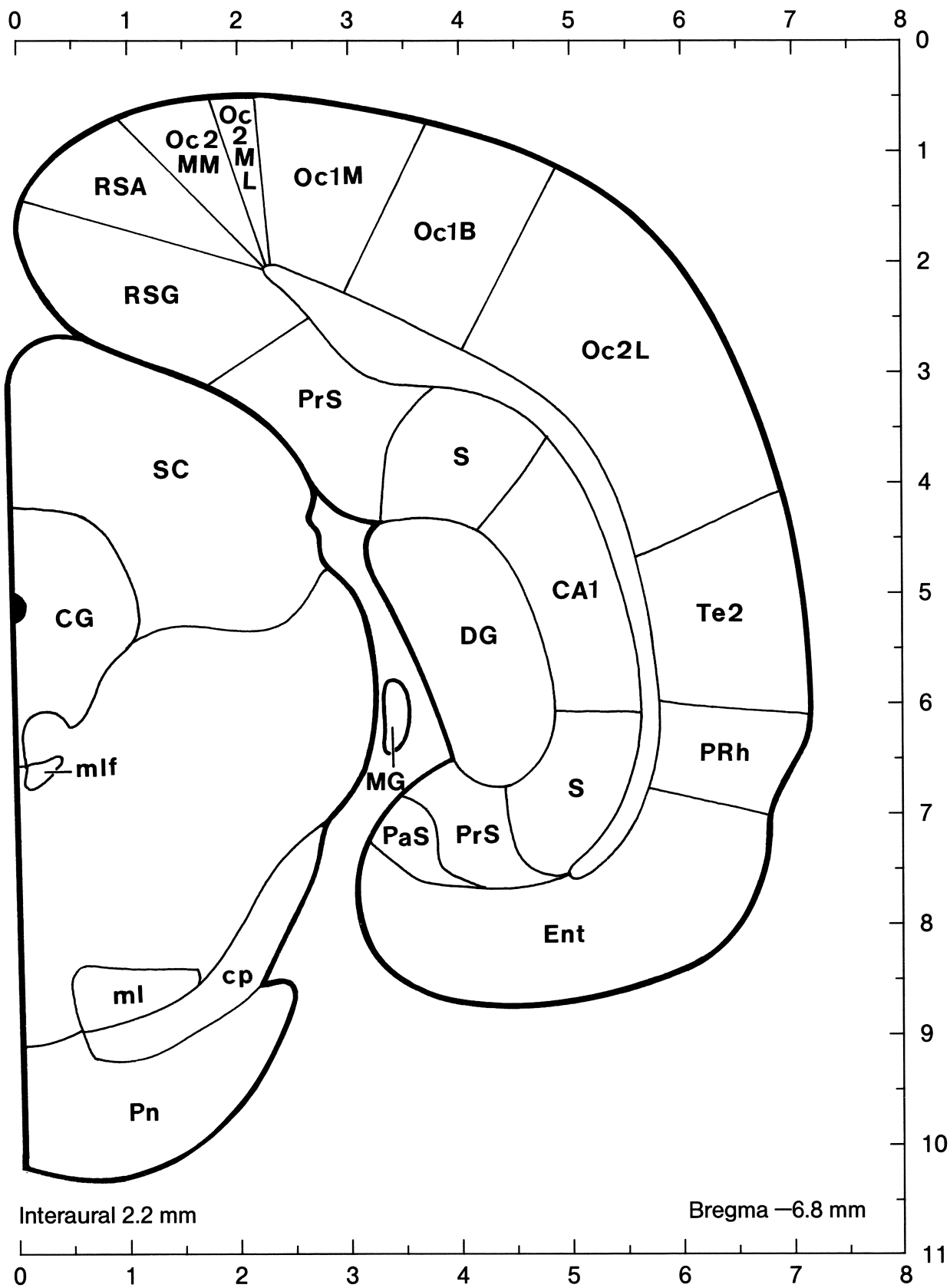


Fig. 26

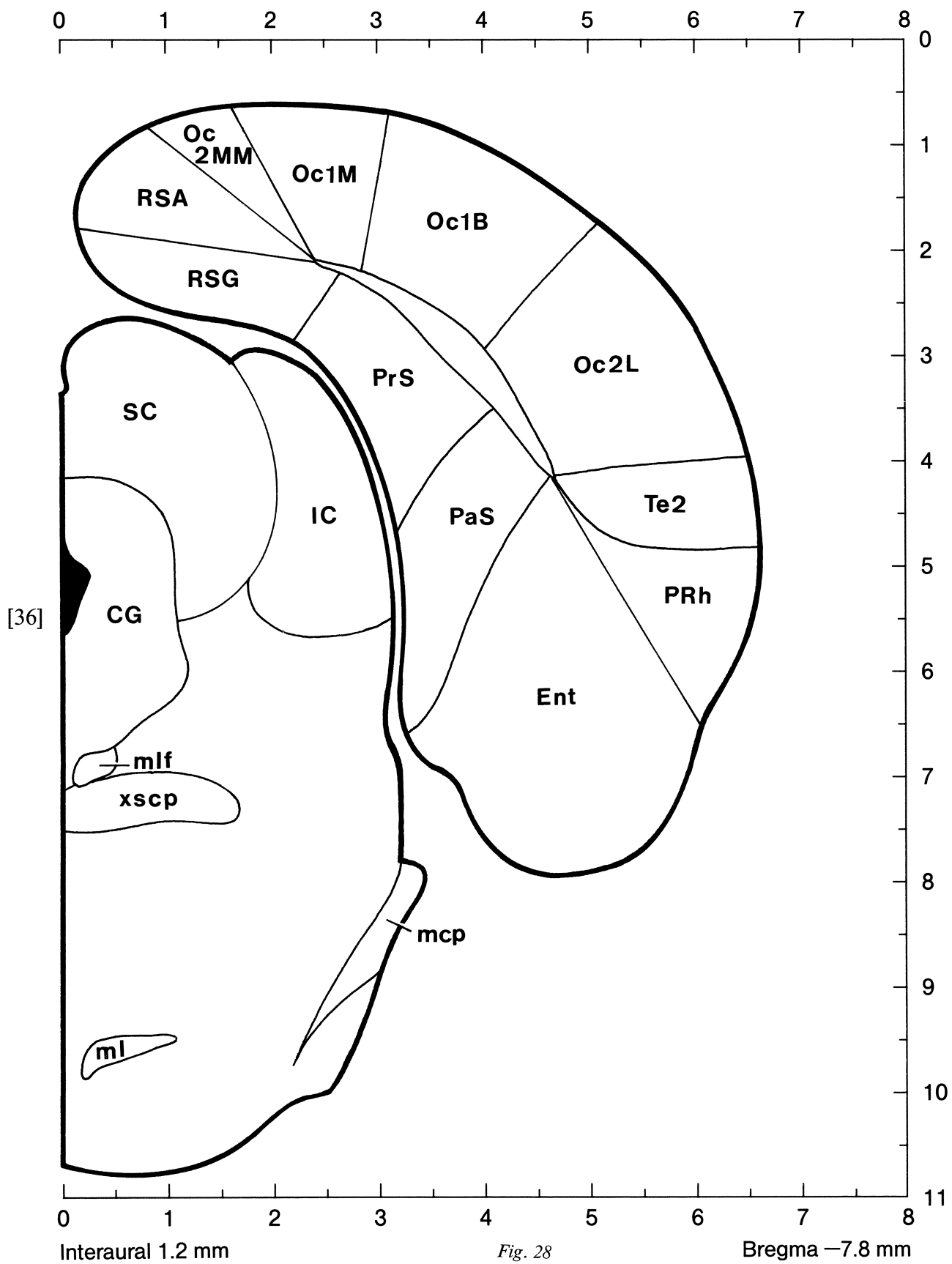
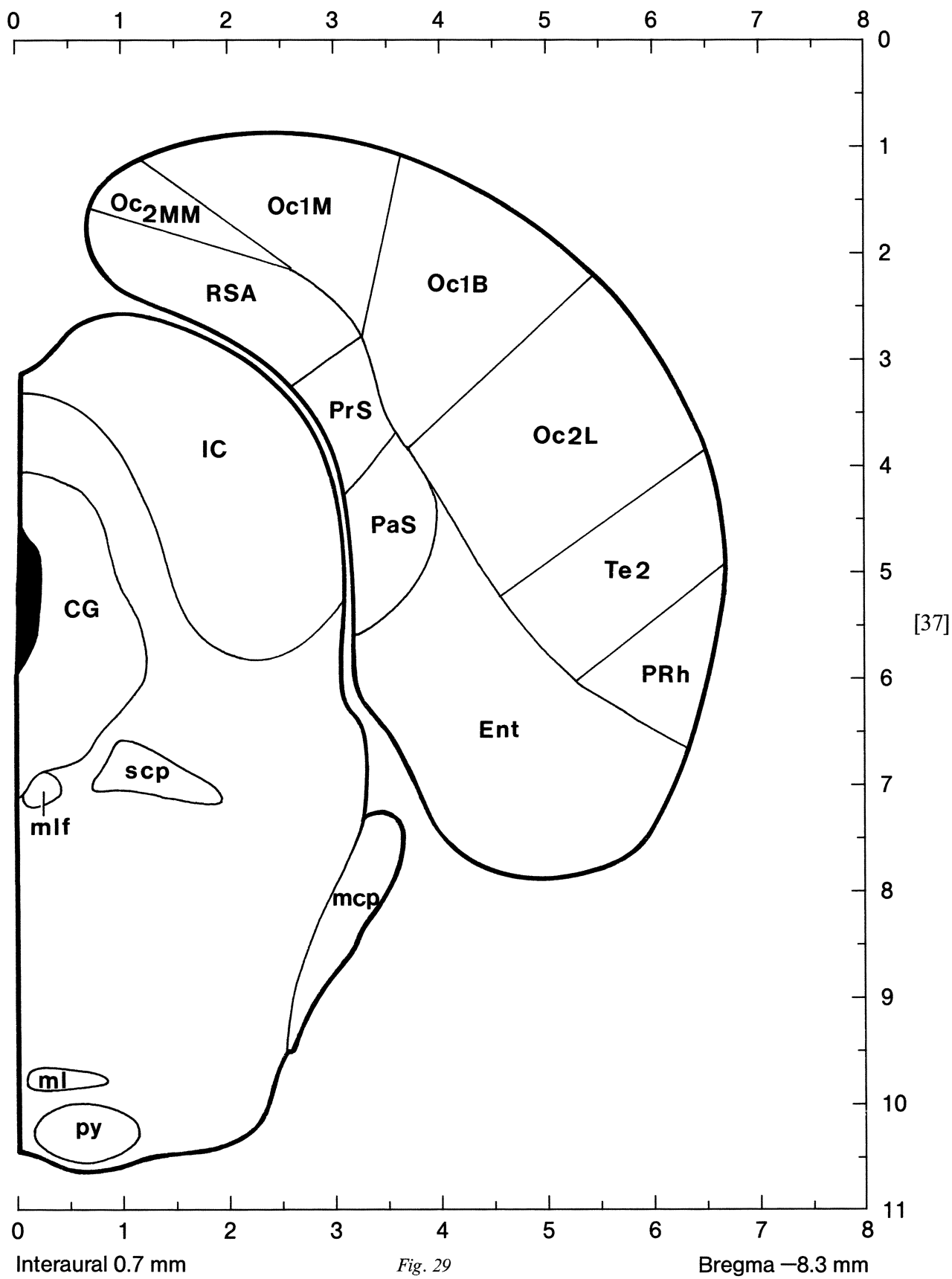


Fig. 28



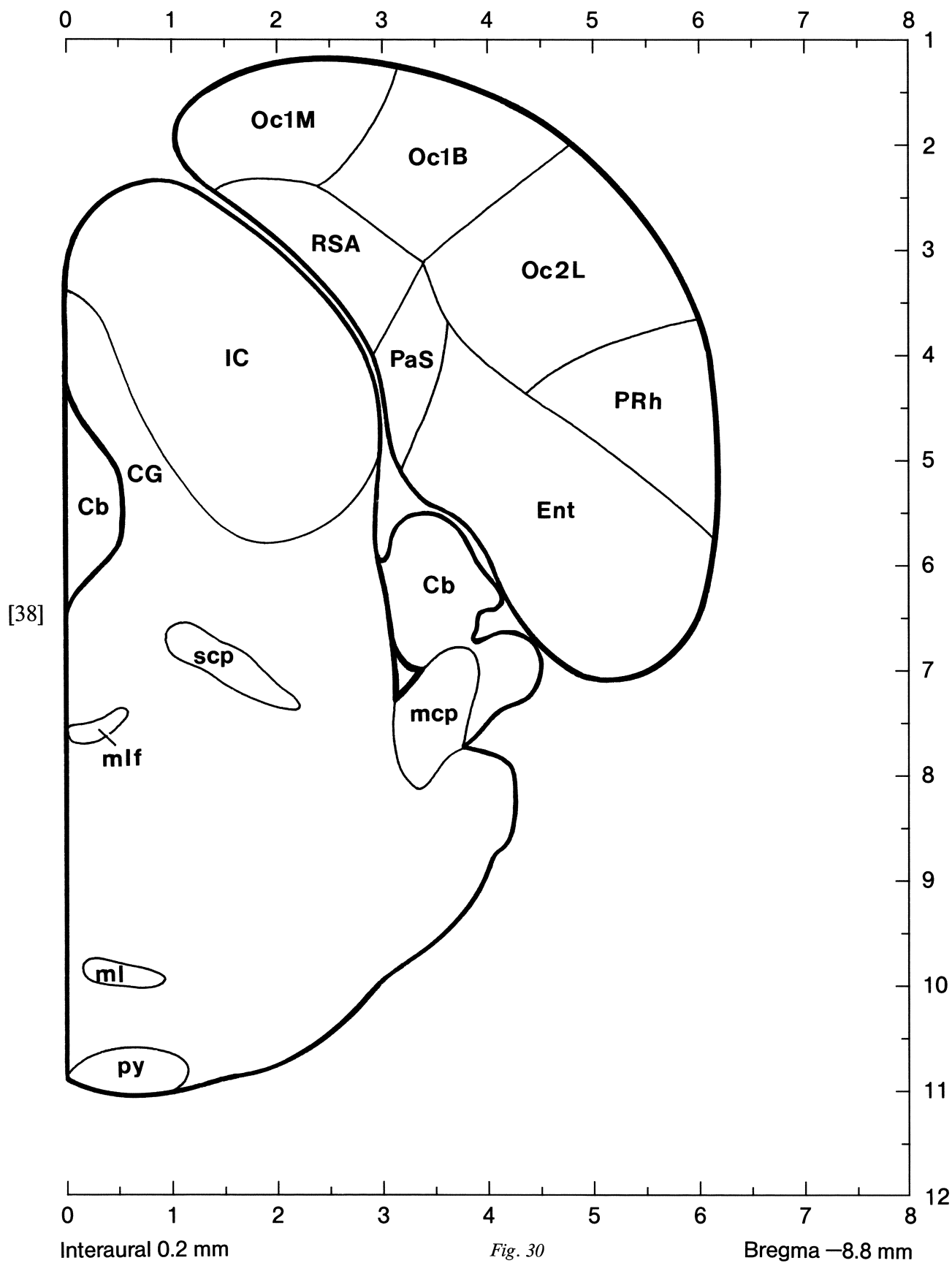


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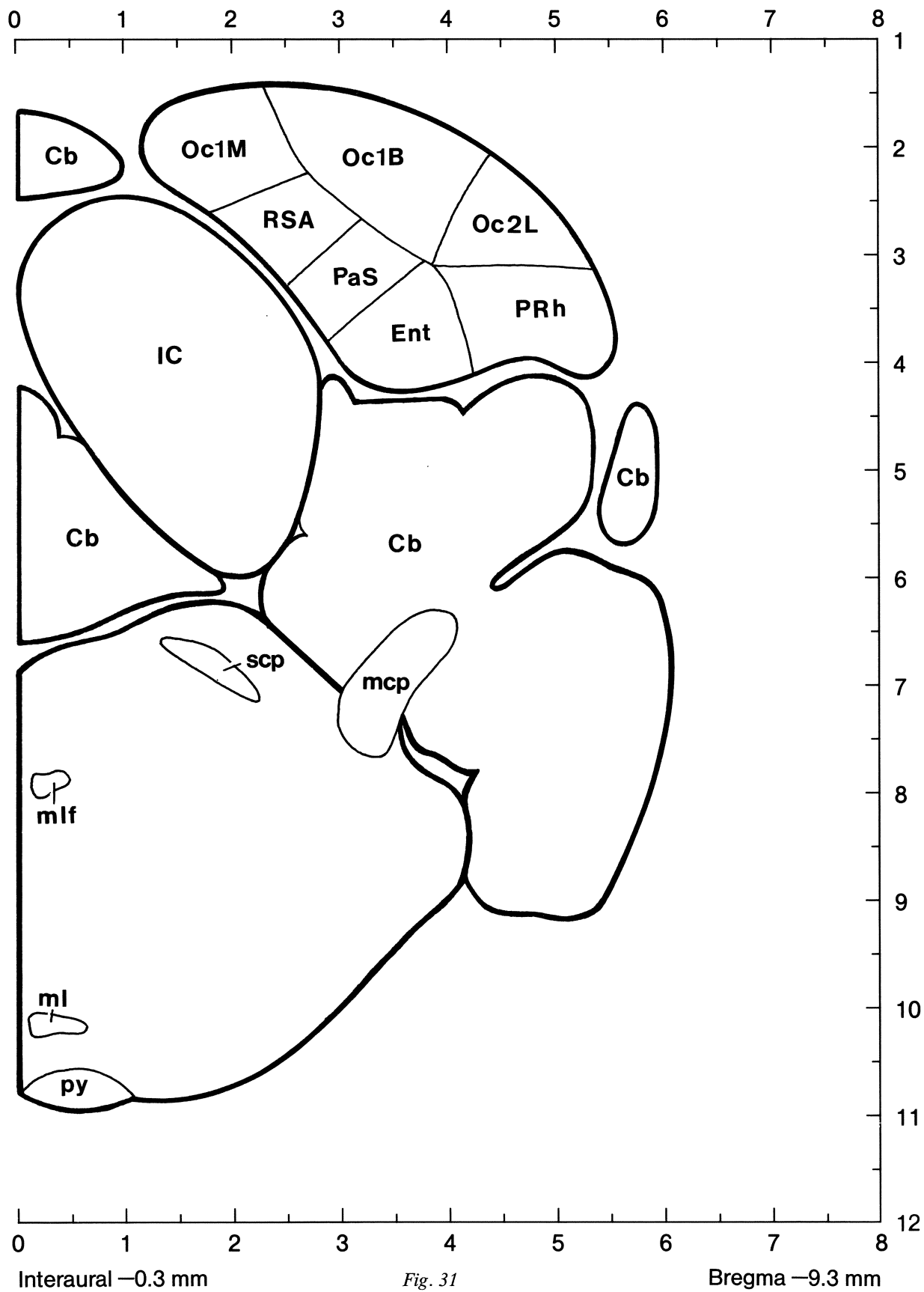


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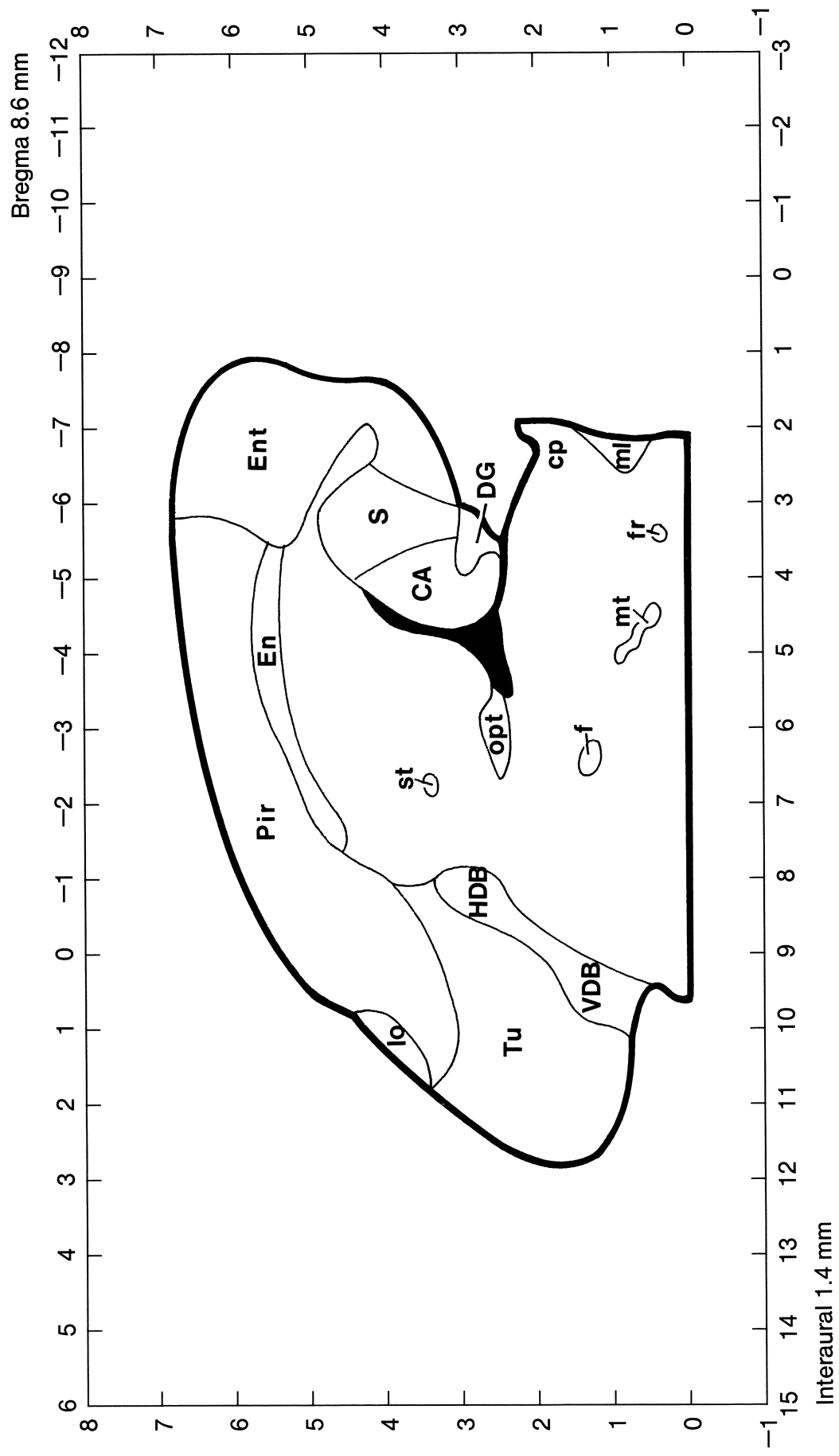


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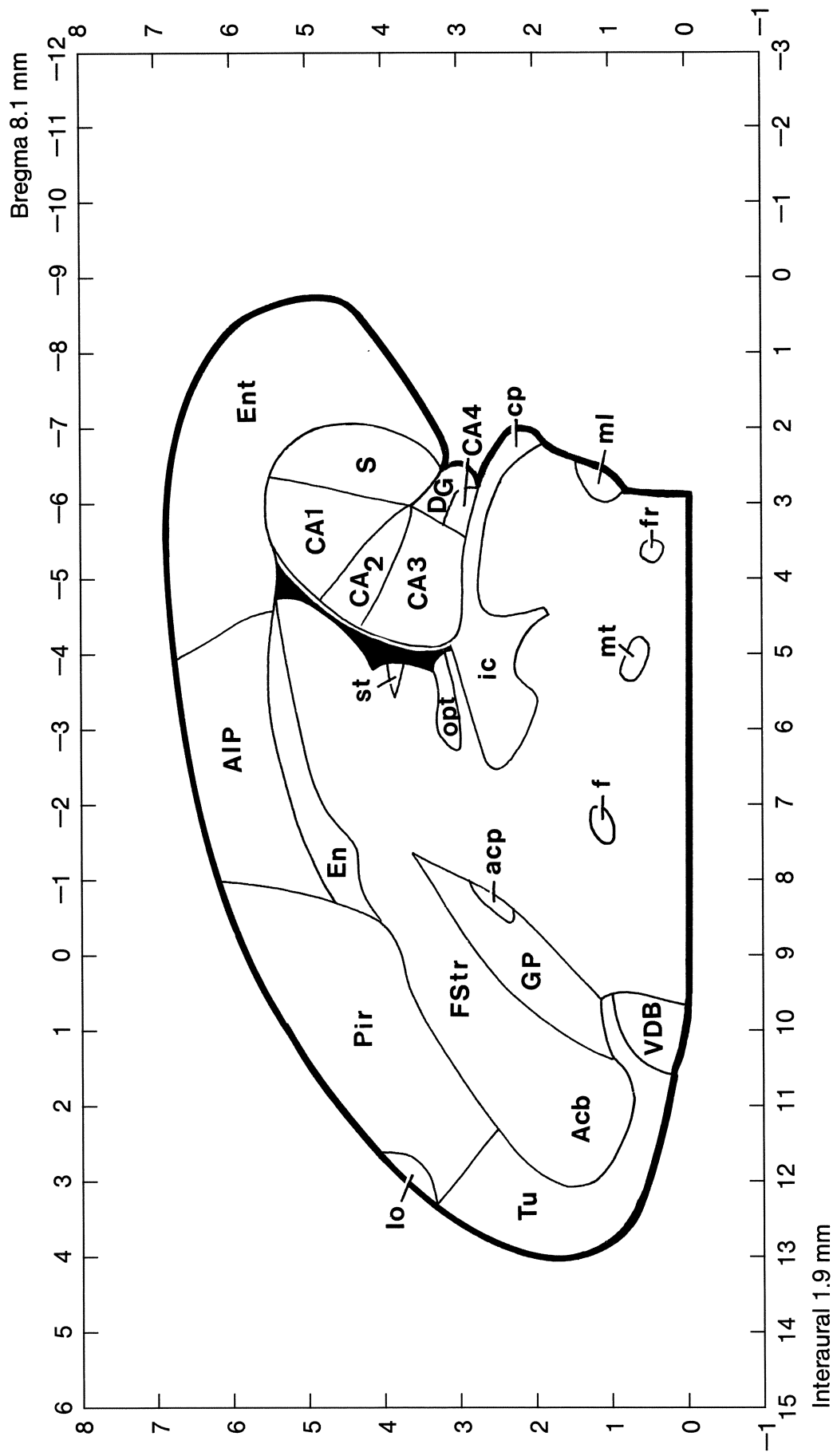


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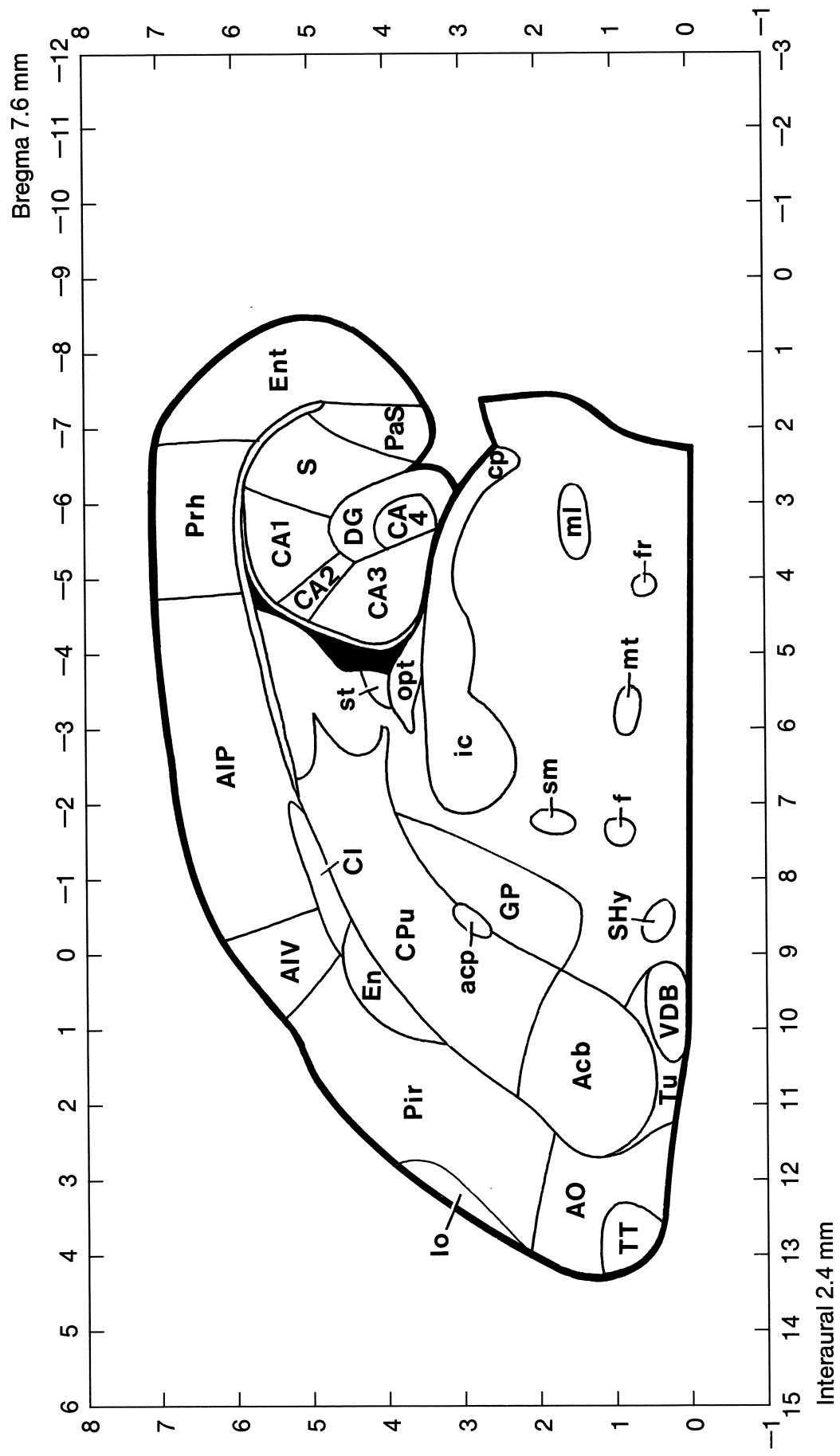


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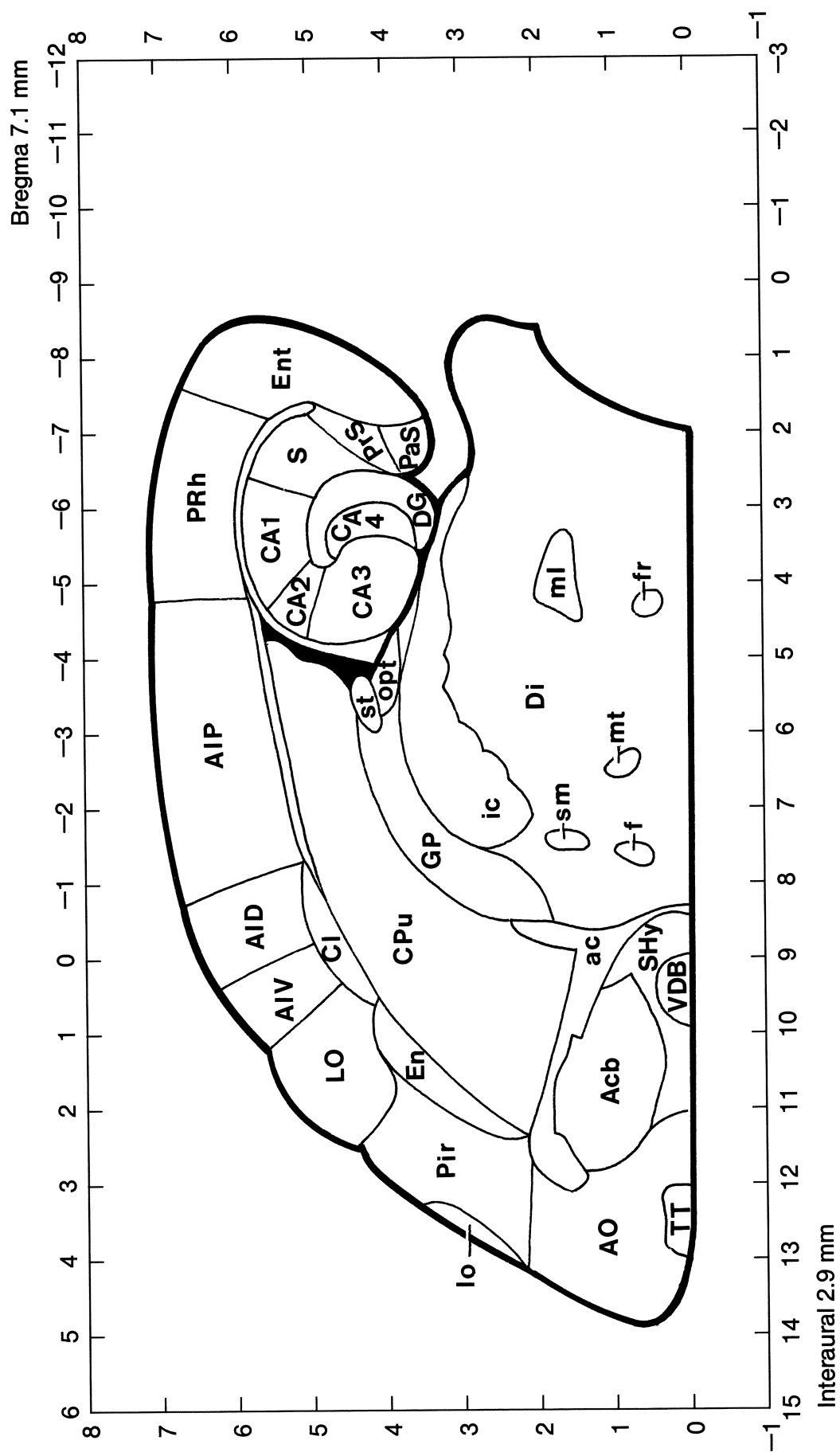


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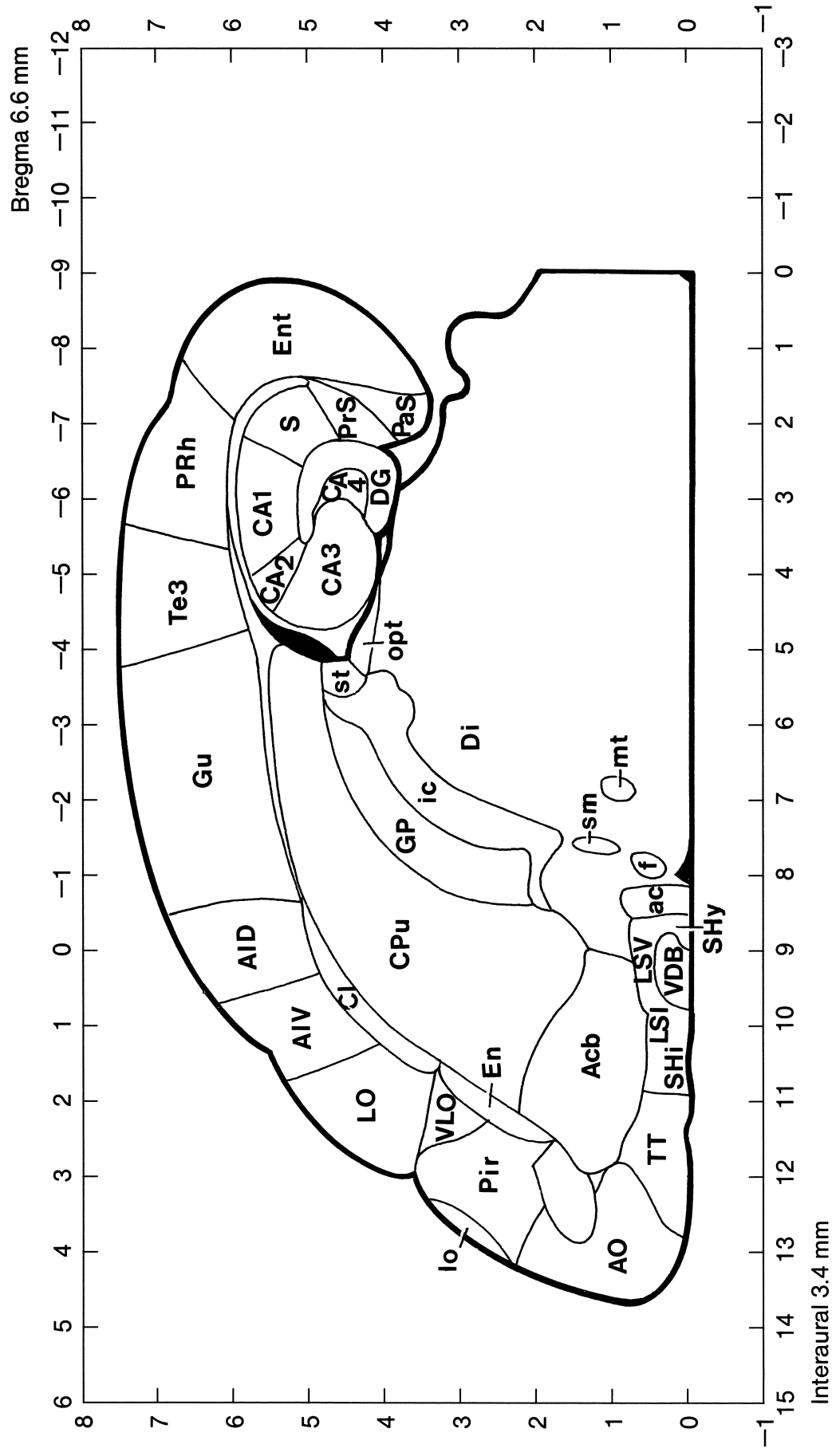


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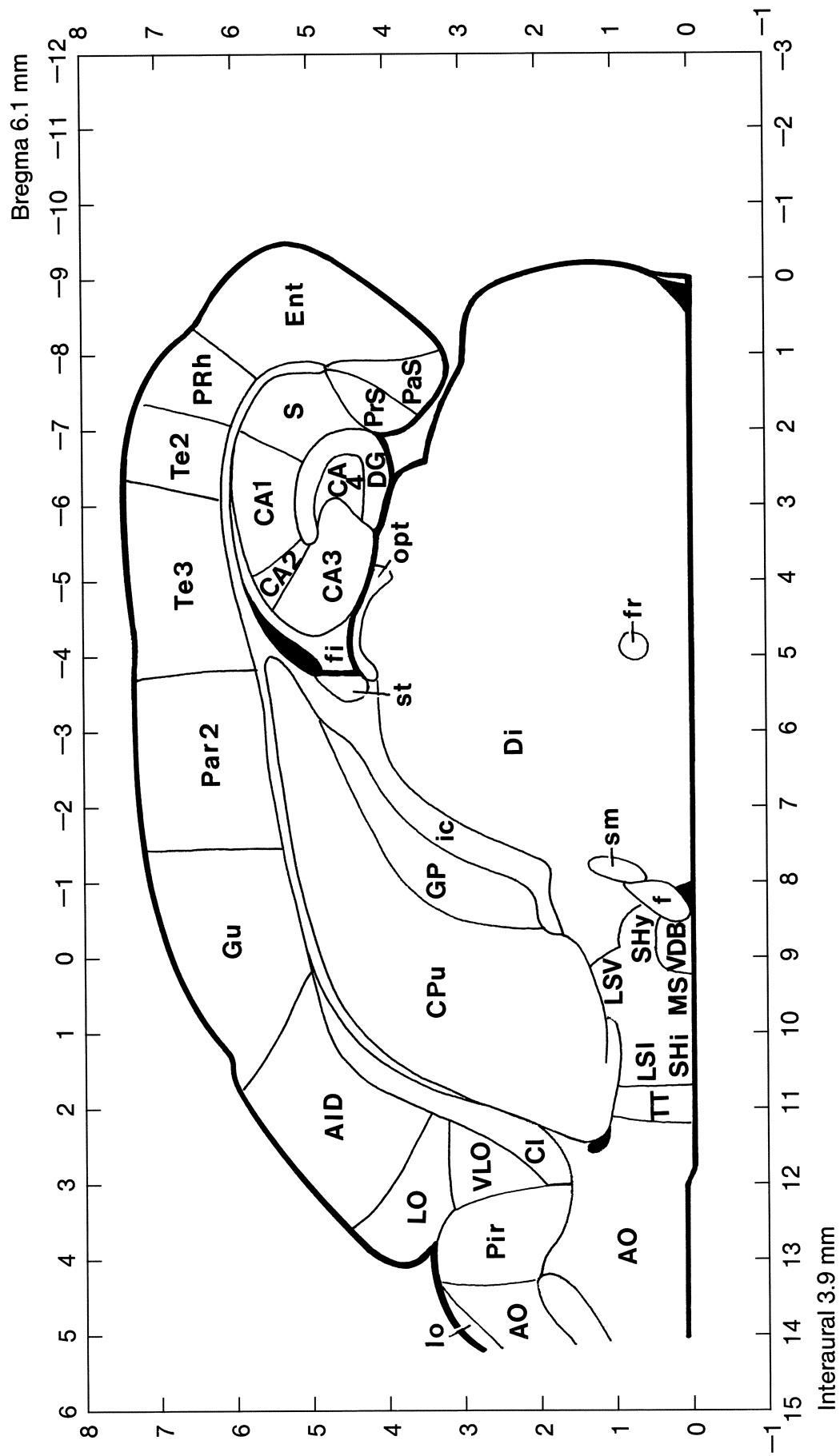


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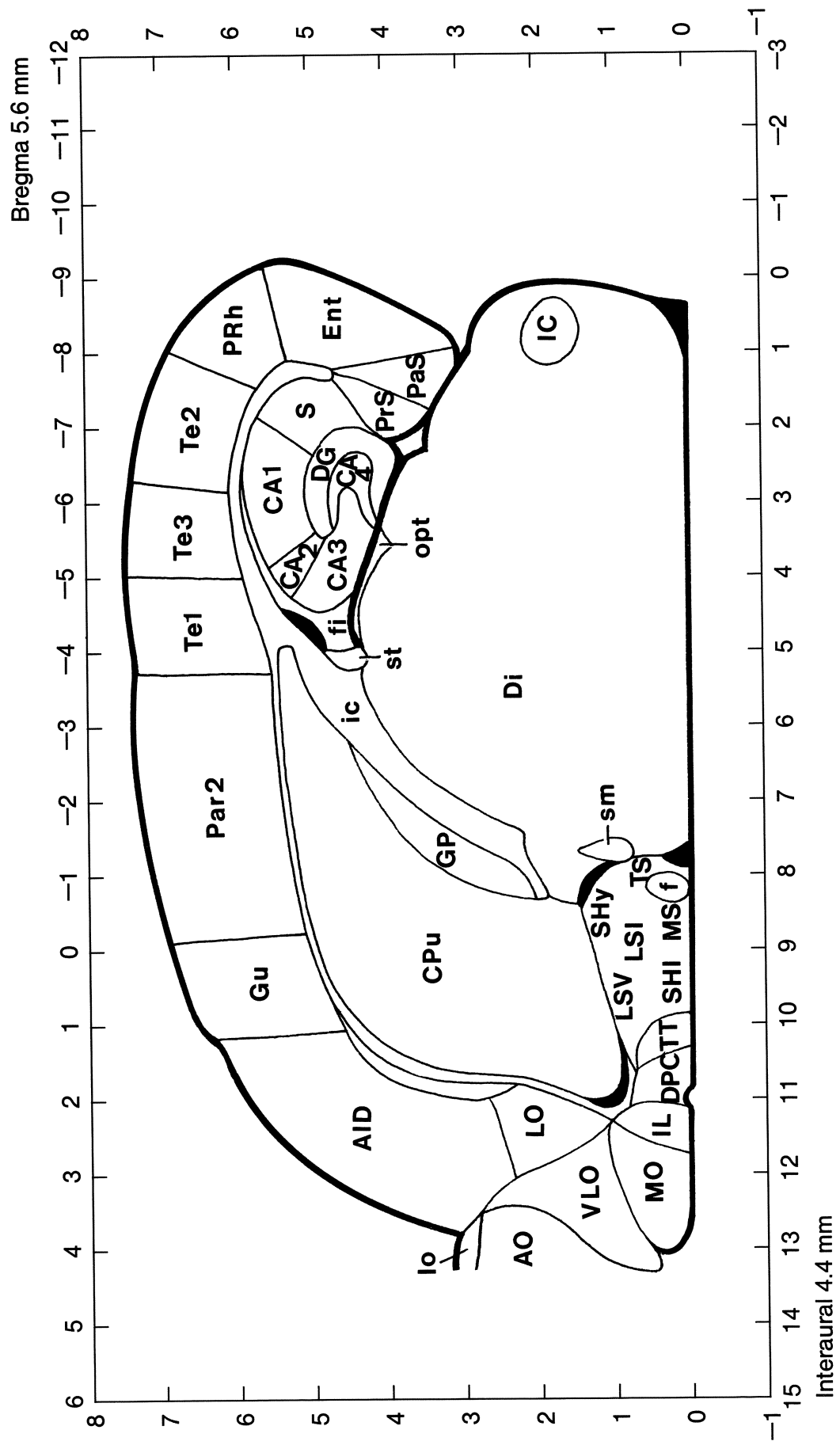


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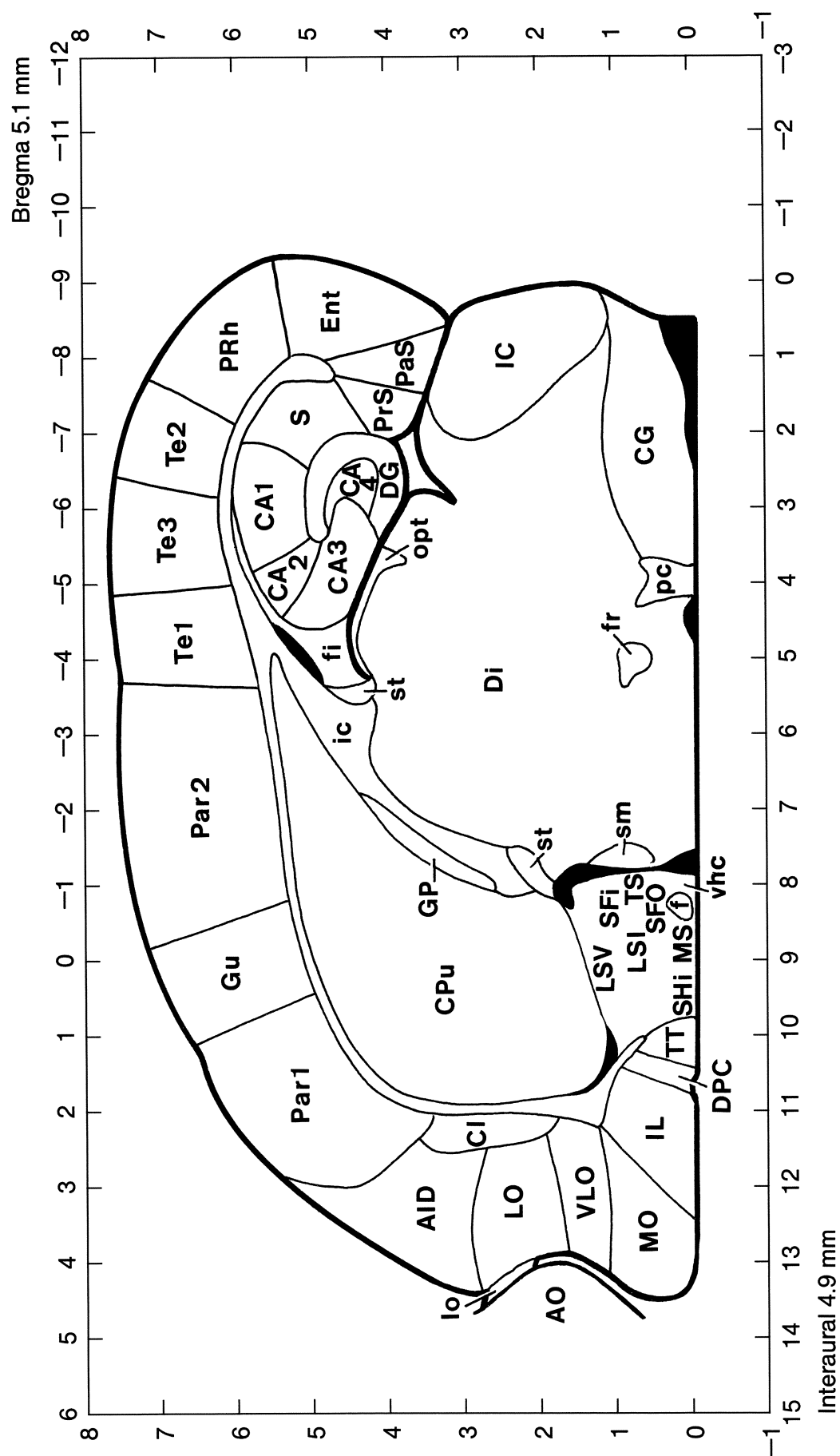


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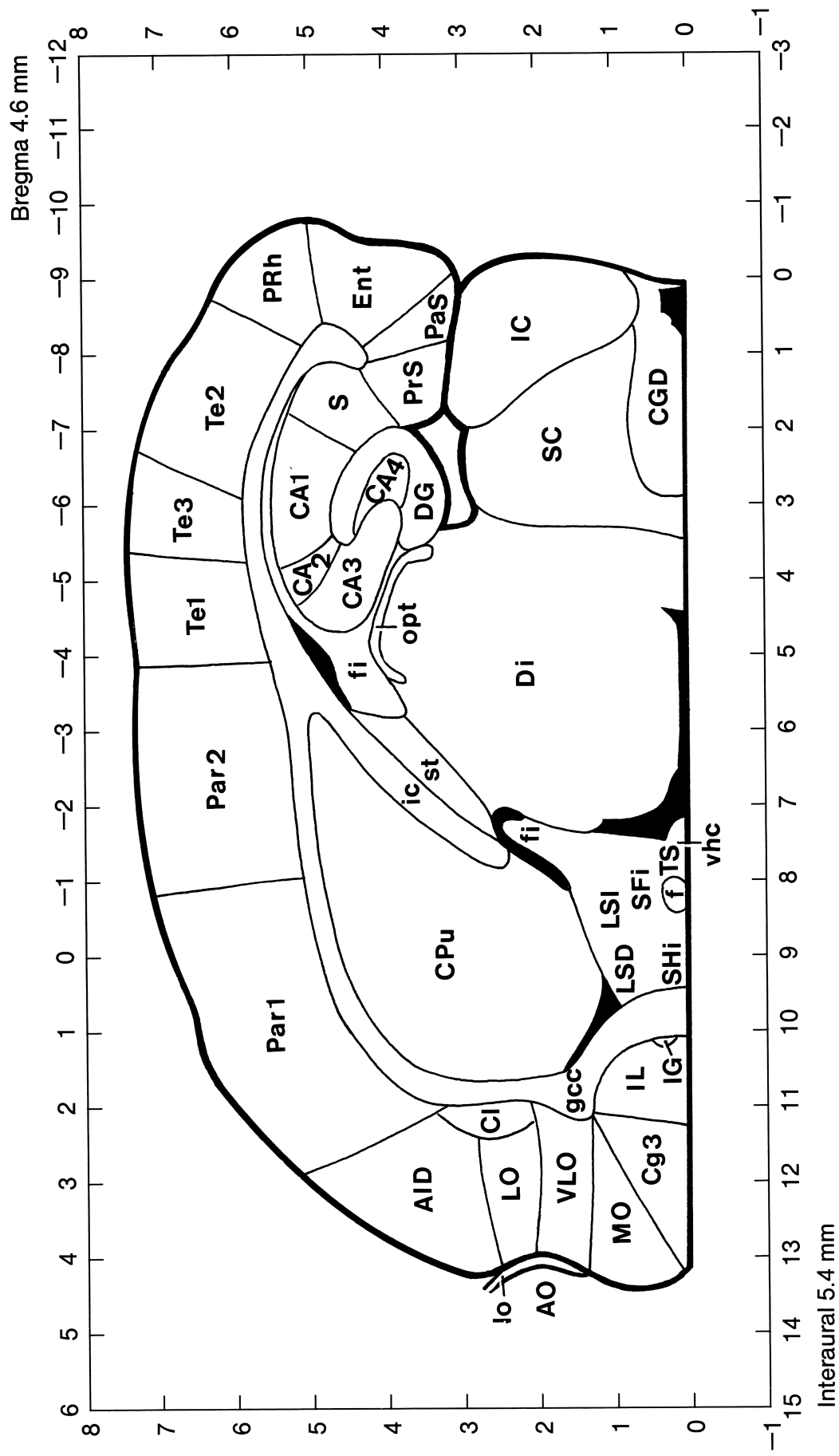


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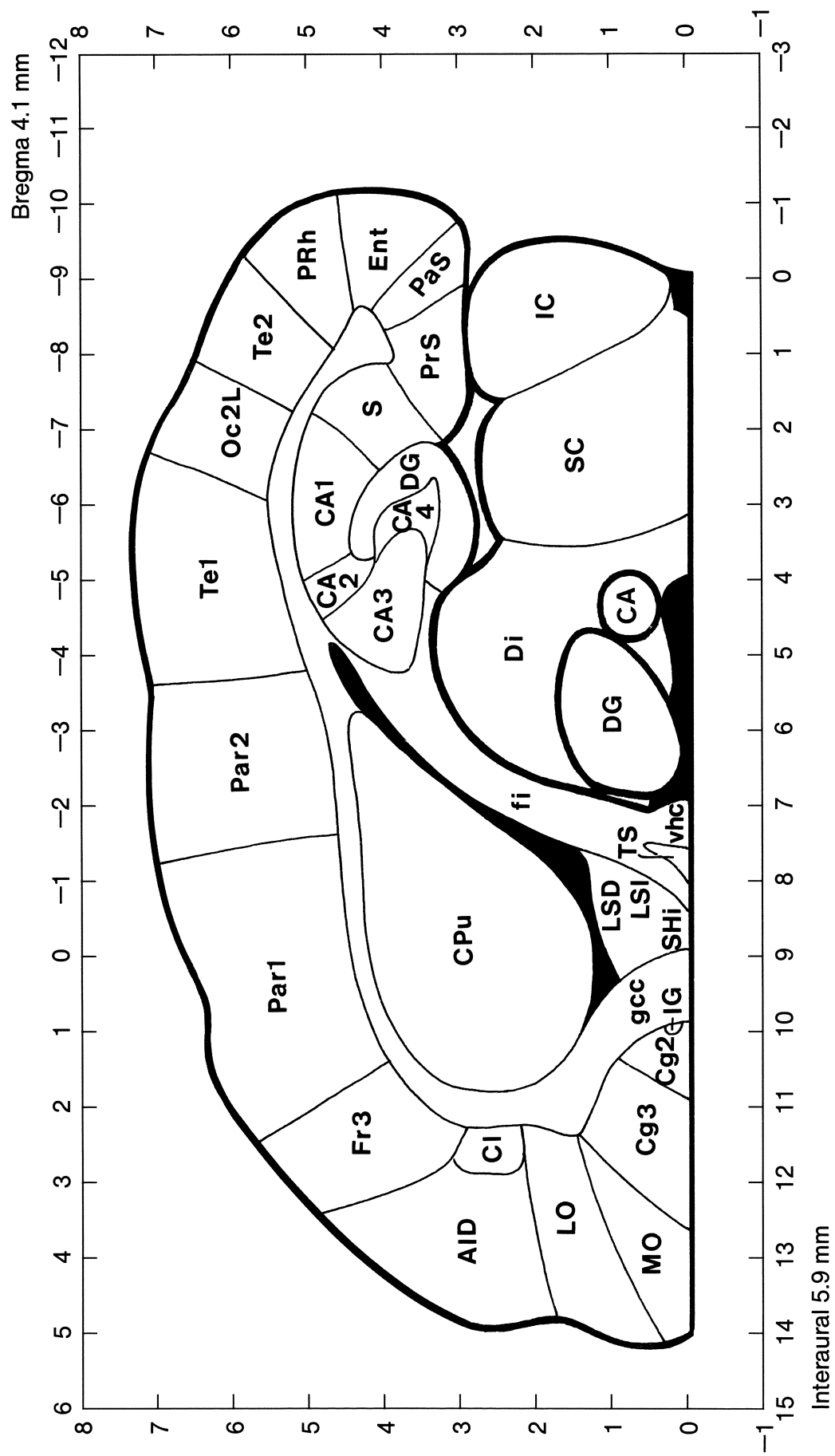


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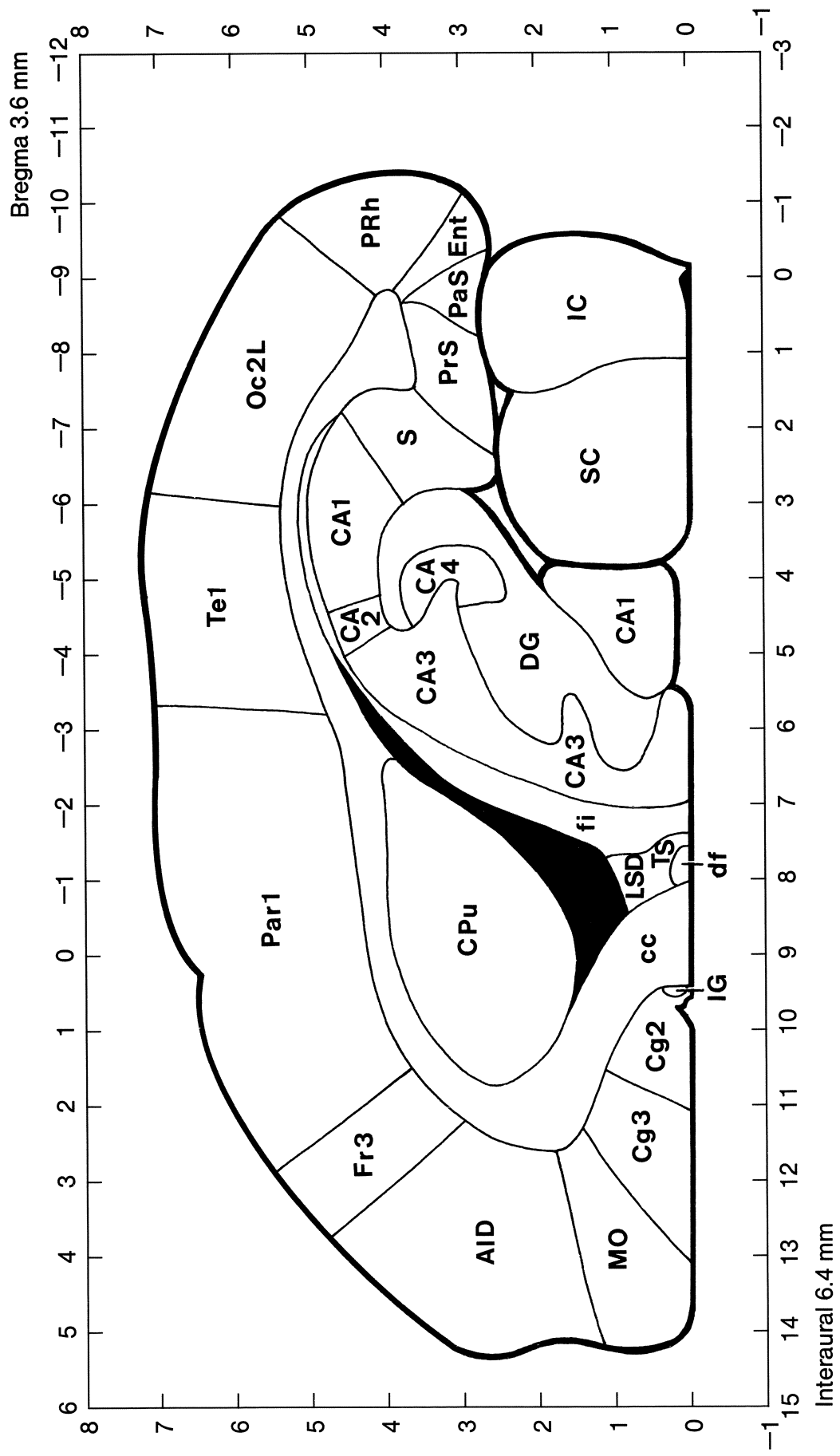


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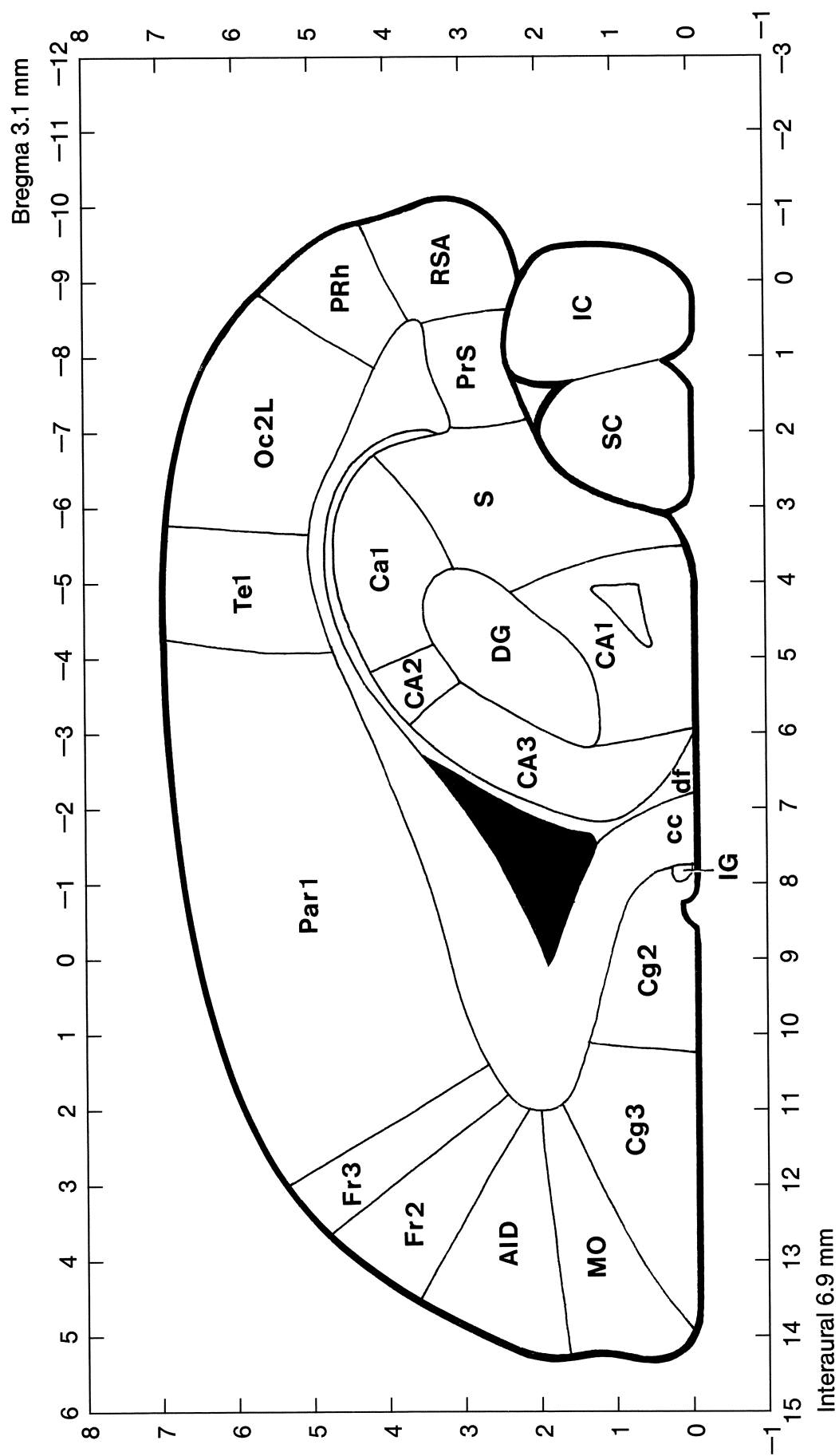


Fig. 43

[51]

[52]

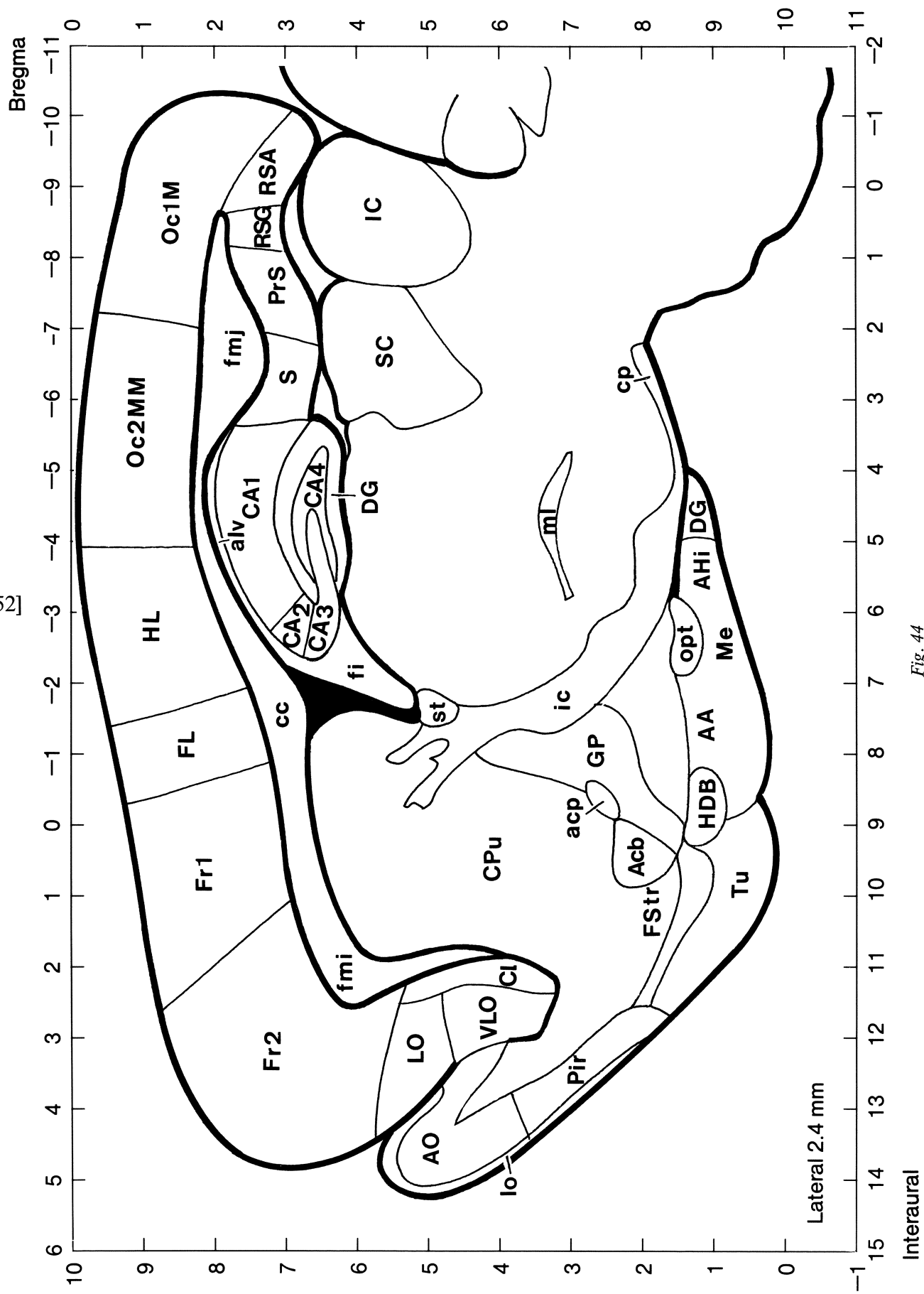


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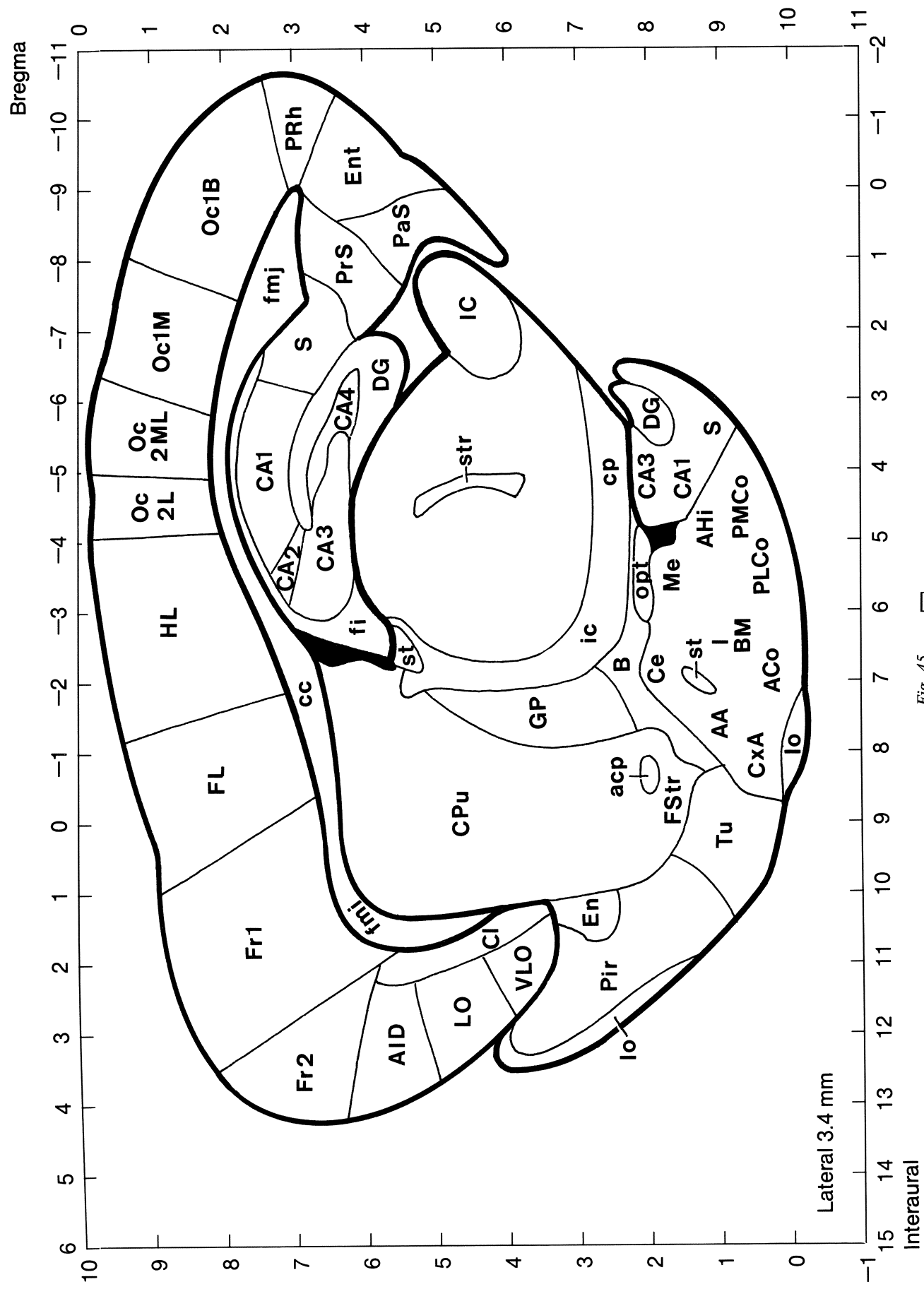


Fig. 45

Cortical Maps in Stereotaxic Coordinates

The cortical maps in Figs. 46–49 have been prepared from the coronal (Figs. 2–31) and horizontal (Figs. 32–43) sections by orthogonal projection. This provides lateral (Figs. 46 and 49), dorsal (Fig. 47), and medial (Fig. 48) views of the rat cortex in stereotaxic coordinates. Figure 49 shows only part of the lateral view, because the most dorsal and basal sections of the horizontal series do not admit of complete reconstruction because of the tangential sections at the top and bottom of the hemisphere. In these tangential sections a reliable delineation of cortical areas is not possible. All these cortical maps show the distances in millimeters from bregma or the interaural line at the top or bottom margins. The distance from the horizontal plane passing through the interaural line and the distance from the horizontal plane passing through bregma and lambda on the surface of the skull are indicated in the left and right margins, respectively, for both the lateral (Figs. 46 and 49) and the medial (Fig. 48) views. The distance in millimeters from the midline between both hemispheres is given for the dorsal view (Fig. 47) in the left and right margins.

[55]

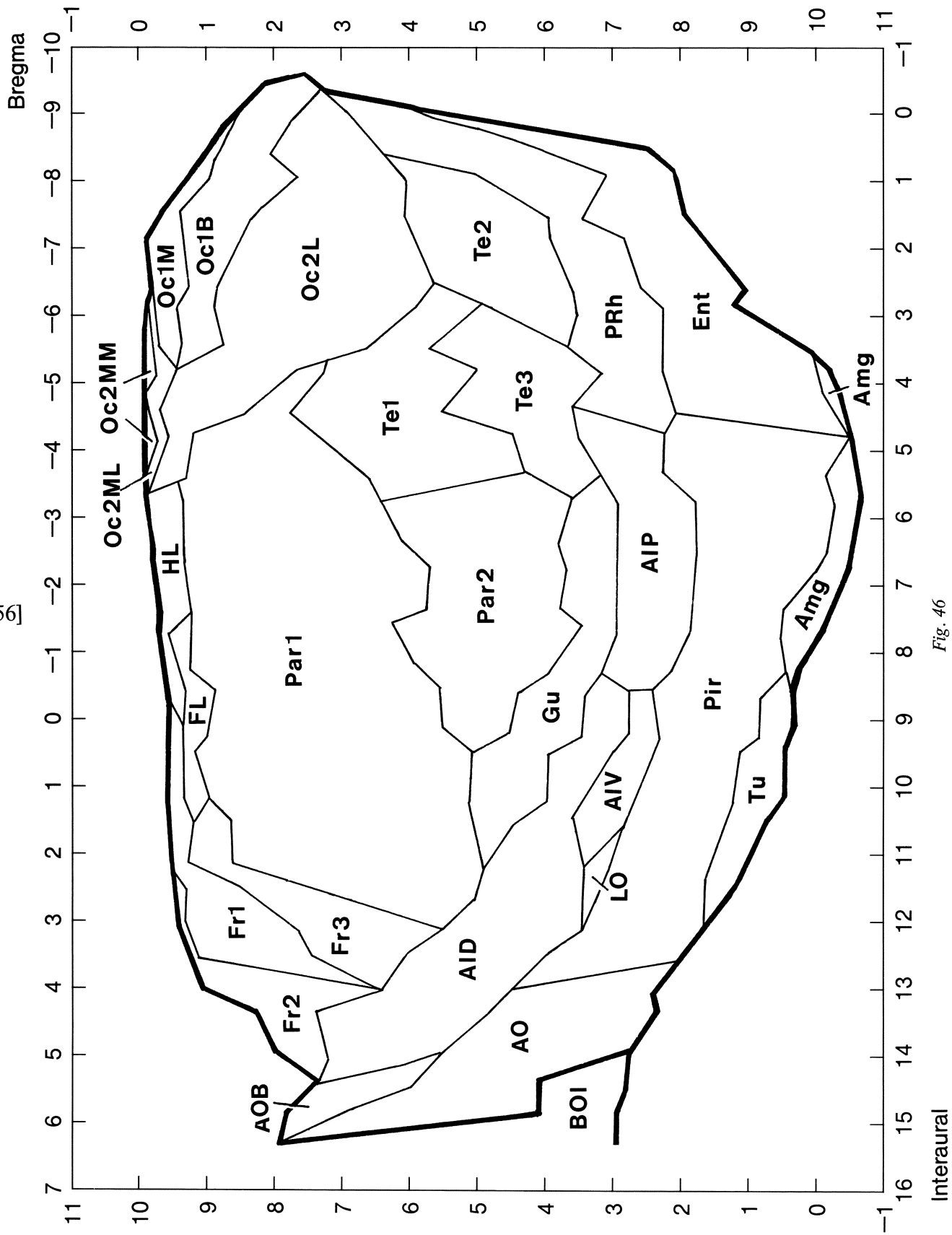


Fig. 46

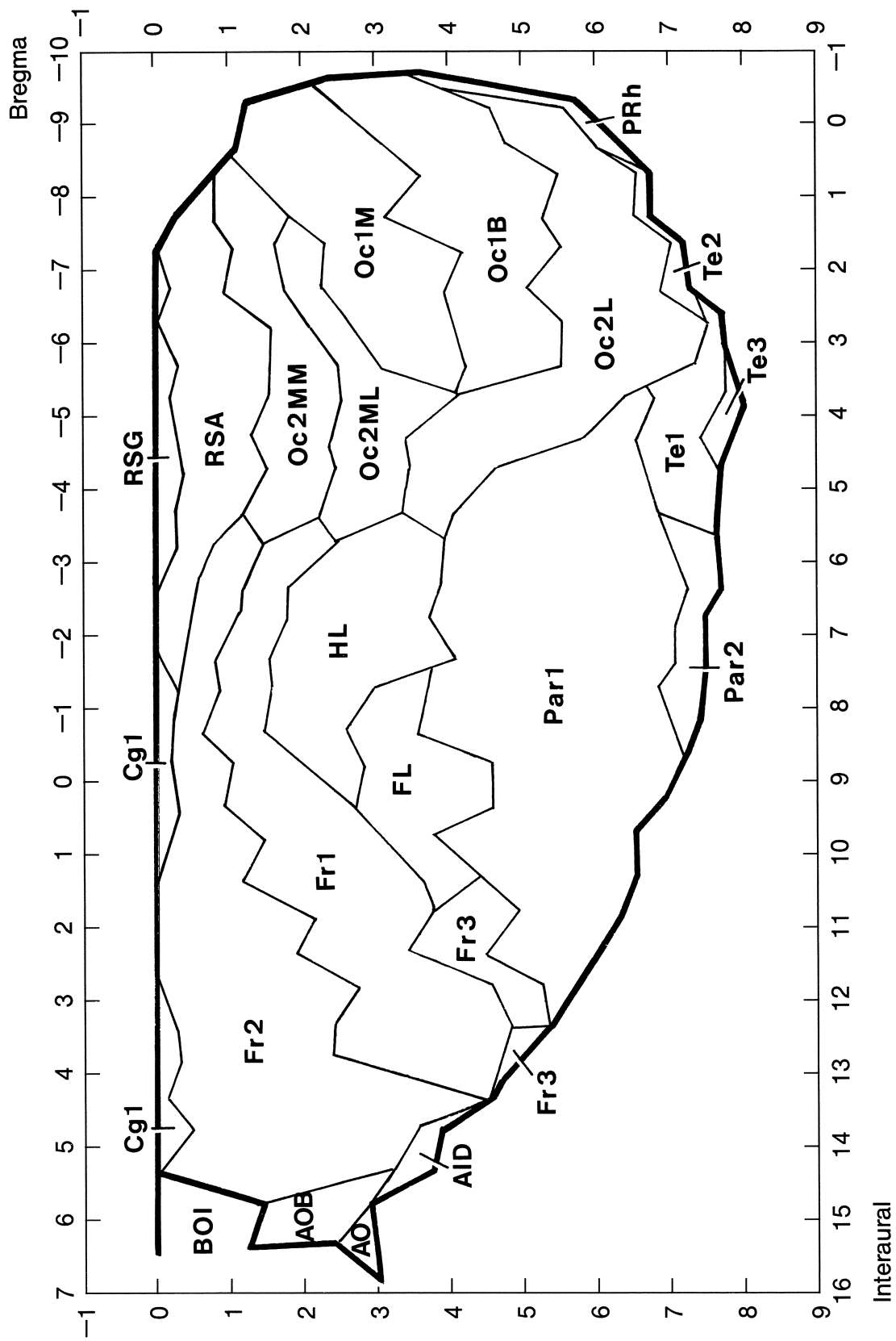


Fig. 47

Fig. 48

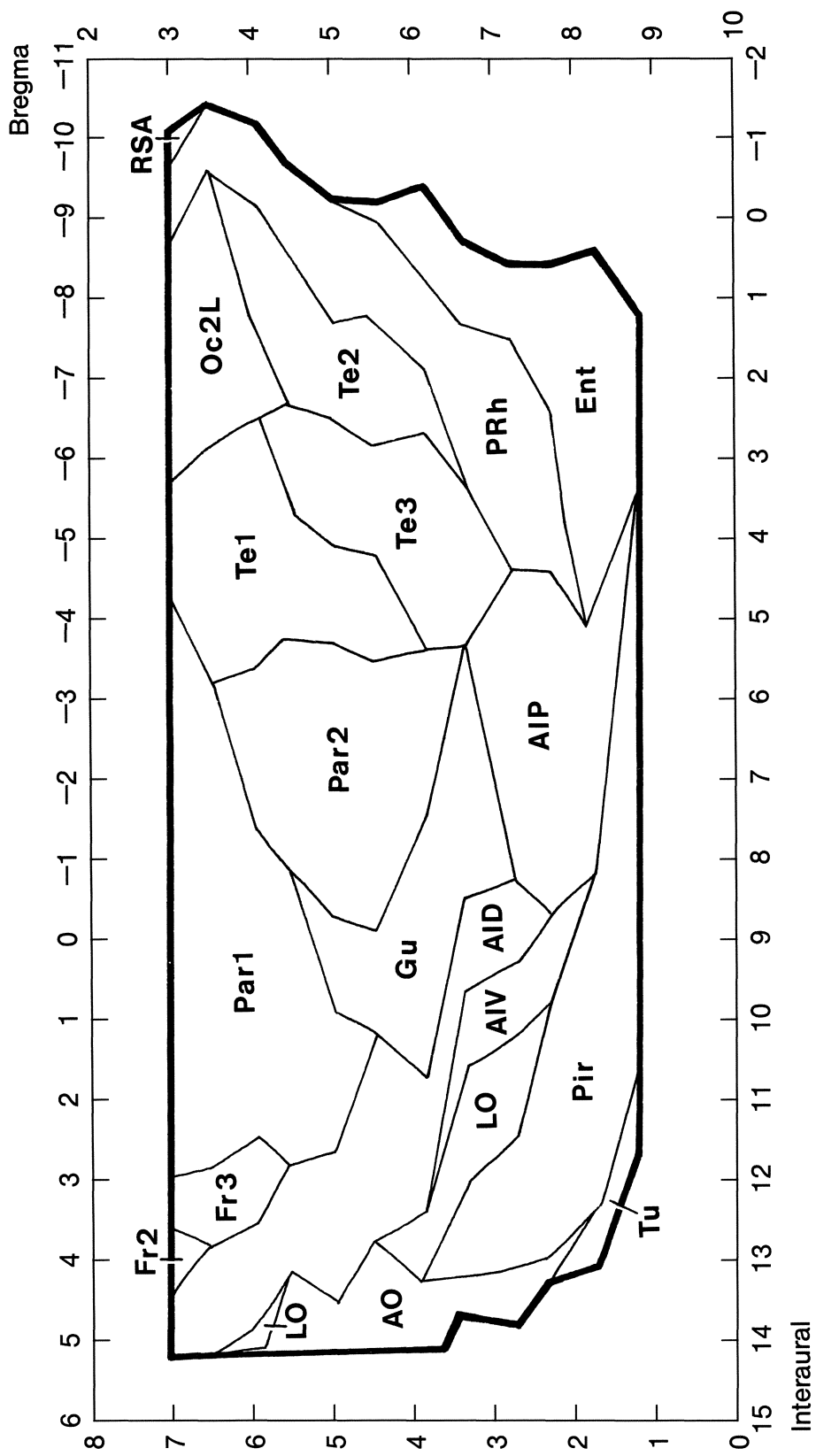


Fig. 49

Frontal and Caudal Aspects of the Hemisphere in Computer Reconstructions

The most frontal and caudal parts of the hemispheres can be only poorly demonstrated in orthogonal cortical maps. The coronal drawings (Figs. 2–31) were therefore used for computer reconstruction with a 3-D program in views at varying angles (Figs. 50–54). The frontal and caudal poles are shown in Figs. 50 and 52, respectively, with no rotation, solely in the frontal and caudal views. In both cases the stereotaxic coordinates are given for the first and last sections as distances from bregma in millimeters. In addition, the distances of structures from the midline are shown at the top and bottom. The numbers in the right margin show the dorsoventral distance from the horizontal plane passing through bregma and lambda on the surface of the skull.

[61]

Figs. 51 and 53 show aspects of the frontal and caudal poles of the hemisphere from a medial and dorsal standpoint. The hemisphere has been rotated clockwise through 80° around a sagittal axis, and this axis has also been tilted through 50° in the horizontal plane.

Figure 54 shows the caudal pole of the hemisphere in a lateral and dorsal view. In this case the hemisphere has been rotated counterclockwise through 60° around the sagittal axis and tilted through 50° in the horizontal plane. The areal borders in these 3-D reconstructions are drawn with thick lines, and the sectioning plane of the first section in rostral or caudal views is cross-hatched.

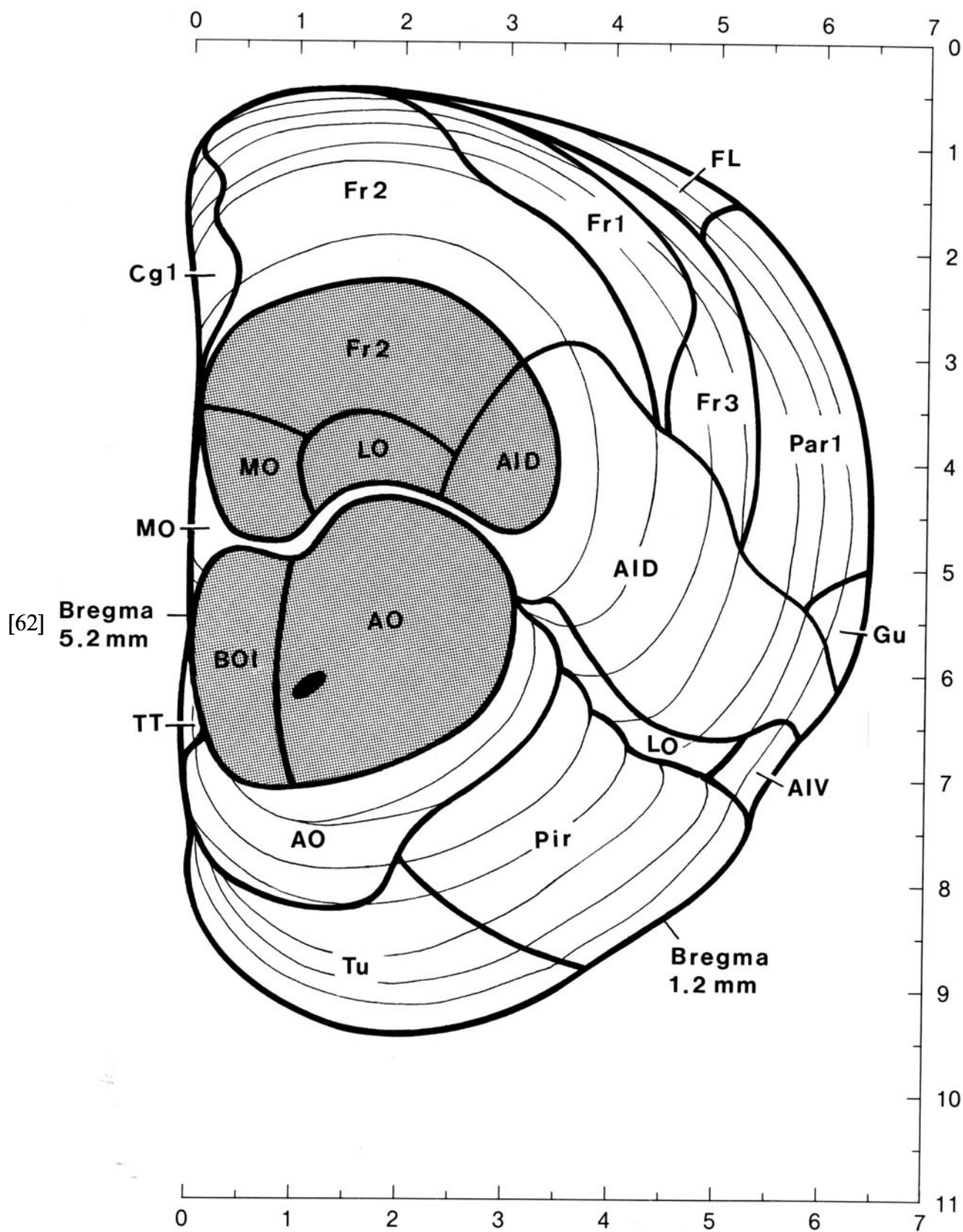
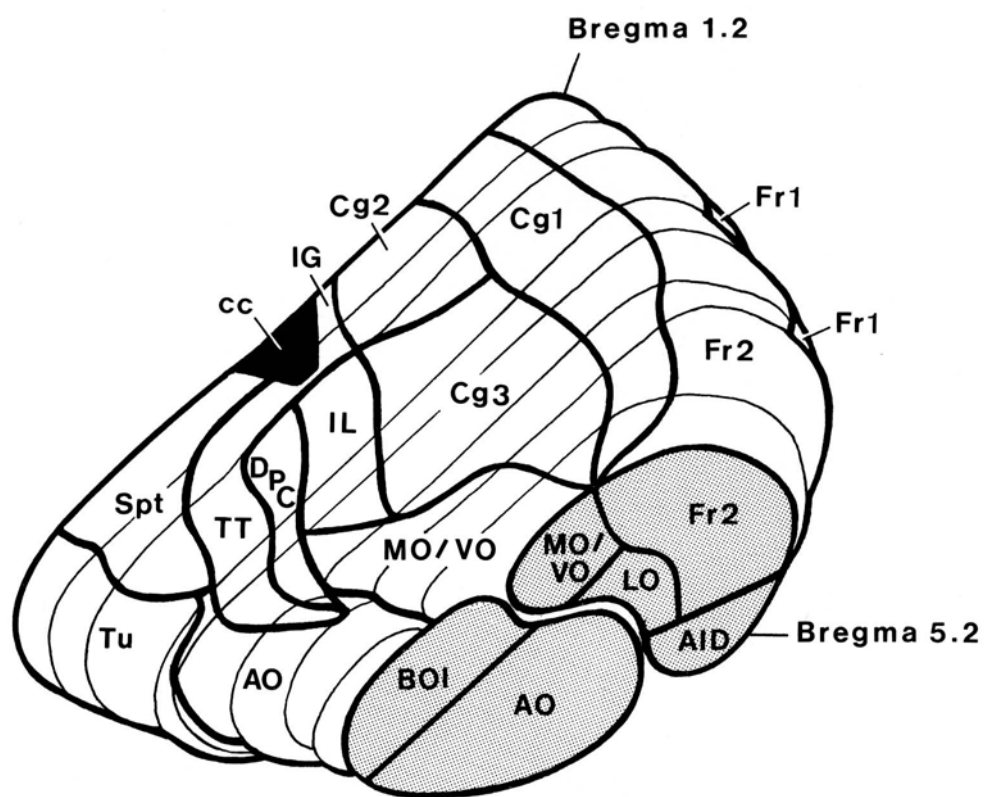


Fig. 50



[63]

Fig. 51

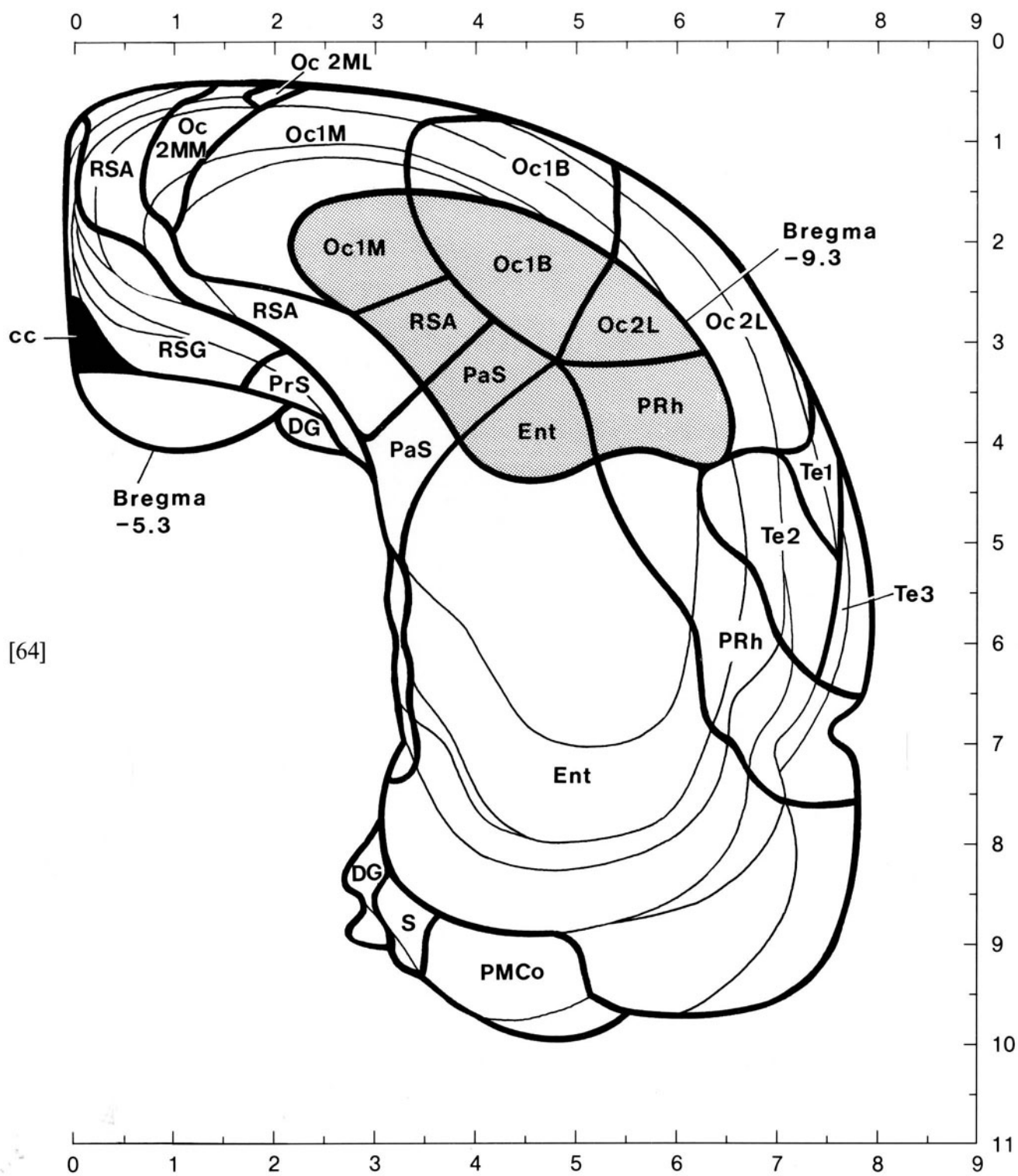
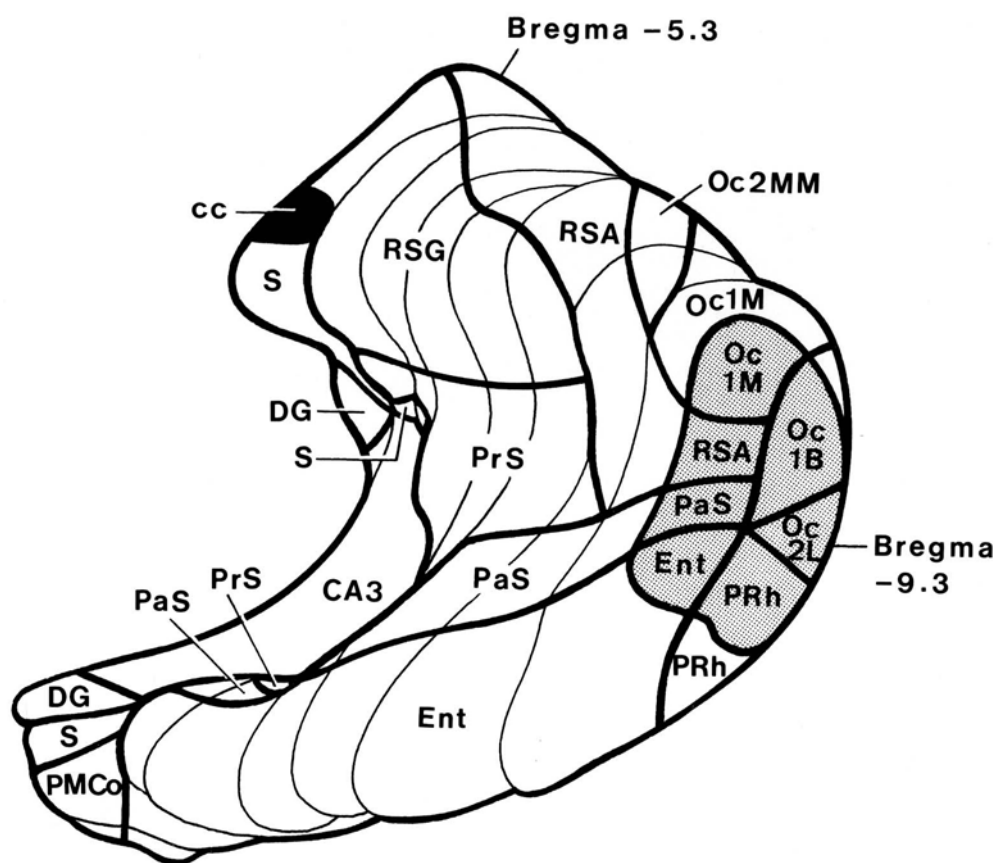


Fig. 52



[65]

Fig. 53

[66]

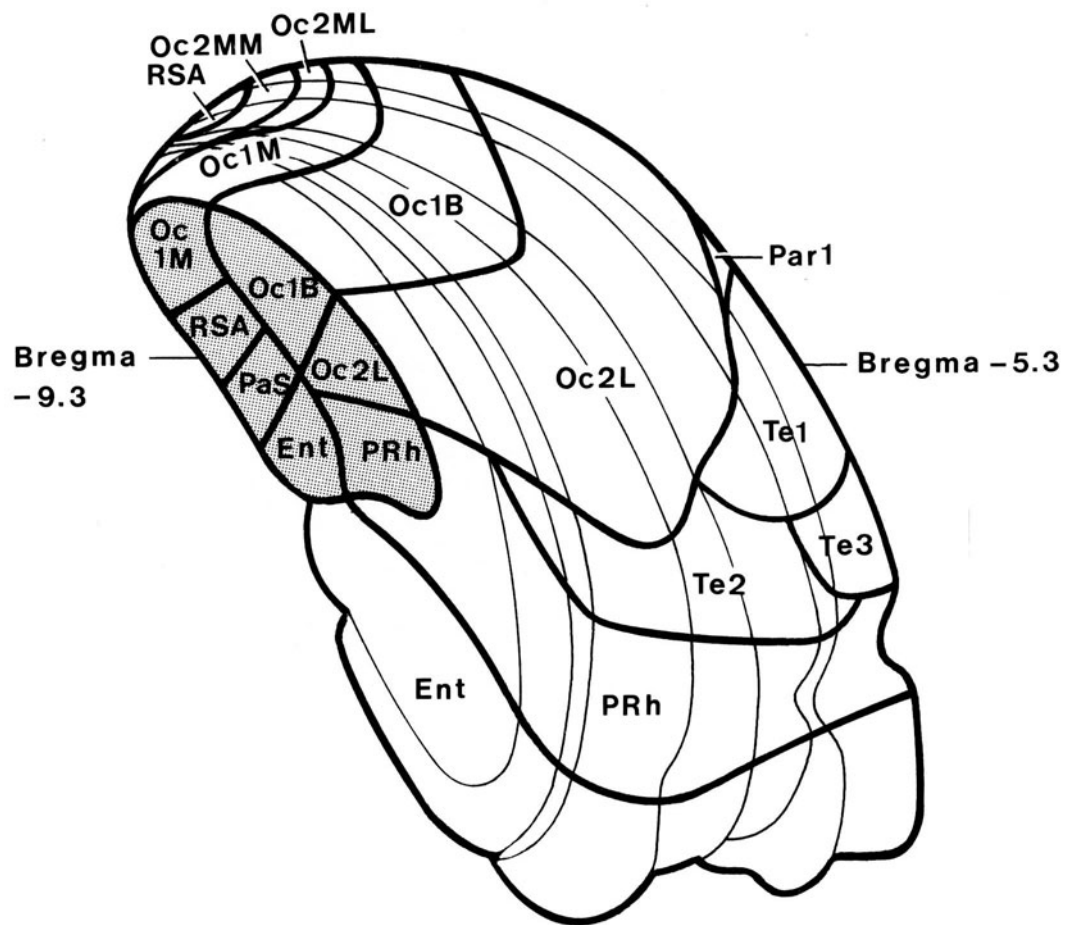


Fig. 54

Delineation of Cortical Areas in Nissl- and Myelin-Stained Sections

The rat brains from which the sections presented in the micrographs in this section (Figs. 55–67) were taken were processed differently than the brains in the Paxinos and Watson (1982) atlas. Most of the brains used in the present atlas came from animals placed in the stereotaxic apparatus after being perfused with fixative.

After perfusion of the animals (cf. p. 12) the skulls were partially removed, but the entire bases of the skulls were left completely intact. Following these preparations, the specimens were placed in a stereotaxic apparatus in the flat-skull position. The brains were then sectioned in a coronal plane with a razor blade fixed to an electrode holder positioned at the interaural line. The two fragments of each brain were embedded in paraffin and serial 20- μ m sections were cut. These sections were stained alternately with Nissl and myelin. The planes in which the sections lie can be deduced from a comparison with the plates in the Paxinos and Watson (1982) atlas, using the bregma and interaural reference points for orientation. This procedure provides sections of good histological quality with known stereotaxic positions. A comparison of the plates in the Paxinos and Watson (1982) atlas with the micrographs of the present material reveals striking compatibility, and in some cases almost complete correspondence. The coronal sections of Paxinos and Watson's specimens show only two distinct deviations from these specimens in Figs. 55–67. The coronally sectioned brain in the Paxinos and Watson (1982) atlas is somewhat broader in the area of the occipital cortex than the Wistar rat brains depicted in Figs. 55–67. This variation may be due to the different stock from which the Wistar strains were bred.

The second variation was in the position of the hippocampus along the rostrocaudal axis. In the brain shown in the Paxinos and Watson (1982) atlas the hippocampus appears to be about 0.5 mm further rostrad than in any other Wistar, Lewis, or

hooded rat brains processed specifically for the present atlas. Apart from these two differences, all cortical and subcortical features have been found to be at comparable positions along the rostrocaudal axis.

Fixation, Embedding, and Sectioning

While in deep anesthesia the rats were perfused with 50–100 ml Ringer's solution through the left ventricle of the heart, followed by Bodian's fixative (900 ml 80% ethanol, 50 ml 37% formaldehyde, and 50 ml glacial acetic acid at 4° C). About 1 liter of this fixative was used per adult rat over 60 min. The brains were removed 1–3 h after perfusion was complete. They were then stored for 1–2 days in the same fixative.

The brains were dehydrated in a graded ethanol/methylbenzoate series and embedded in paraffin. Once the brains had been carefully blocked, serial 20- μ m sections were cut. Gelatin-coated slides were used to mount alternate sections in Nissl and myelin series.

[68]

Nissl Staining

The slides were treated in the following baths: xylene, 100% ethanol, 96% ethanol, 70% ethanol, distilled water. They were then stained for 30 min in cresyl fast violet solution at 60° C. To make this solution, a buffer consisting of 100 ml distilled water, 0.544 g sodium acetate, and 0.1 ml 99%–100% acetic acid is added to 2 mg cresyl fast violet. The slides were differentiated by the following procedure: rinsing in distilled water and 70% ethanol (rapid contrast increase) or 96% ethanol (slower contrast increase). Following immersion in xylene the slides were covered with Depex.

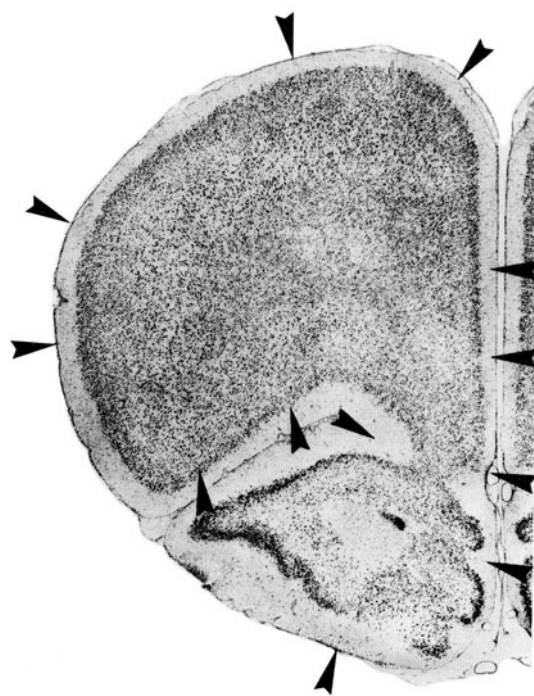
Myelin Staining

The slides were prepared for staining in xylene, 100% ethanol, 96% ethanol, 70% ethanol, and distilled water. The following staining procedure uses a slightly modified version of the Gallyas myelin stain (Gallyas 1971, 1979), and myelin fibers are stained black to contrast with the completely unstained background.

1. Immersion for 30 min in a solution of 80 ml pyridine and 40 ml acetic acid
2. Three rinses in distilled water (5 min each)
3. Immersion for 30 min in a solution of 1 g ammonium nitrate, 1000 ml distilled water, 1 g silver nitrate, and 3 ml 4% NaOH [added in the given order]
4. Three rinses in 0.5% acetic acid (3 min each)
5. Immersion for 20–30 min in the physical developer. Four stock solutions are used to make this developer: Solution A: 50 g sodium bicarbonate in 1000 ml distilled water. Solution B: 2 g ammonium nitrate, 2 g silver nitrate, 10 g tungstosilicic acid dissolved in 1000 ml distilled water. Solution C: 2 g ammonium nitrate, 2 g silver nitrate, 10 g tungstosilicic acid, 7.3 ml 37% formaldehyde, 1000 ml distilled water. Solution D: Equal parts of distilled water and Agfa bleach fixer. The physical developer is prepared by combining 50 ml A, 15 ml B, 35 ml C, and 0.16 ml D
6. One 5-min rinse in distilled water
7. Immersion for 10 min in a photographic fixer, e.g., 20 ml Kodak Ektaflo in 70 ml distilled water
8. One 5-min rinse in distilled water
9. Graded ethanol series (70%, 96% 100%)
10. Xylene bath
11. Coverslipping with Depex

[69]

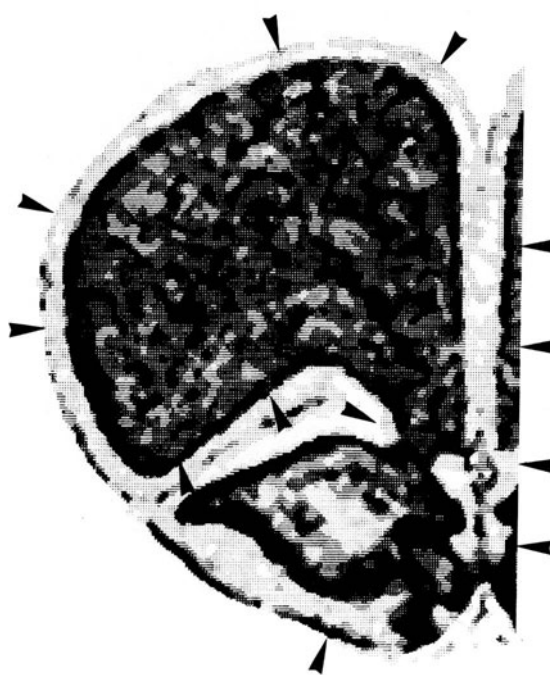
Following step 8 a differentiation procedure (increase in contrast) can be carried out. This consists in a 5-min rinse in 0.5% acetic acid, a 15- to 30-s differentiating bath (bleach fixer as in solution D), and two rinses in 0.5% acetic acid.



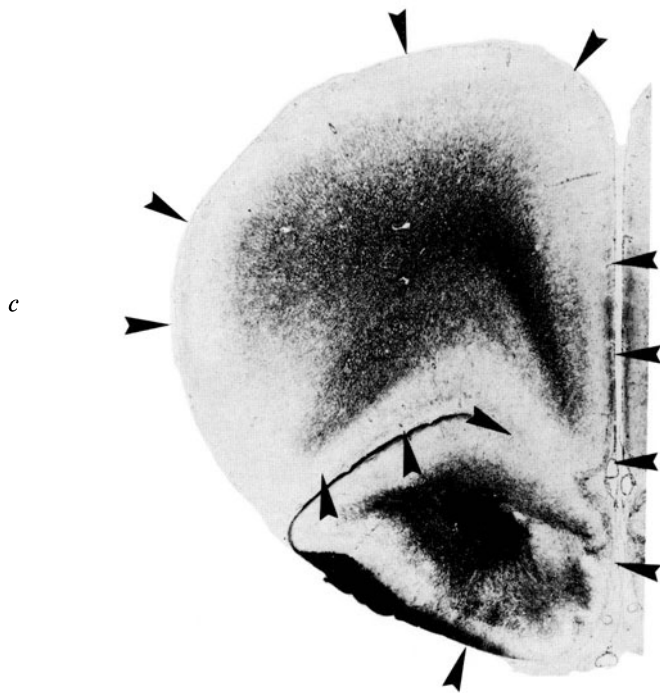
a

[70]

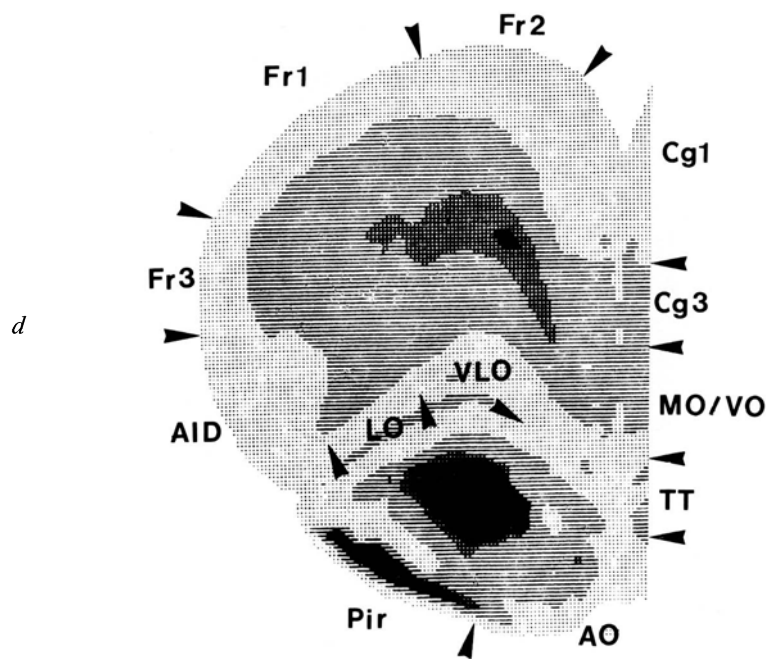
Fig. 55 a-d

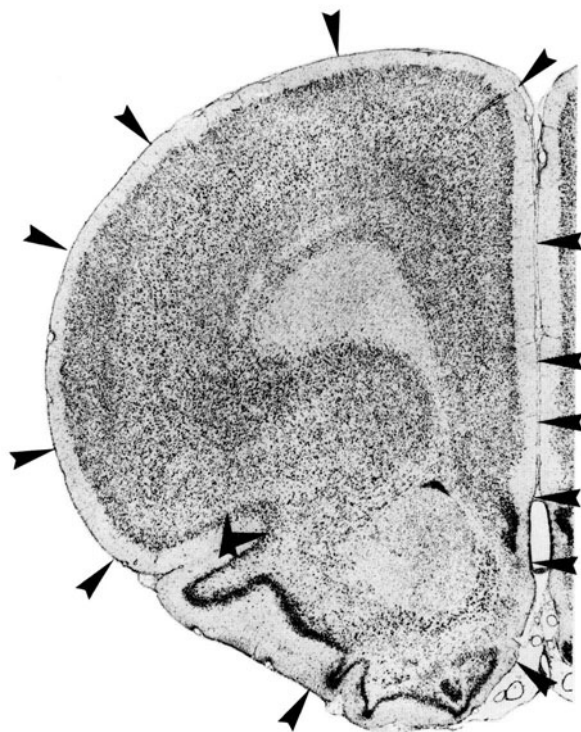


b



[71]





a

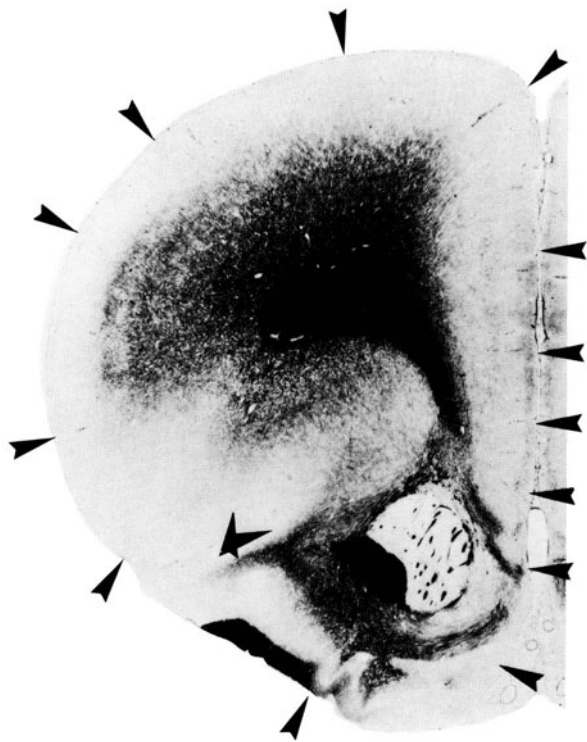
[72]

Fig. 56 a-d



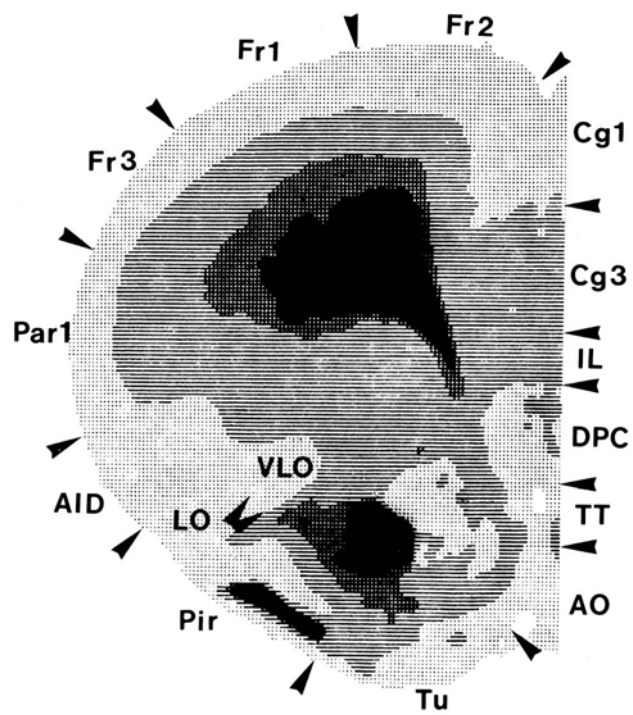
b

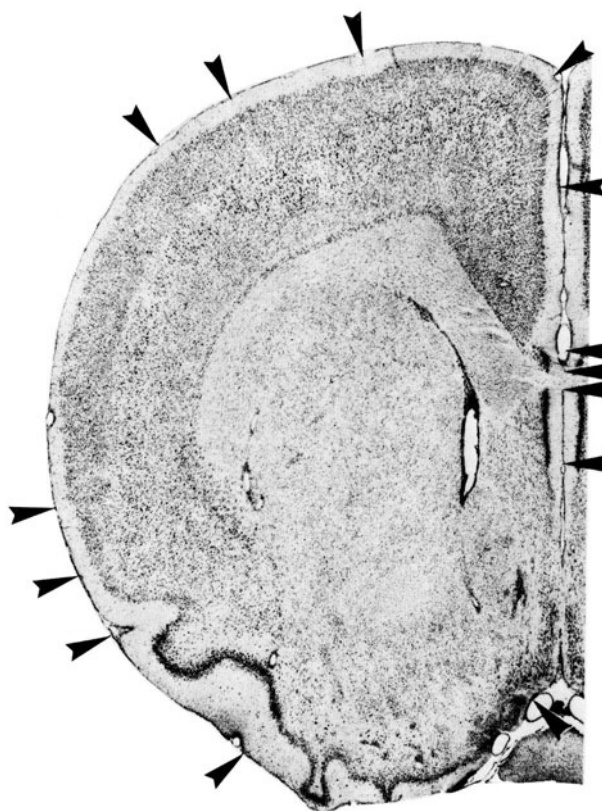
c



[73]

d





a

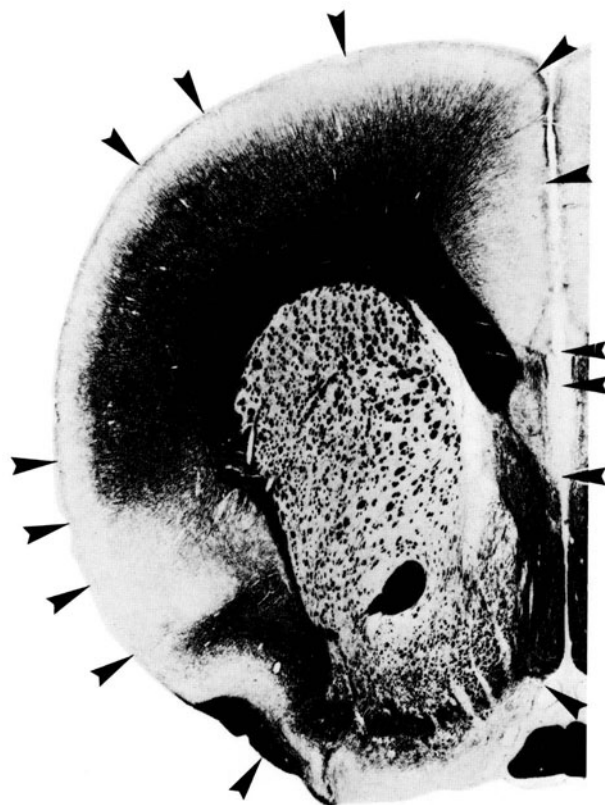
Fig. 57 a-d

[74]



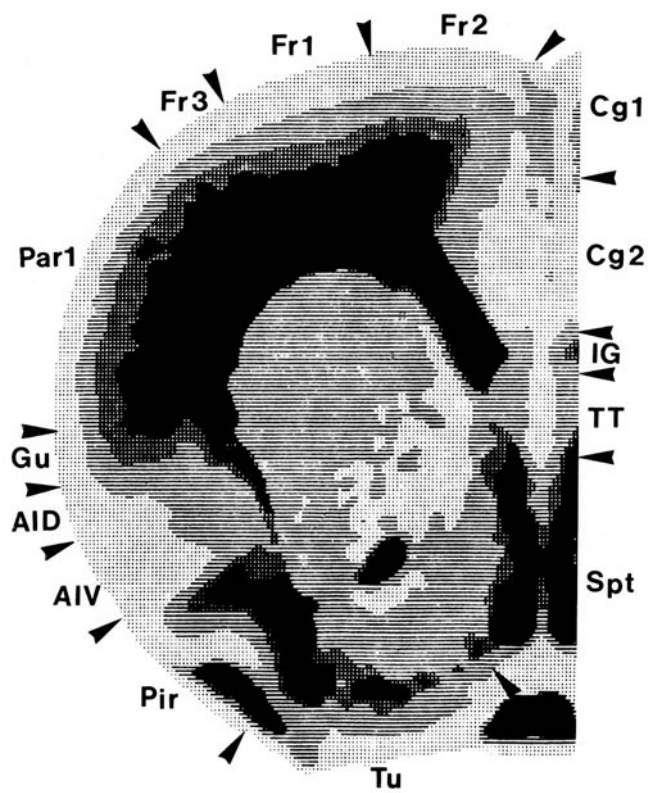
b

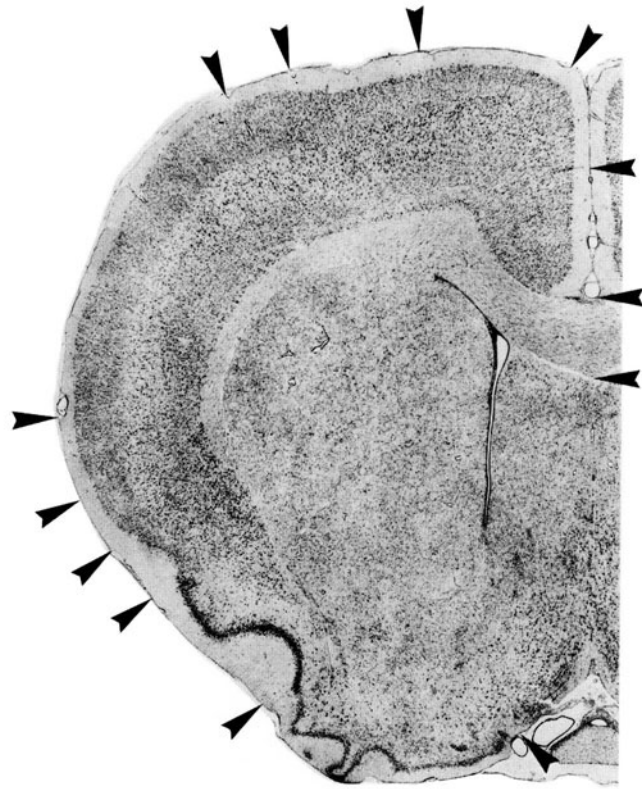
c



[75]

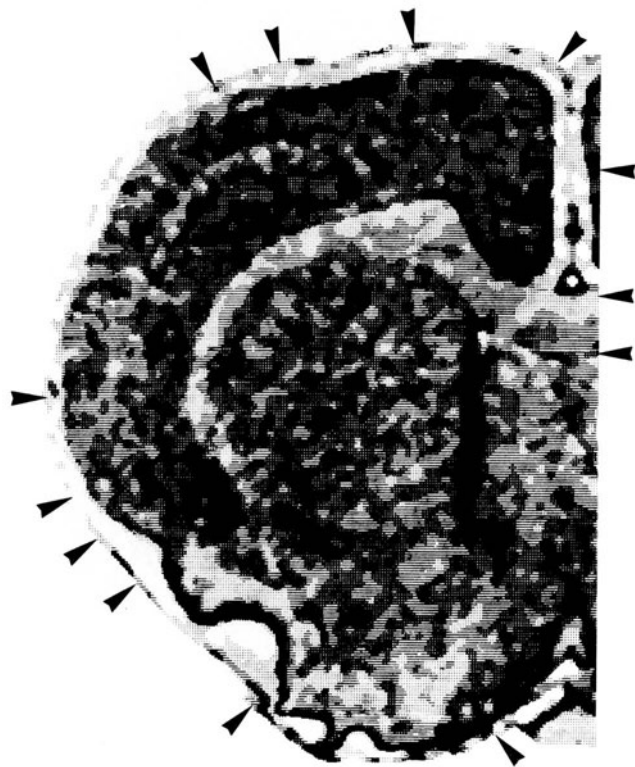
d





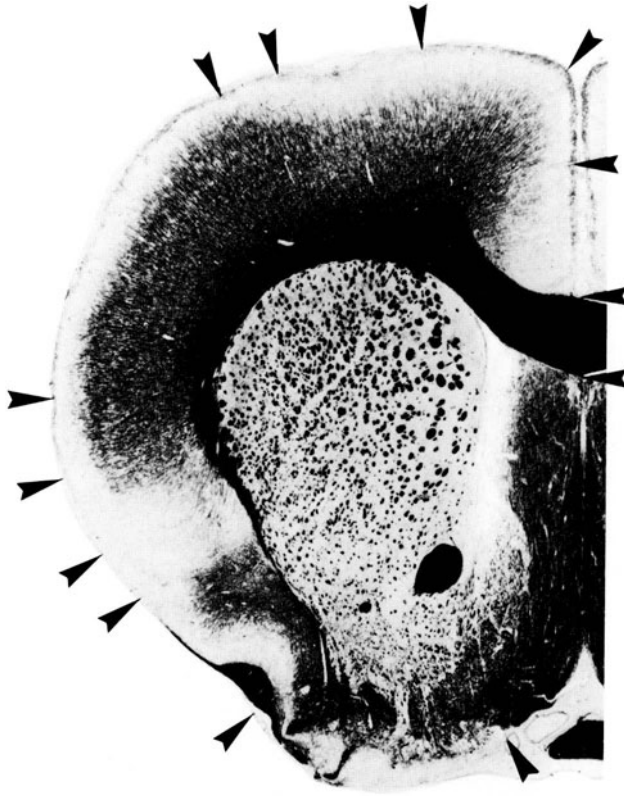
a

Fig. 58a-d



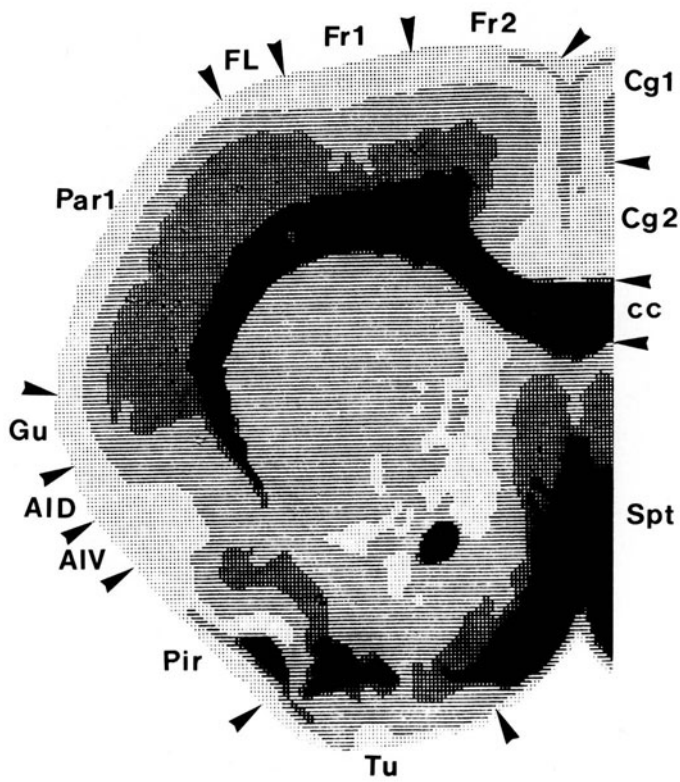
b

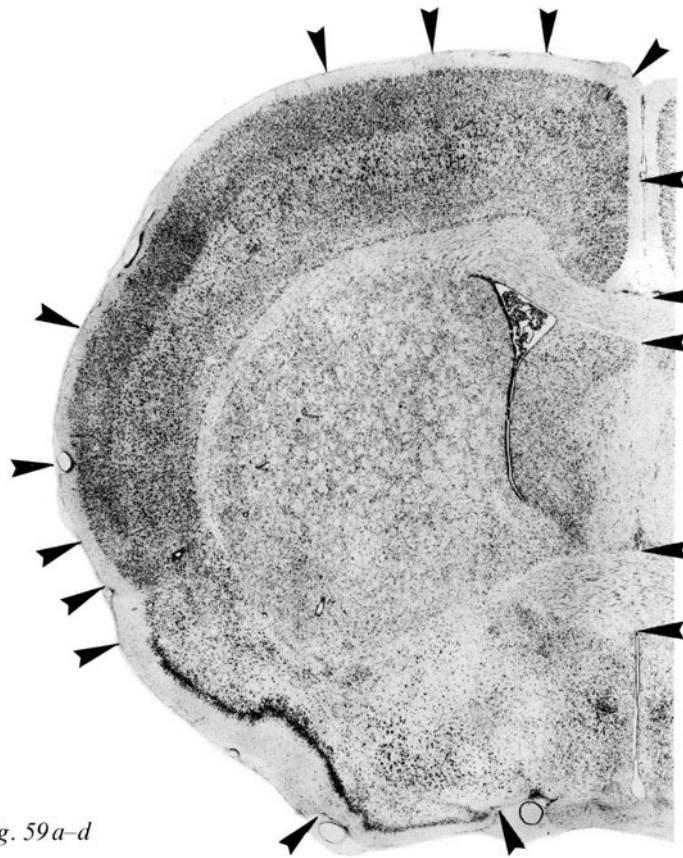
c



[77]

d





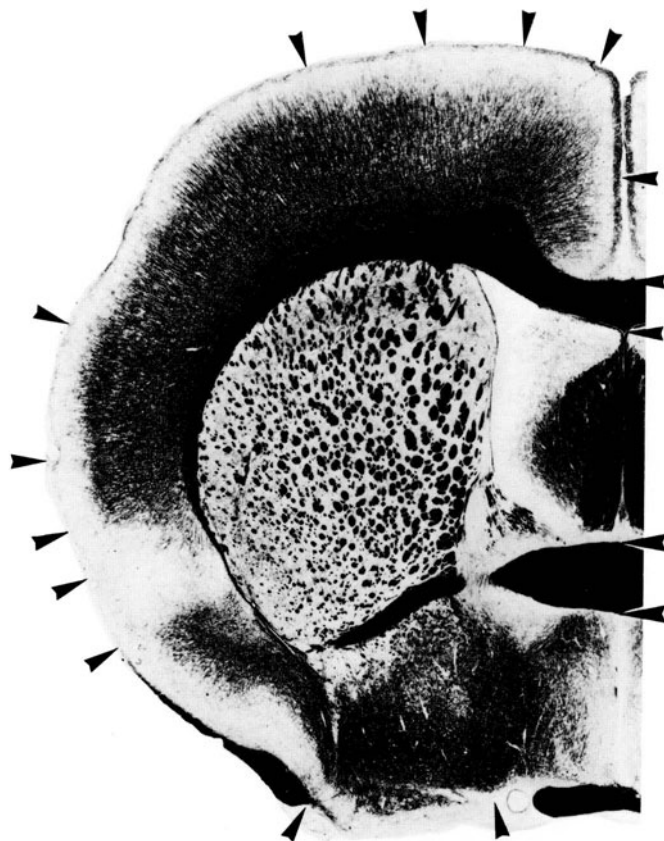
a

Fig. 59 a-d



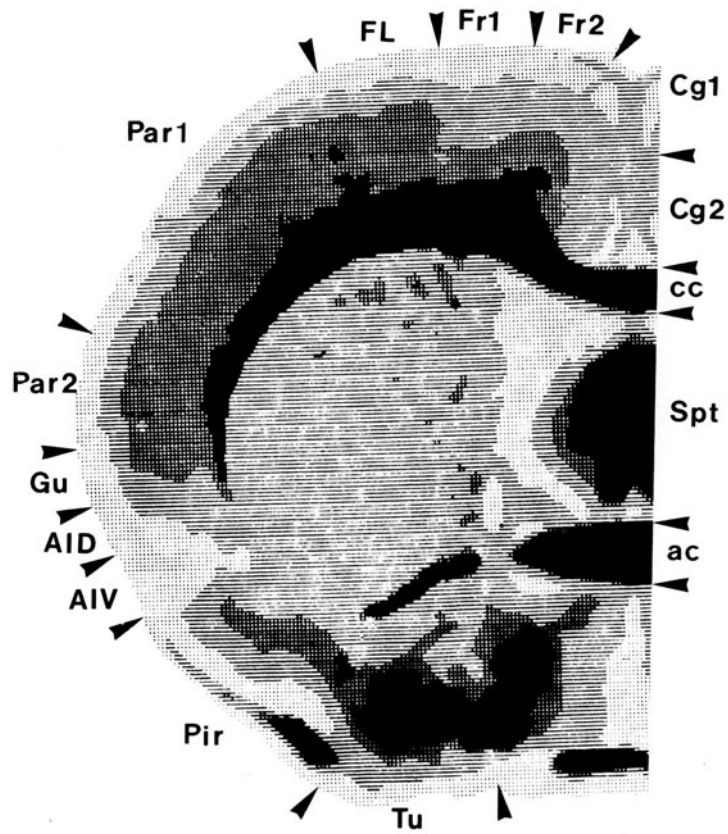
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c



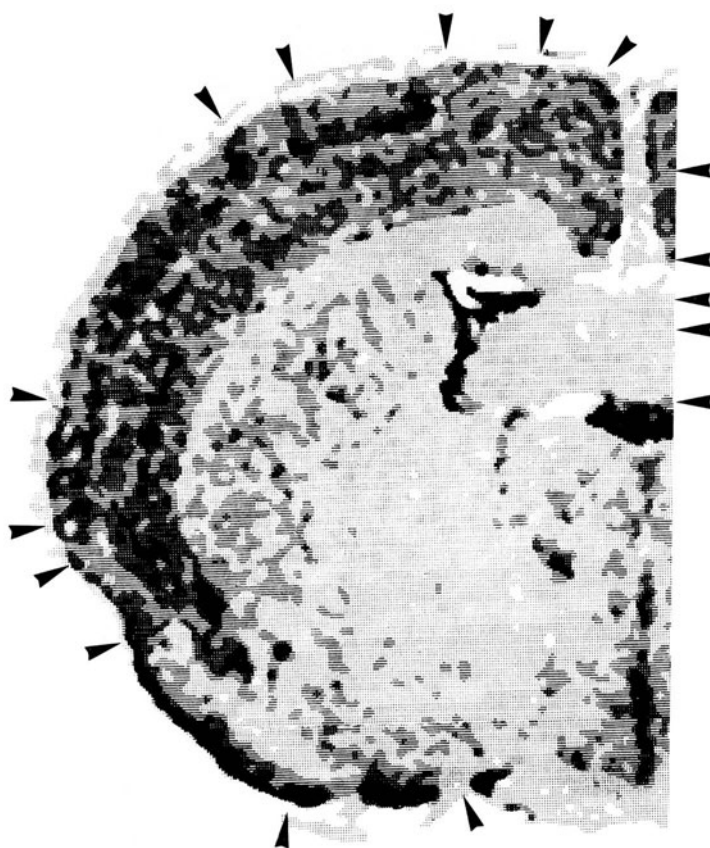
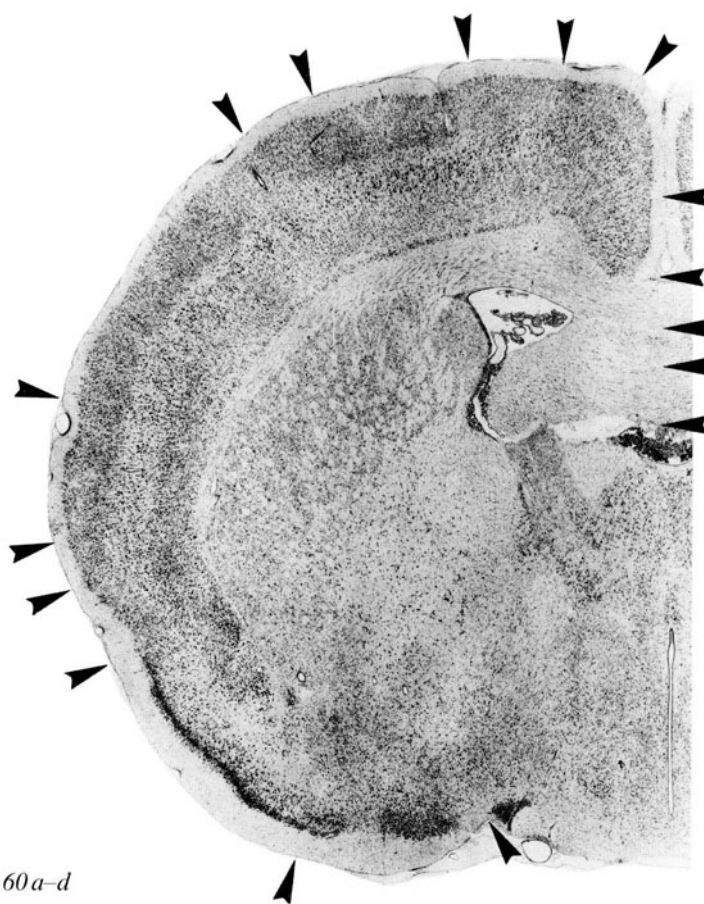
[79]

d

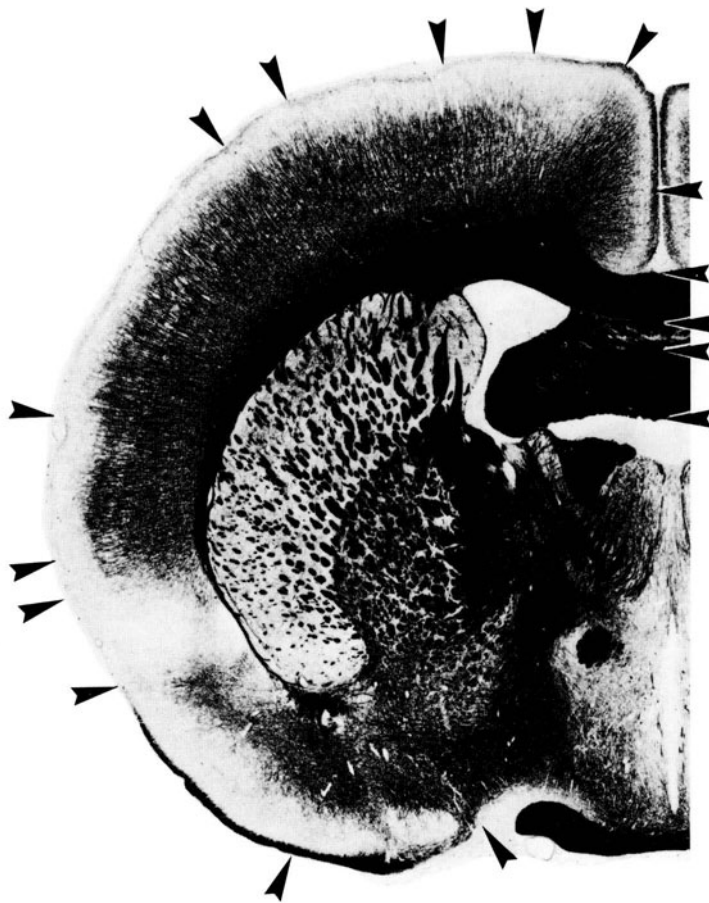


[80]

Fig. 60 a-d

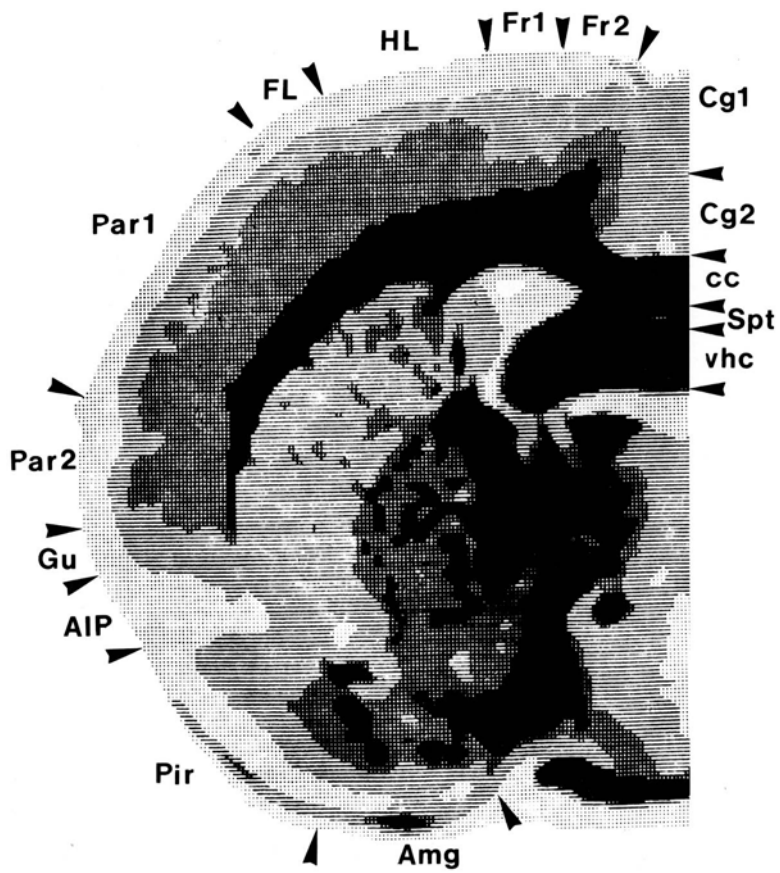


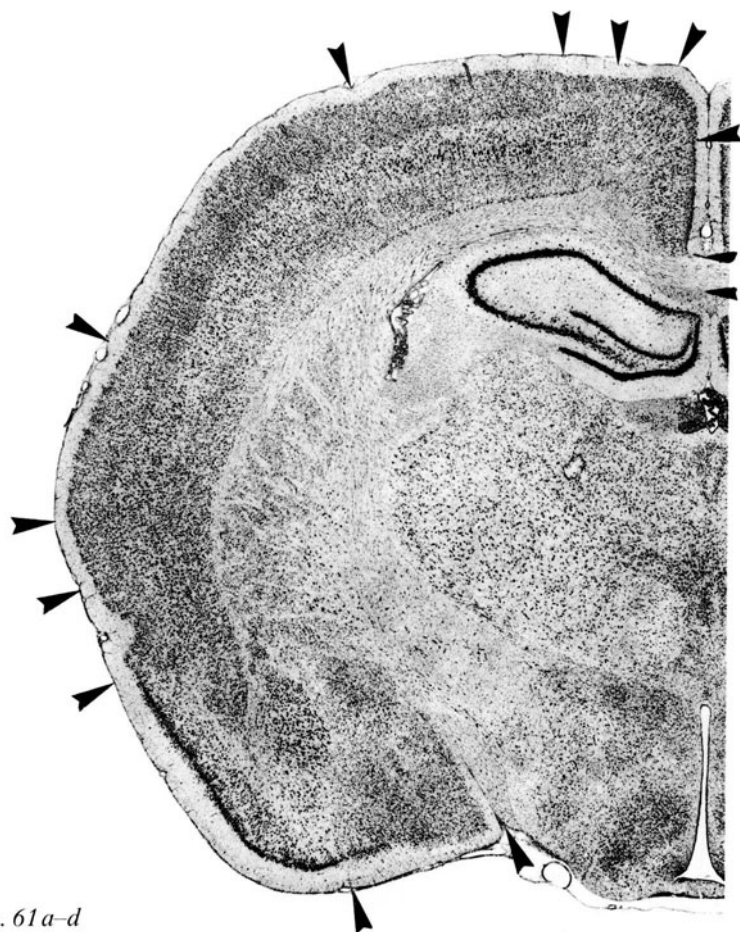
c



[81]

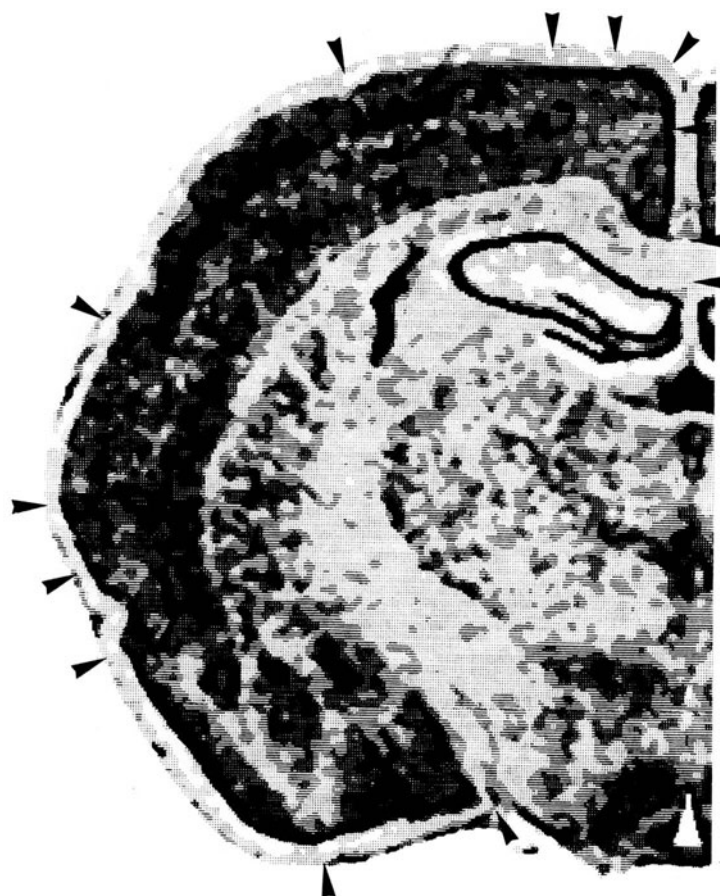
d





a

Fig. 61 a-d



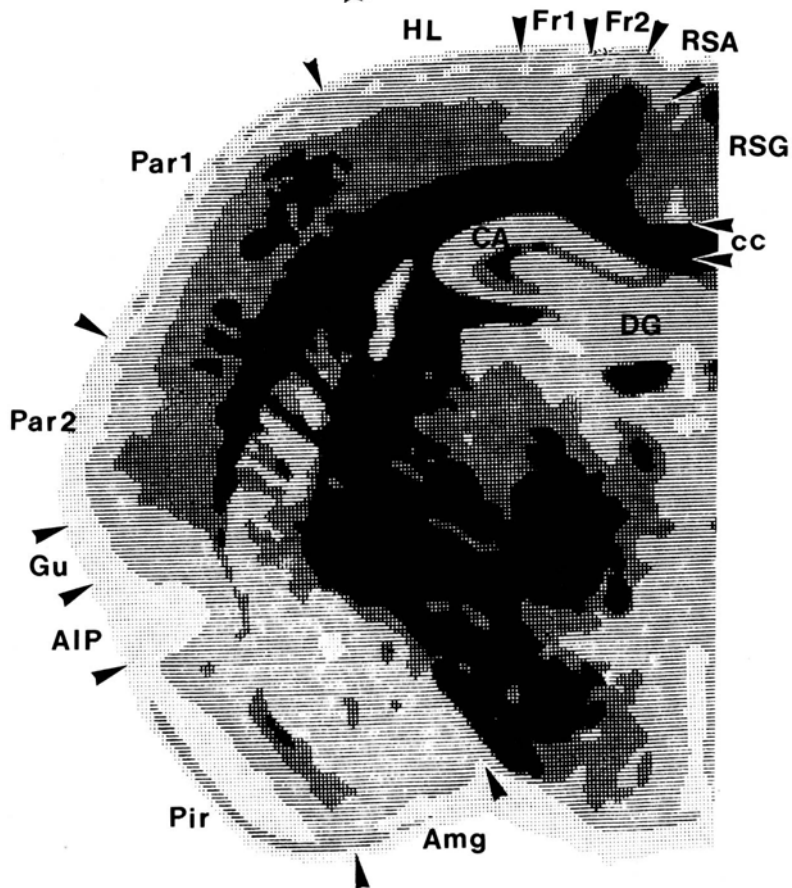
b

c



[83]

d



[84]

Fig. 62a-d



a



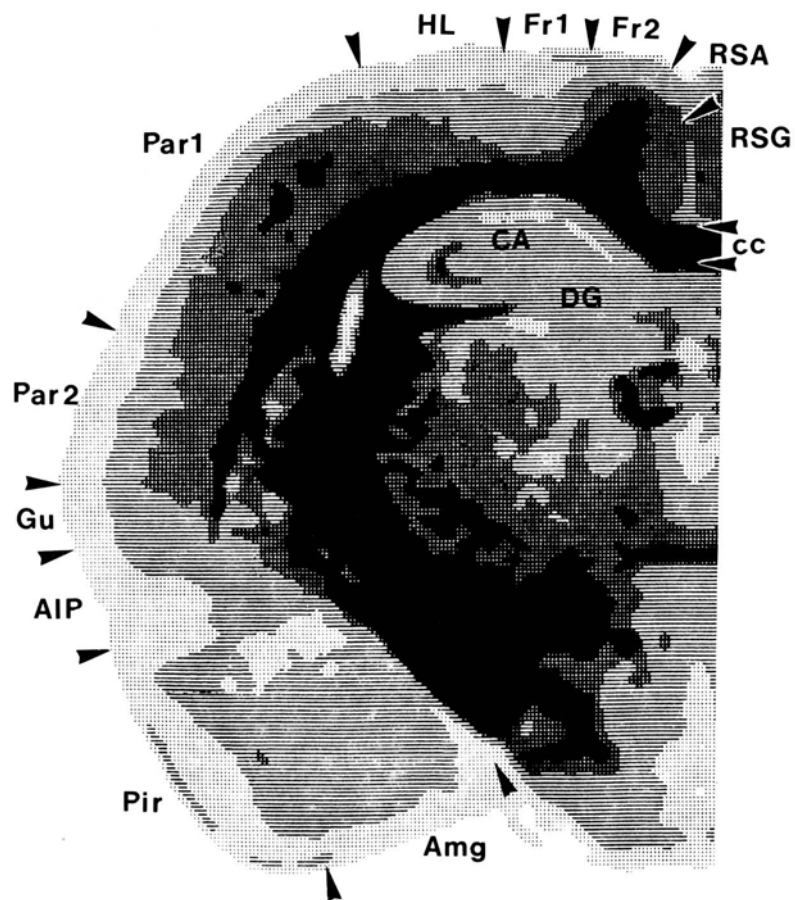
b

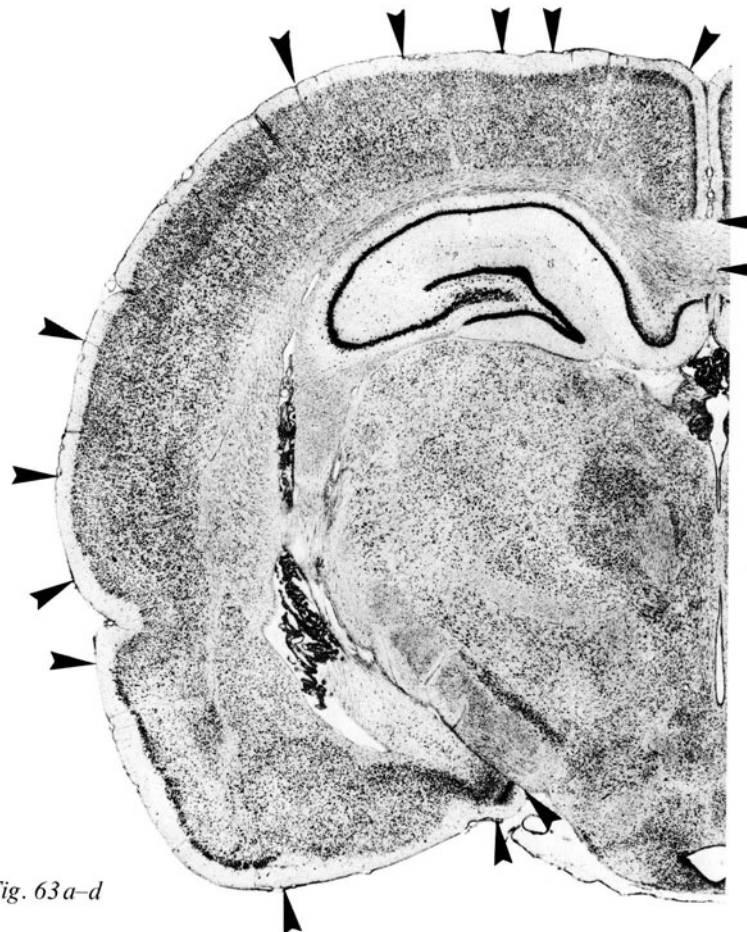
c



[85]

d





a

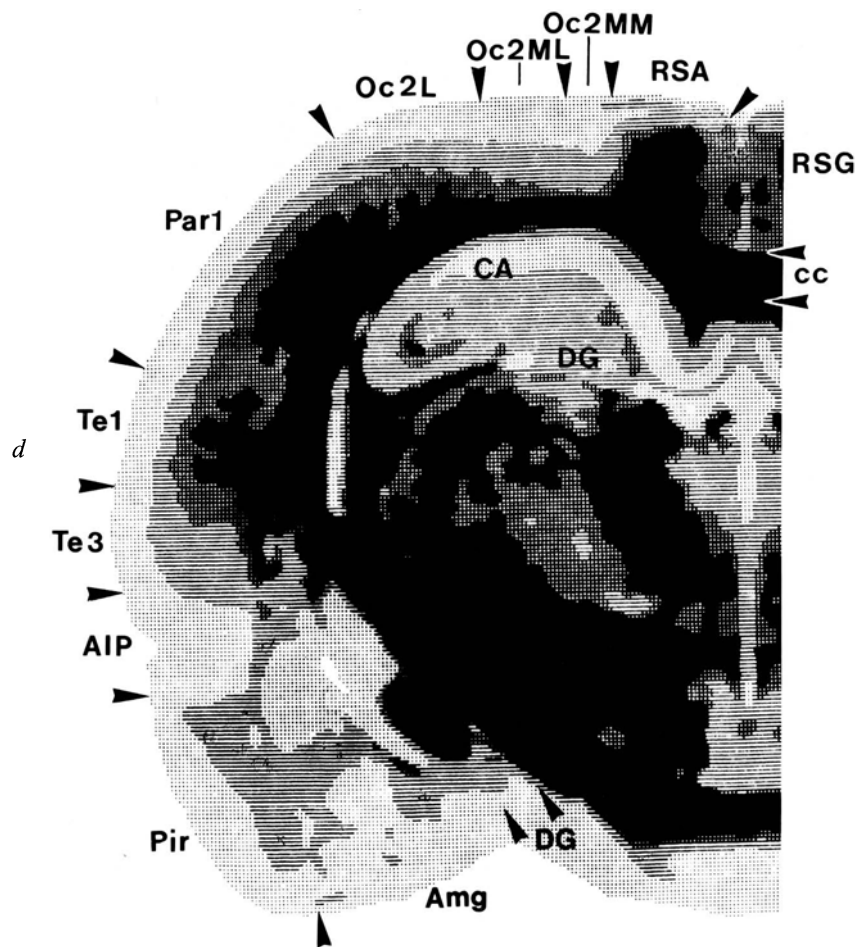
Fig. 63 a-d

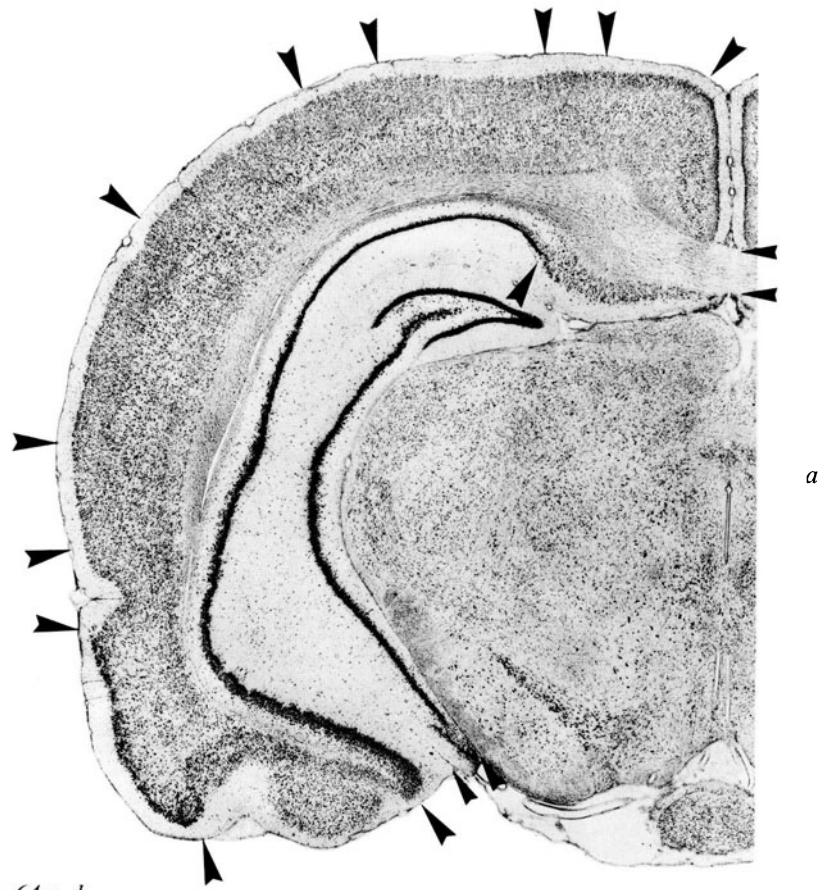


b



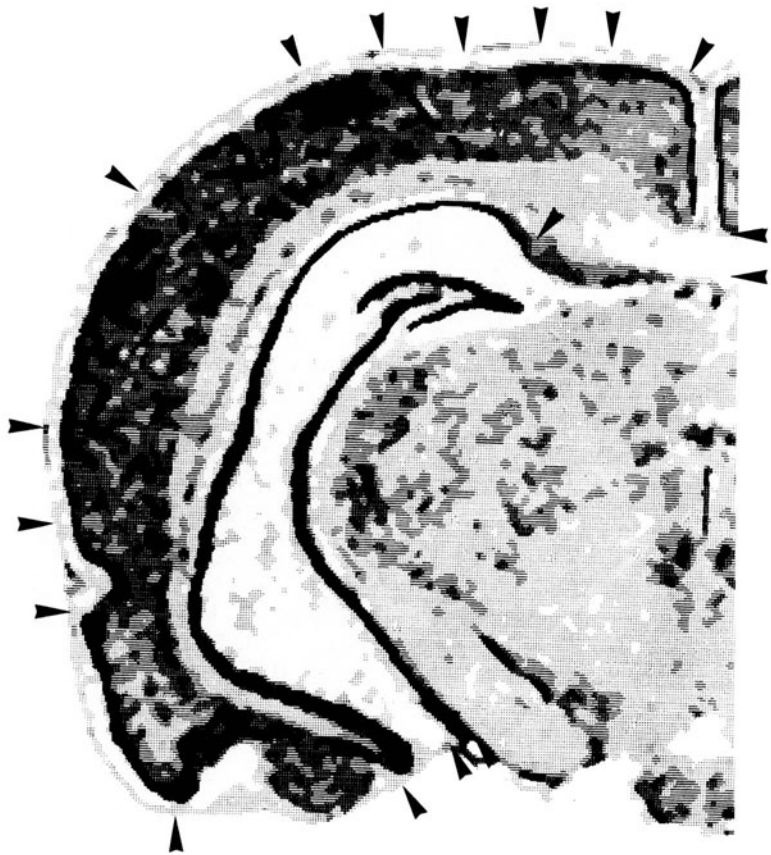
[87]



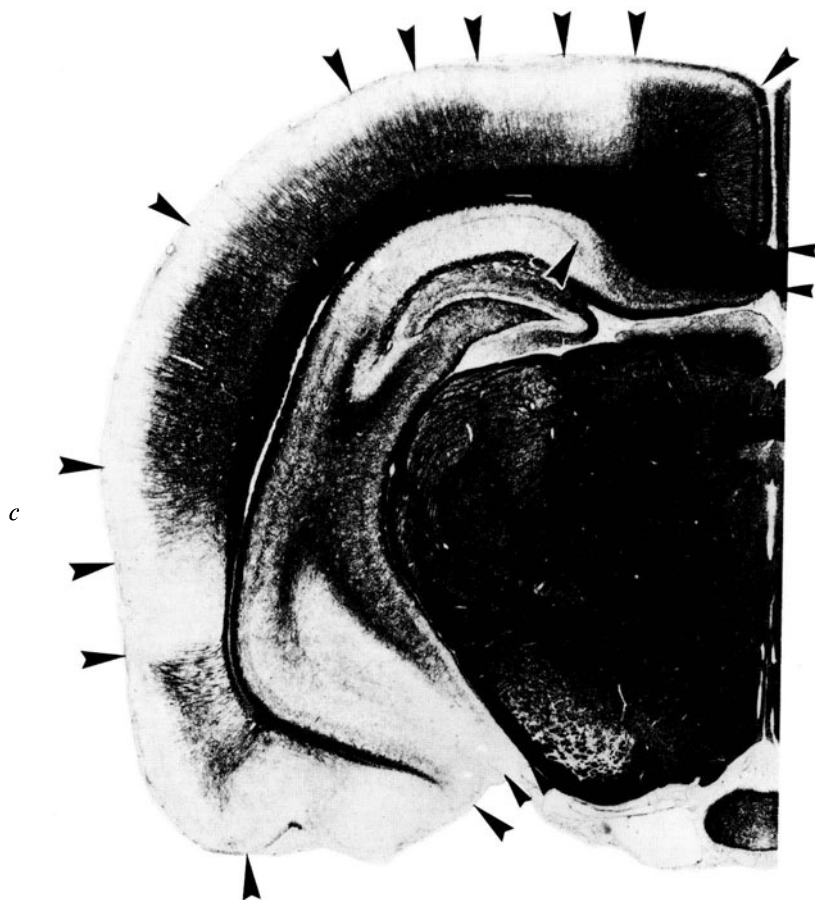


a

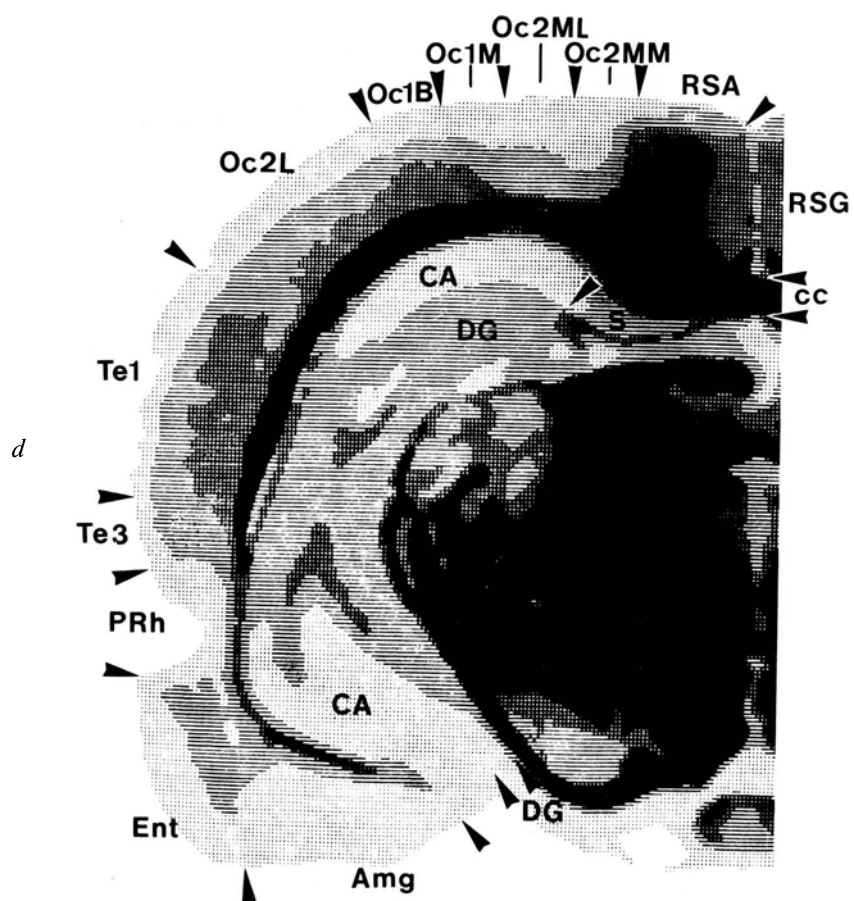
Fig. 64 a-d

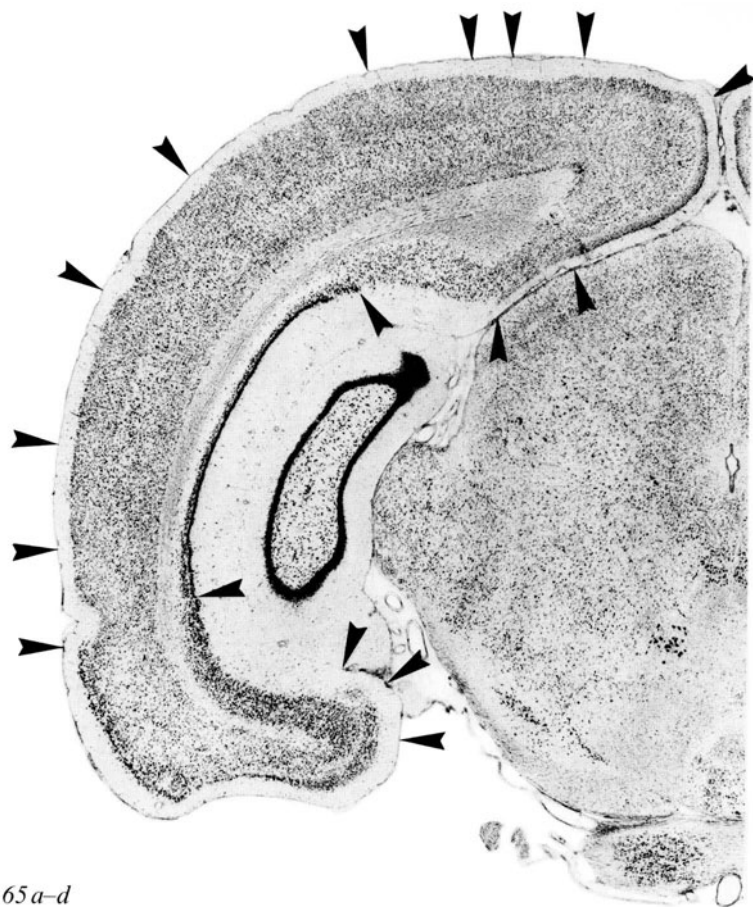


b



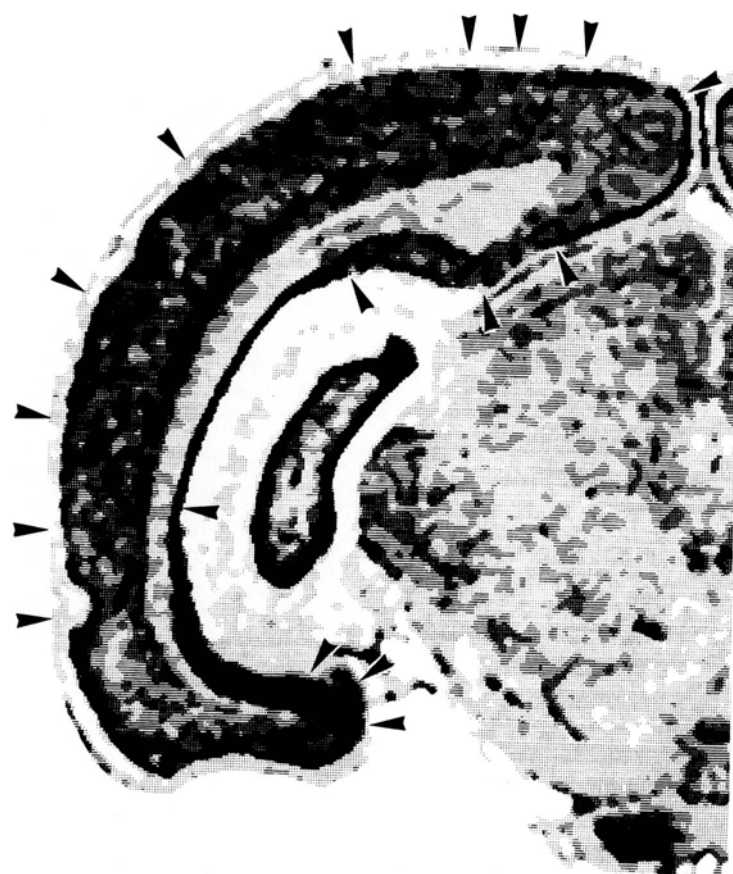
[89]



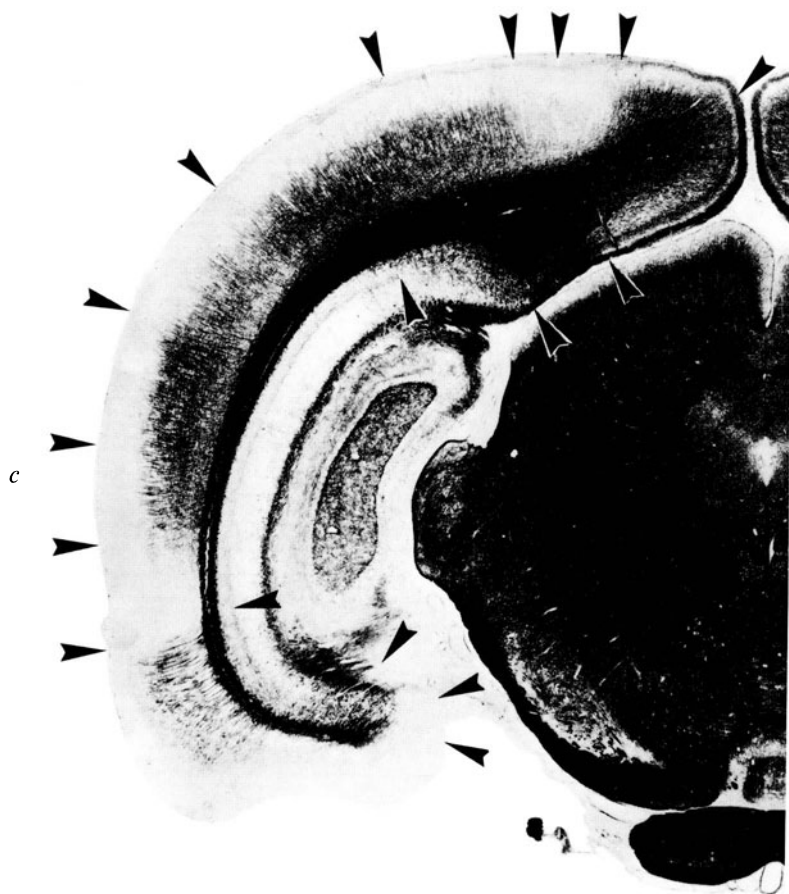


a

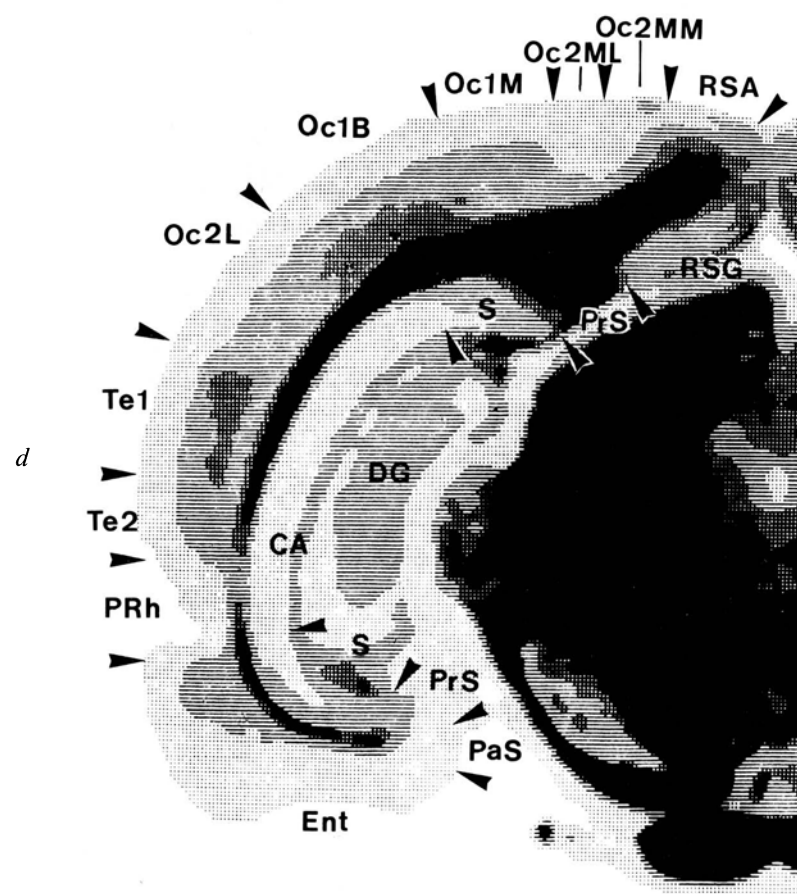
Fig. 65 a-d

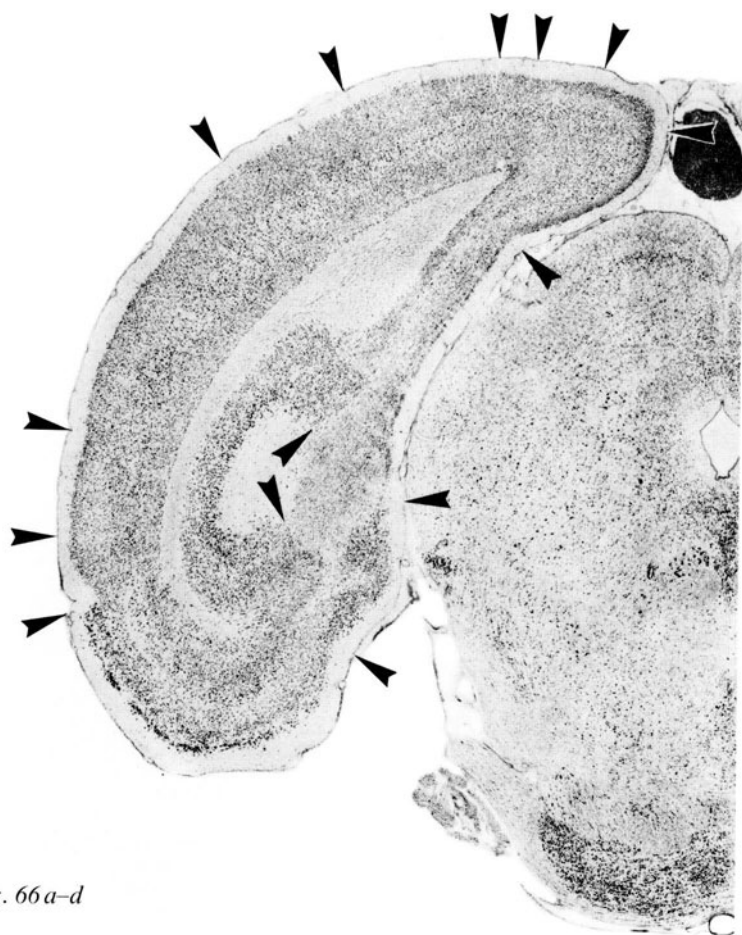


b



[91]





a

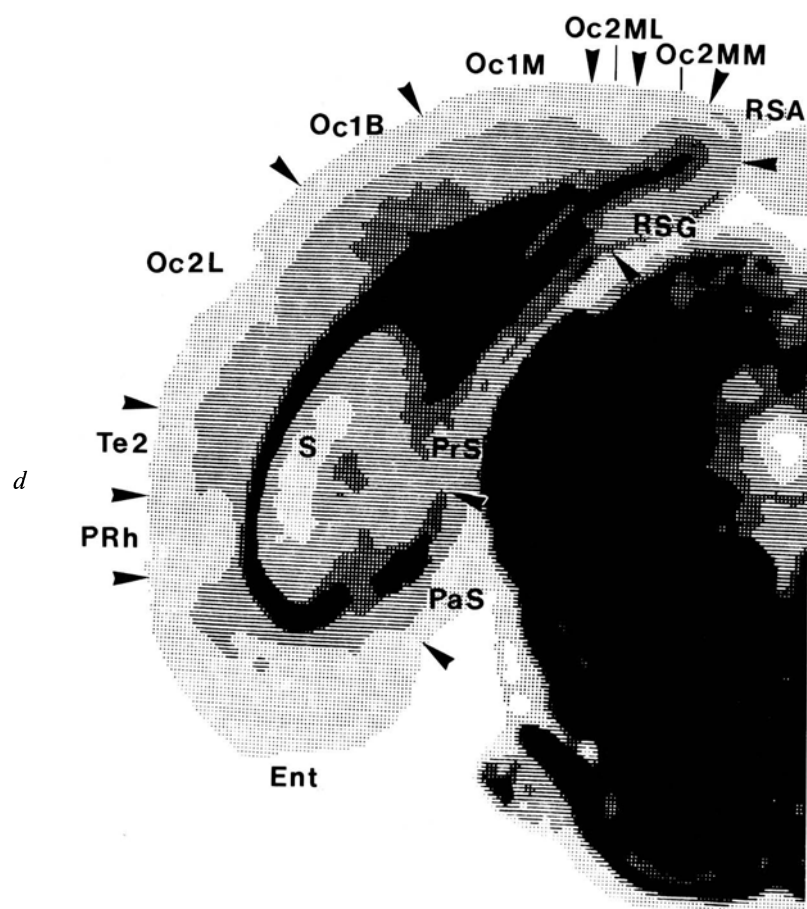
Fig. 66 a-d

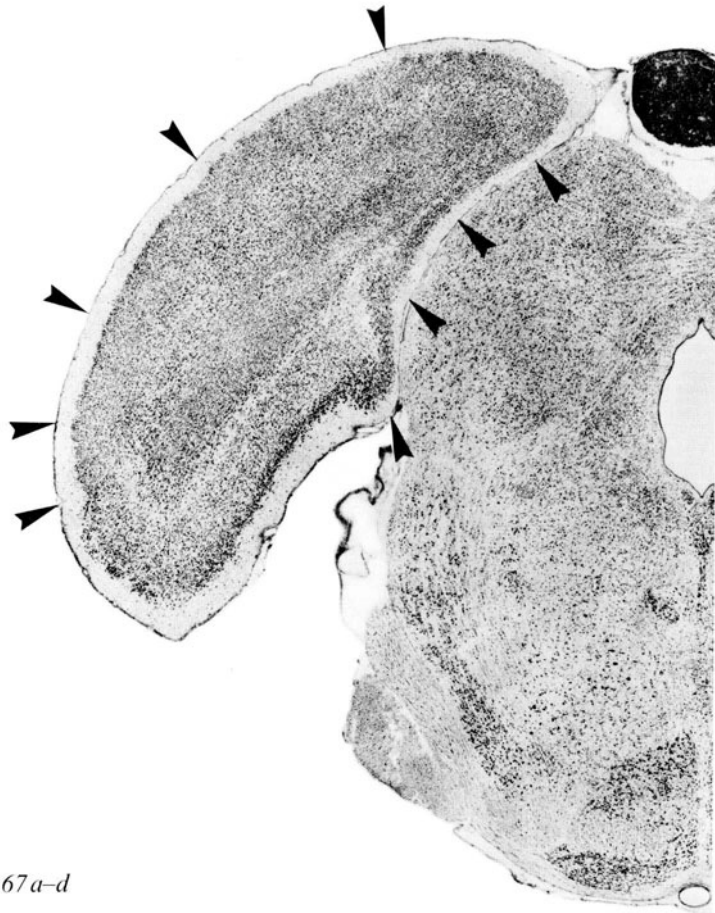


b



[93]





a

Fig. 67 a-d

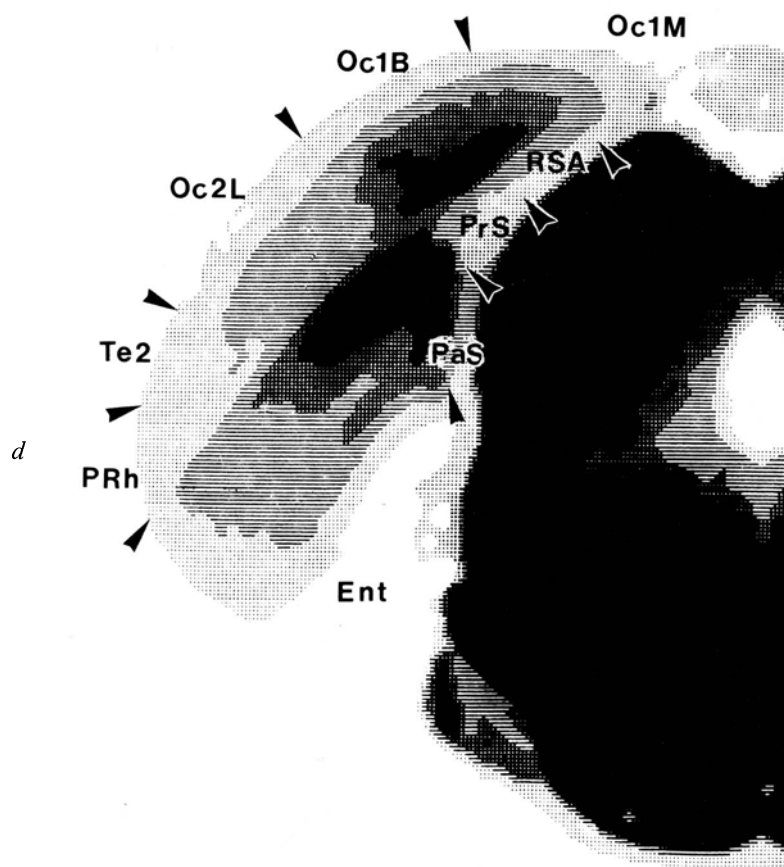
[94]



b



[95]



Laminar Structure of Cortical Areas in the Nissl- and Myelin-Stained Sections

Boundaries of cortical areas (Figs. 68–72) were determined in Nissl- and myelin-stained sections by the quantitative method described above (p. 6). In most cases the areal borders delineated in Nissl- and myelin-stained specimens from the same plane coincided precisely. In a few cases the areal boundaries shown are based on analysis by only one of the methods.

This atlas presents the stereotaxic position, the laminar pattern, and the Nissl and myelin features of those cortical areas and subfields which are often discussed in morphological, physiological and psychological studies, as revealed by Nissl and myelin staining.

[97]

AChE-stained sections of the primary visual cortex have been included because this histochemical method is a valuable tool for definition of the extension of lamina IV of the primary visual cortex (Zilles et al. 1984). This easily recognizable feature agrees with experiments involving transneuronal transport of intraocularly injected ^3H -proline (Zilles et al. 1984).

Fig. 68a–d. Laminar pattern of cortical areas in the frontal cortex. Areas Fr1 (a), Fr2 (b) and Fr3 (c) are shown at bregma 2.7 (cf. Fig. 7) and areas Cg1 and Cg2 (d) at bregma –0.3 (cf. Fig. 13). Nissl; 64 ×

Fig. 69a–d. Laminar pattern of cortical areas in the parietal cortex. Area Par1 (a) is shown at bregma 1.7 (cf. Fig. 9), areas Par2 (b) and HL (d) at bregma –3.3 (cf. Fig. 19) and area FL (c) at bregma –0.3 (cf. Fig. 13). Nissl; 64 ×

[98]

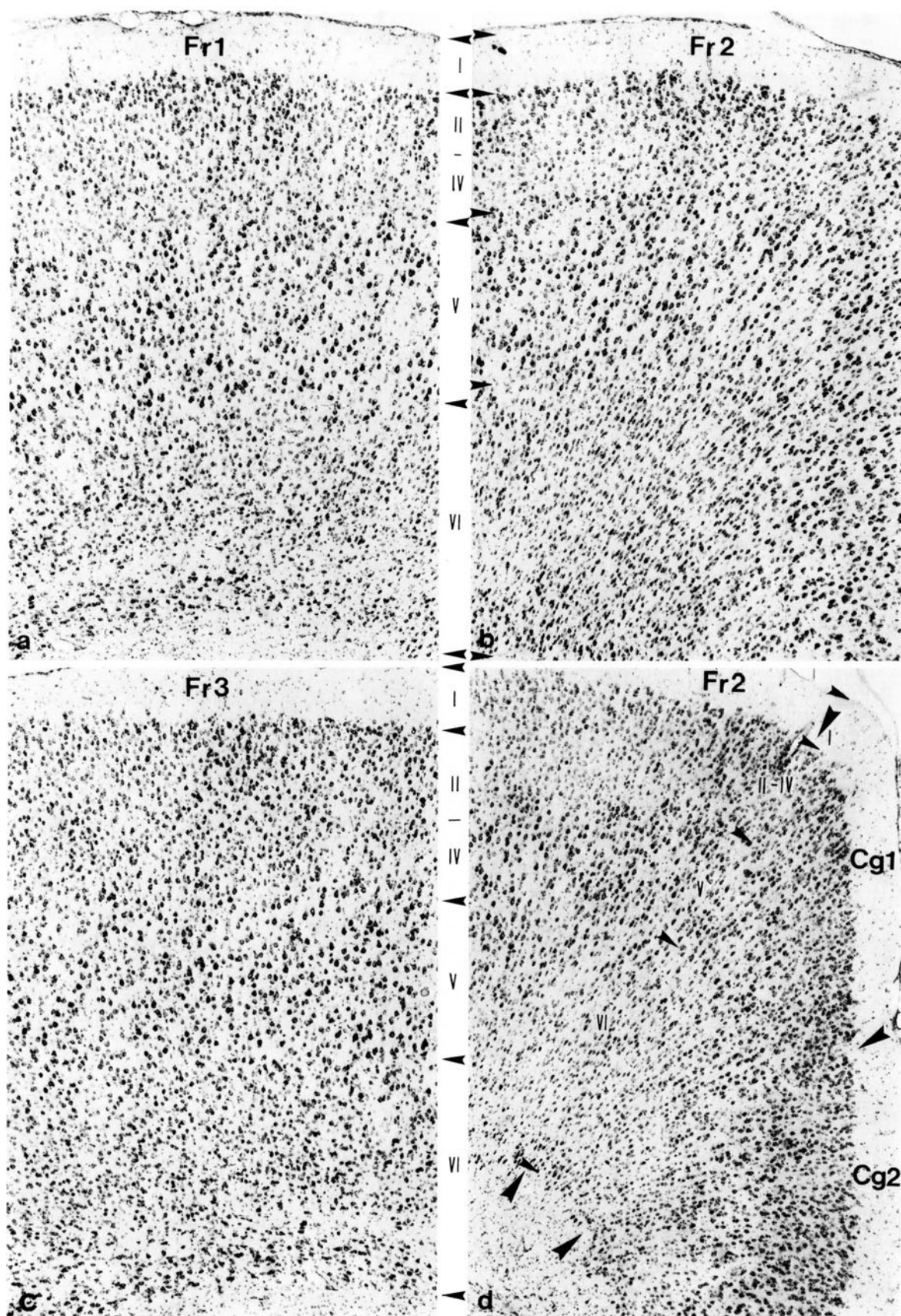
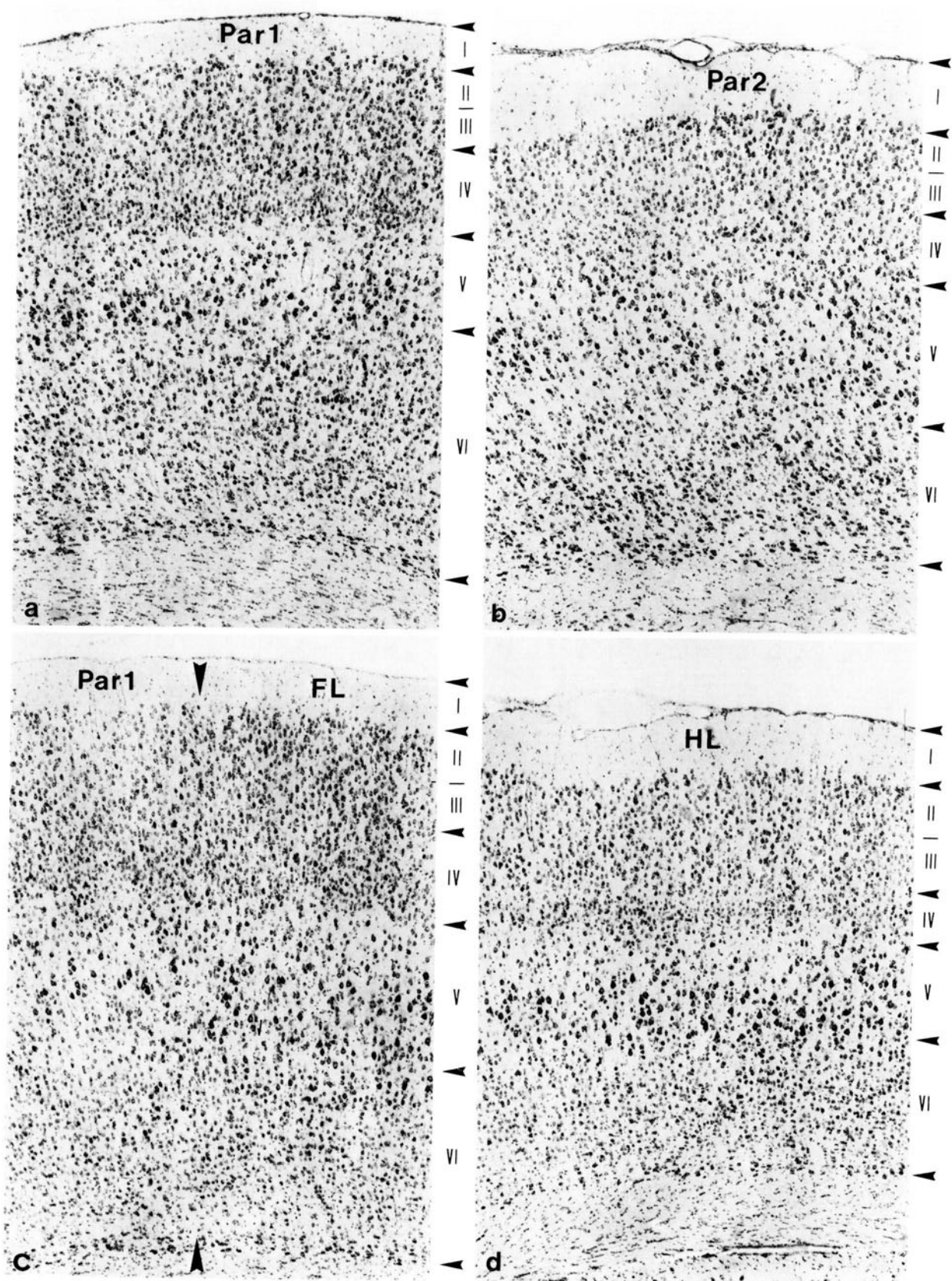


Fig. 68a-d



[99]

Fig. 69a-d

[100]

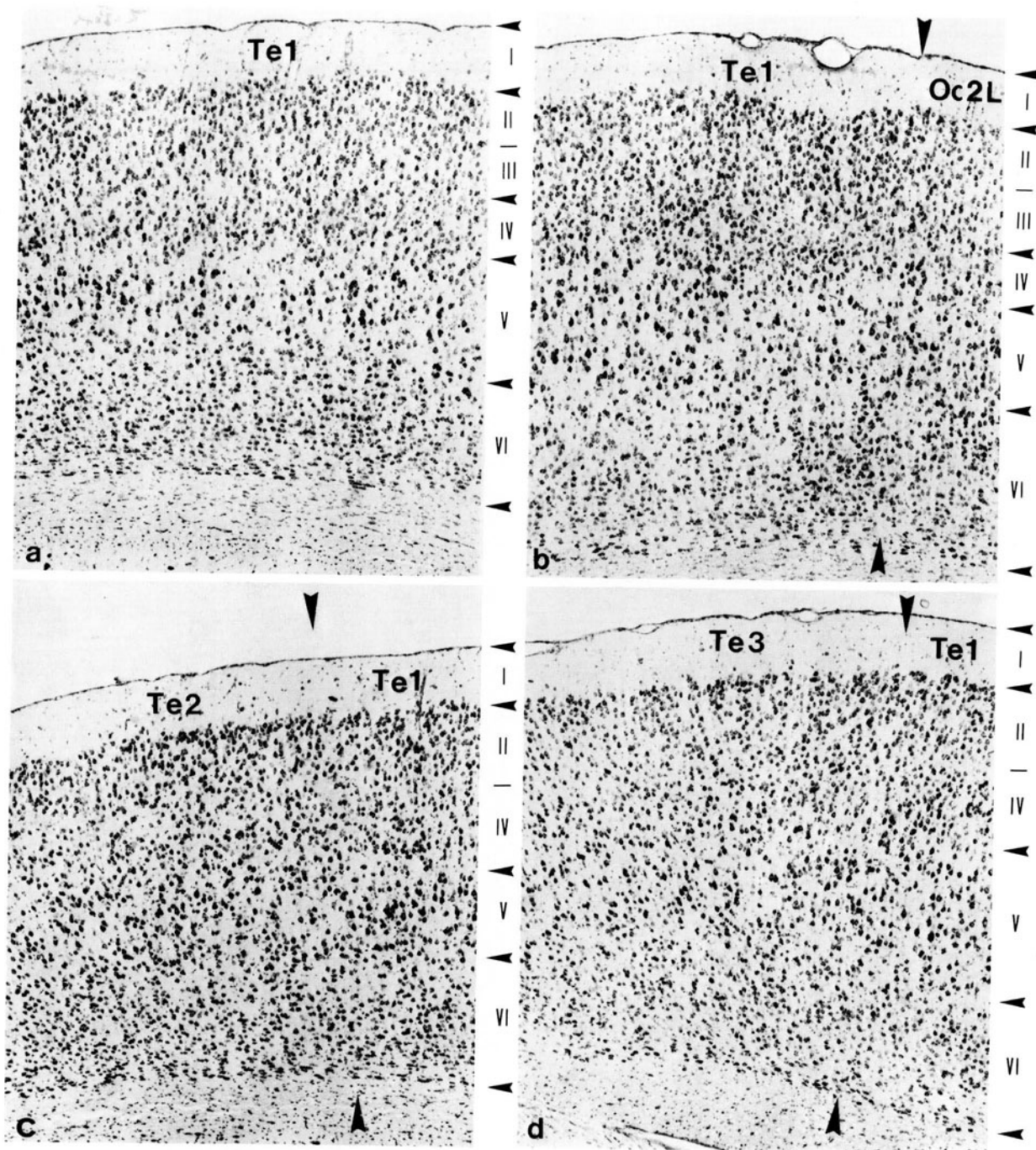


Fig. 70a-d. Laminar pattern of cortical areas in the temporal cortex. Area Te1 (*a, b*) is shown at bregma -6.3 (*a*) and -5.3 (*b*) to demonstrate the structural inhomogeneity of this area (cf. Figs. 25 and 23). Area Te2 (*c*) is shown at bregma -6.3 (cf. Fig. 25) and area Te3 (*d*) at bregma -5.3 (cf. Fig. 23). Nissl; $64\times$

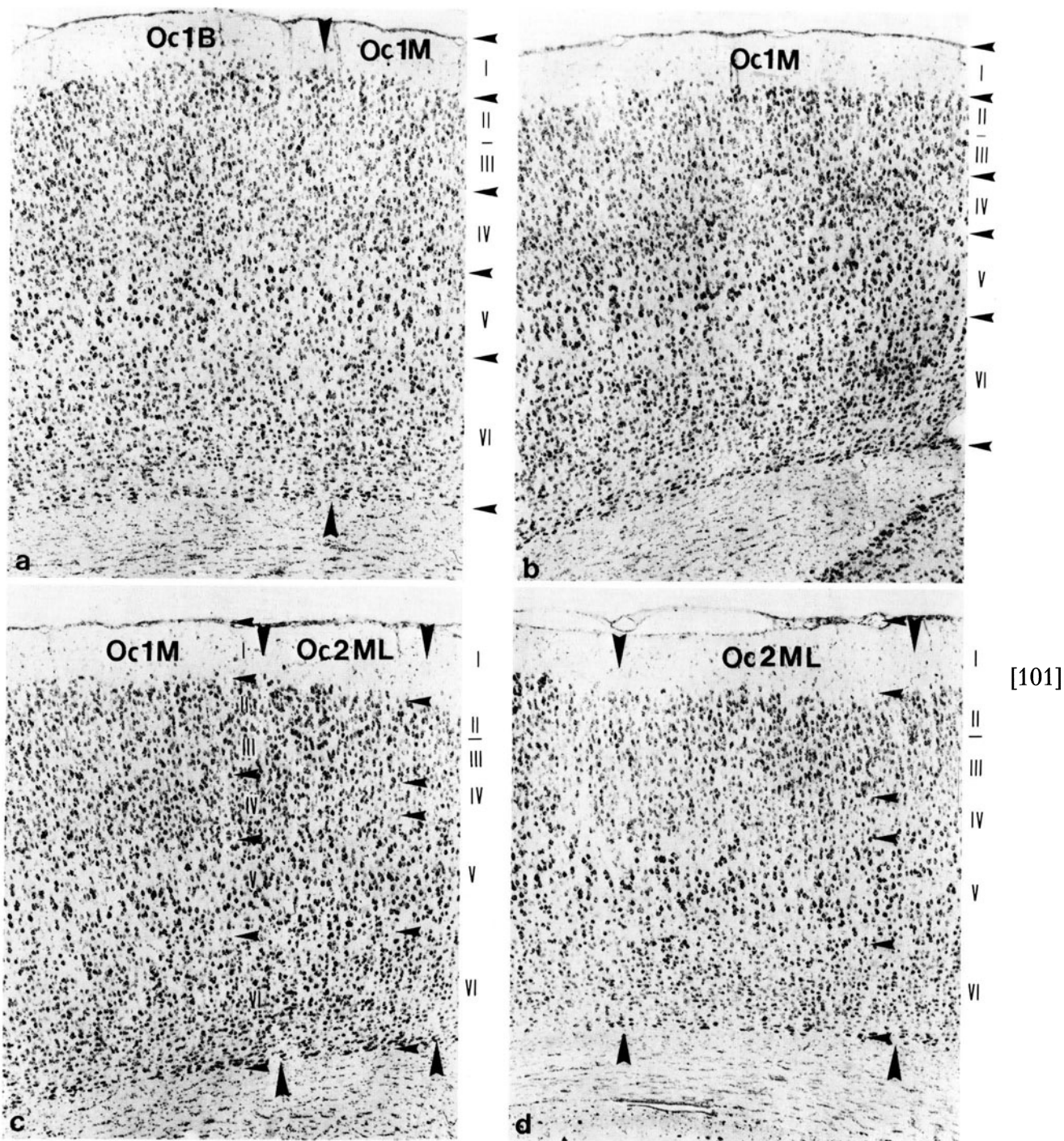


Fig. 71a-d. Laminar pattern of cortical areas in the occipital cortex. Areas Oc1B (*a*) Oc1M (*a*, *b*, *c*) and Oc2ML (*c*) are shown at bregma -6.3 (cf. Fig. 25) area Oc1M (*b*) at bregma -7.3 (cf. Fig. 27), and area Oc2ML (*d*) at bregma -5.3 (cf. Fig. 23). Nissl; $64\times$

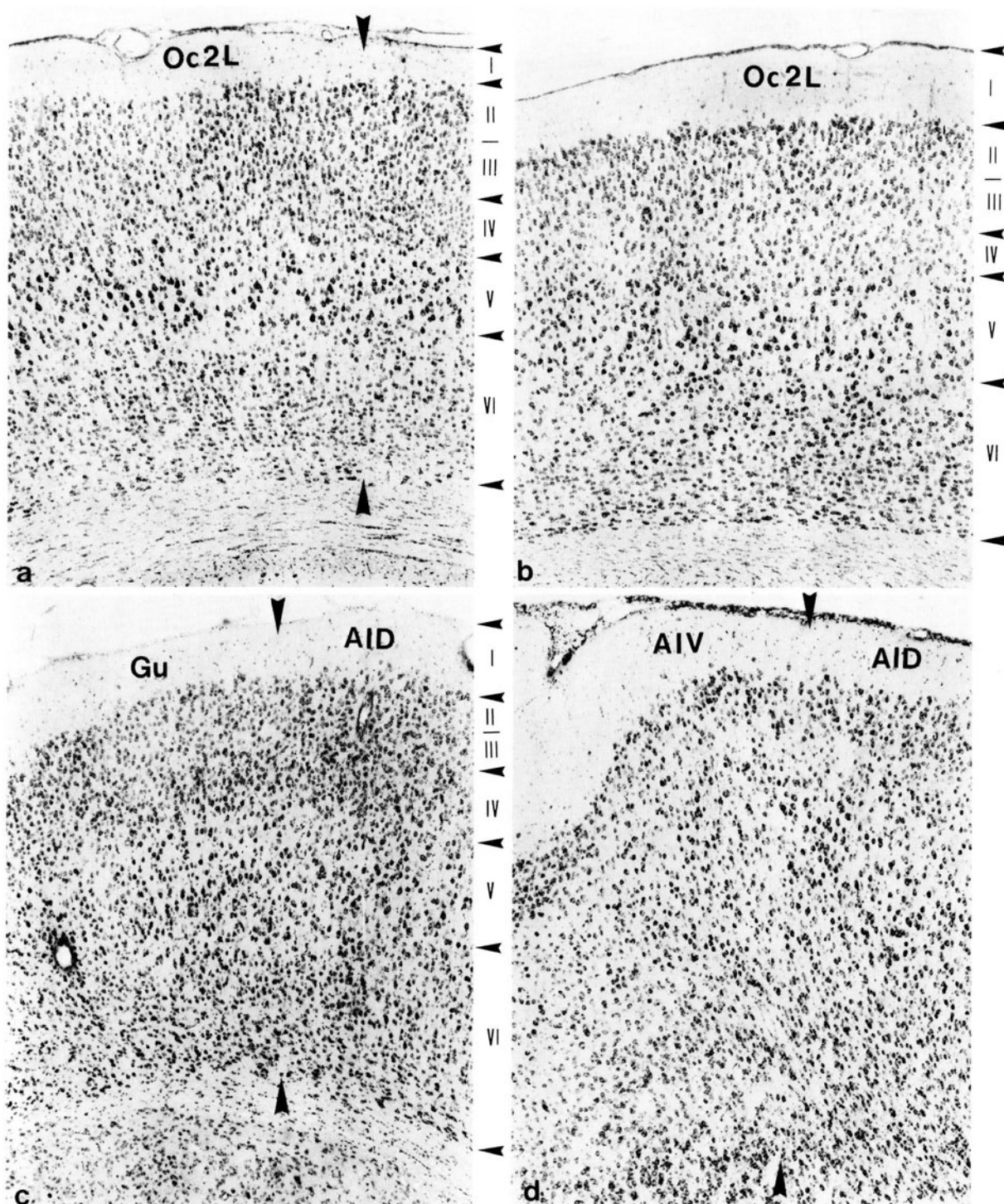


Fig. 72a-d. Laminar pattern of cortical areas in the occipital cortex. Area Oc2L (*a, b*) is shown at bregma -5.3 (*a*) and -7.3 (*b*) to demonstrate the structural inhomogeneity of this area (cf. Figs. 23 and 27). Areas Gu and AID (*c*) are shown at bregma -0.3 (cf. Fig. 13) and areas AID and AIV (*d*) at bregma 1.7 (cf. Fig. 9). Nissl; $64\times$

Aspects of Cortical Variability

This atlas is not intended to be a detailed study of interindividual variations, but efforts have been made to present some important aspects of this problem arising from the use of different rat strains. Figures 73–76 present the results of cortical mapping of different individuals from the same strain and of individuals from different strains (Han: Wistar strain, Lewis strain, hooded rats BDE/Han).

Figure 73 shows two cortical maps of the lateral aspect of the hemispheres superimposed. One cortical map is that of the rat presented in Figs. 2–43 and the other is that of two other male Wistar rats almost identical with the earlier rat in age and body weight. The basic topology of the cortical areas is identical in both brains. Moreover, the position and extension of each of these cortical areas is so well comparable that the areas in which the same field is found in both brains (white areas) are very much larger than the areas in which no identical fields are found (black areas). Upon superimposition of more brains from animals of the same strain the black areas presumably would increase in size though the preponderance of overlapping areas over nonoverlapping areas would persist. This emphasizes the interindividual variability within *one* rat strain, which is wide enough to lead us to doubt the significance and reliability of experiments based on uncritical confidence in stereotaxic coordinates mostly derived from only one brain, especially if the experiments involve regions near the borderline of a cortical area. On the other hand, Fig. 73 demonstrates that stereotaxic coordinates give important and reliable information about the position of a large proportion of a cortical area, which is not influenced by interindividual variation. Personal experiments with stereotaxic injections of HRP into the areas Oc1M, Oc1B, Oc2ML, and the border-region between Oc1M and Oc2ML have shown successful placement in all cases. These were checked by subsequent histological identification of the

sites of these injections. In 2 of 11 cases the Oc1M injections were nearer the Oc1M/Oc2ML border than had been expected from the stereotaxic coordinates derived from Fig. 47. Nevertheless, the injections were confined to Oc1M.

Figures 74–76 present cortical maps resulting from the superimposition of the cortical maps of a Lewis rat, a hooded rat, and a Wistar rat brain. The maps show lateral, dorsal, and medial views of a model brain constructed from mean values for all these three brains. The outer contours of these brains were superimposed to allow graphic generation of the outer contour of the “average” brain. Analogous treatment of the areal borders of several cortical areas in these three brains yielded the delineation of core regions as stereotaxically definable minimal common areas in all three brains from these different strains. This type of map could prove valuable when a relatively reliable stereotaxic position of a cortical area is needed, e.g., at the beginning of tracer studies.

Acknowledgements

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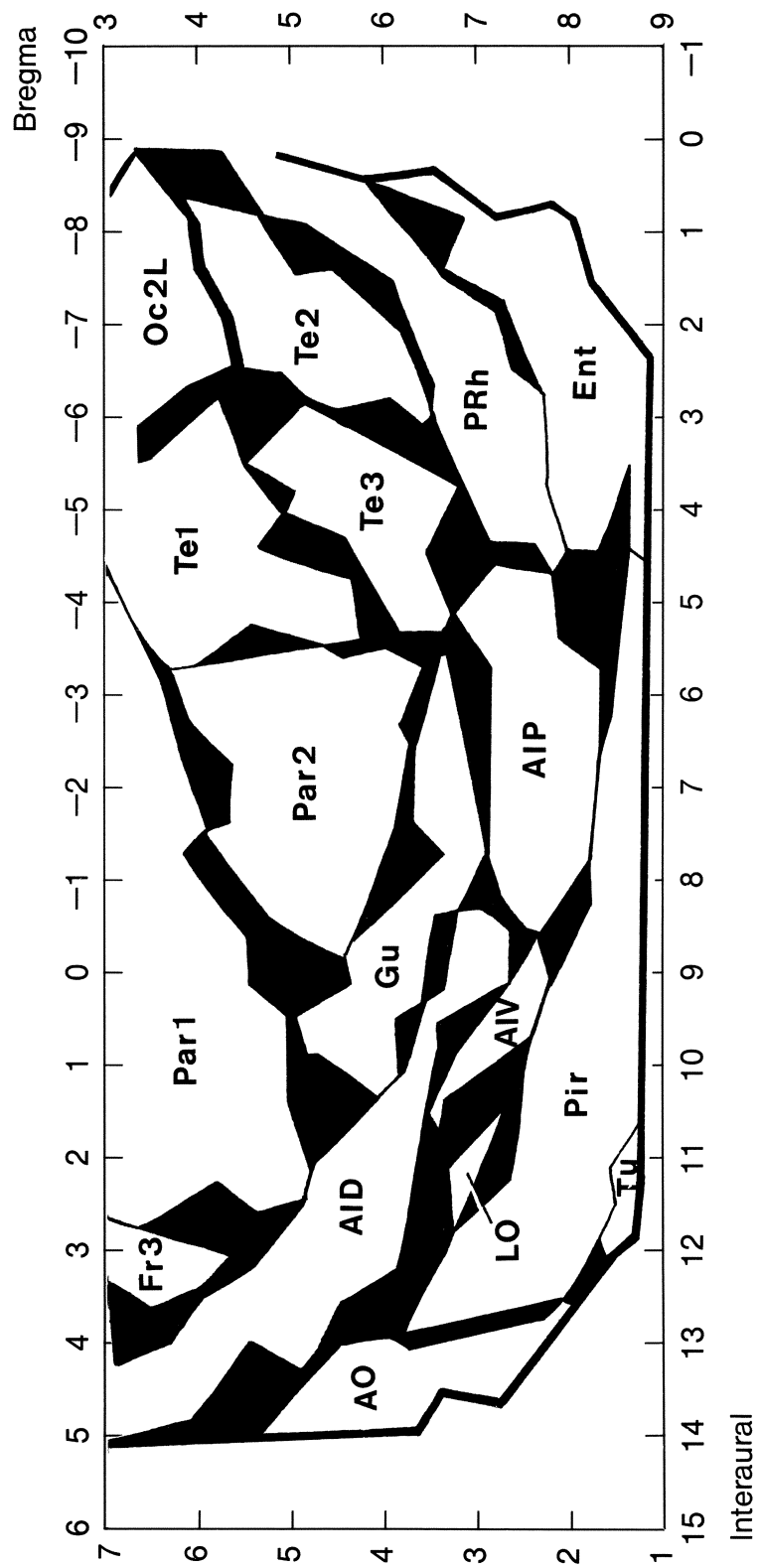


Fig. 73

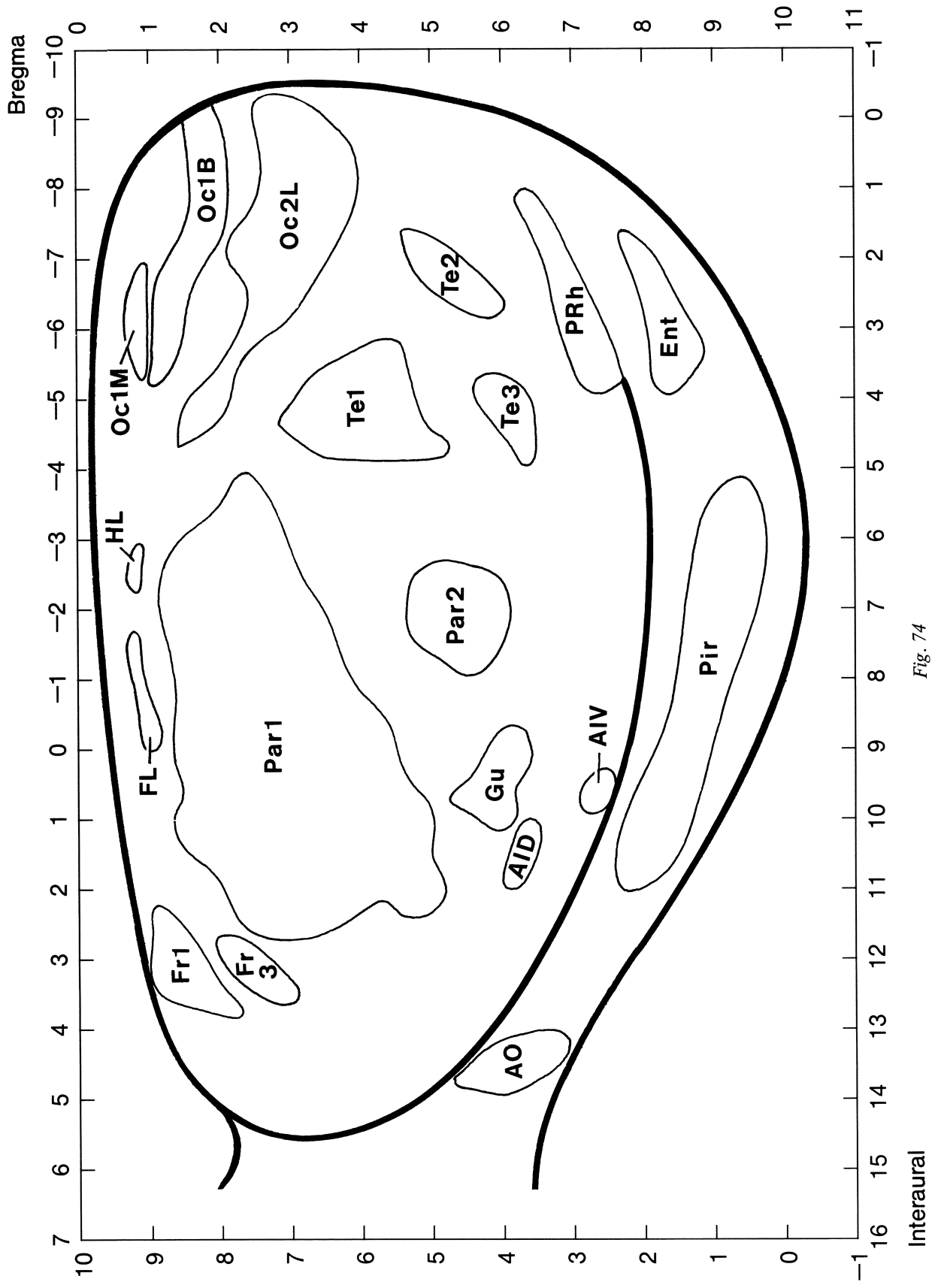


Fig. 74

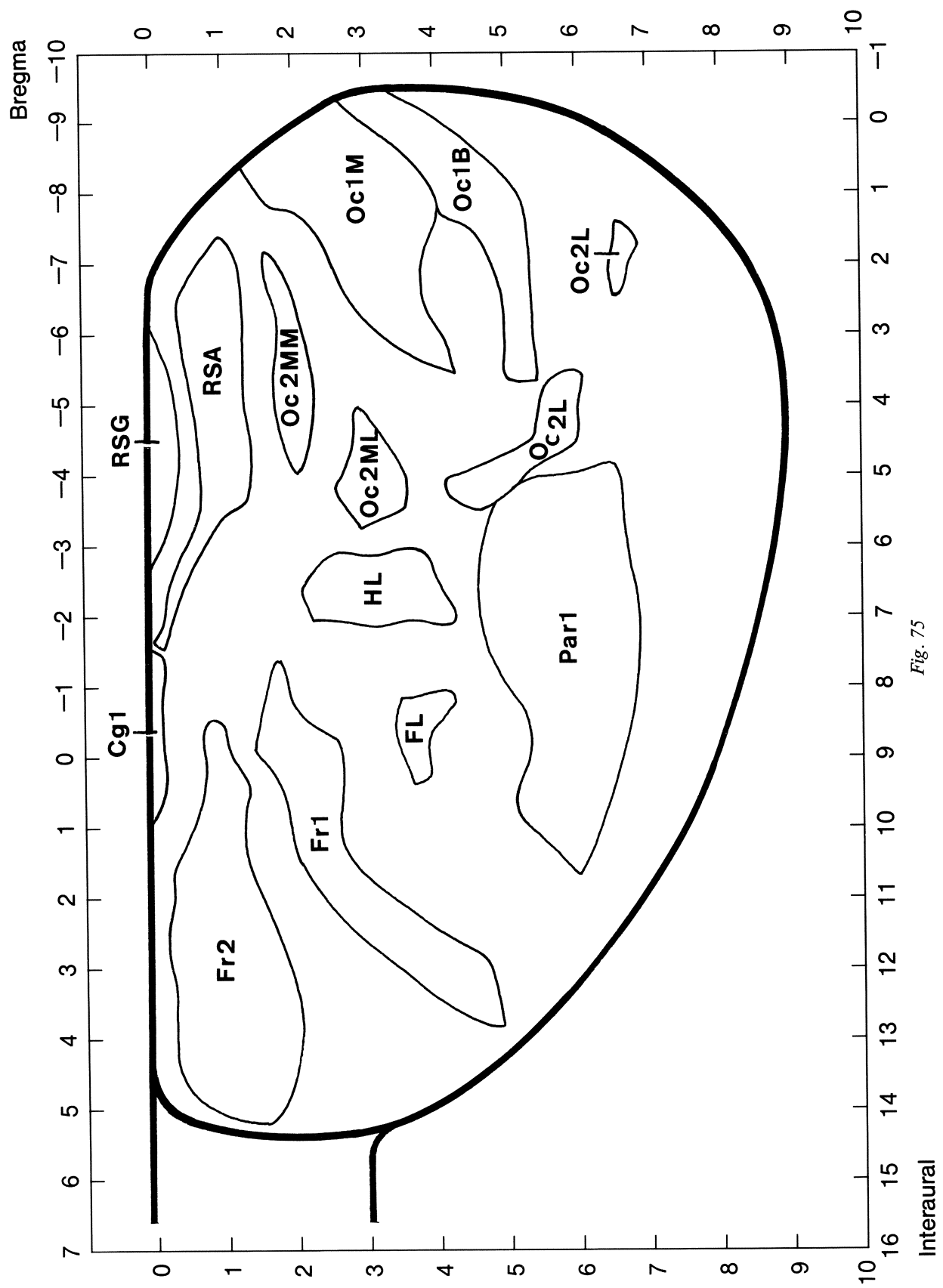


Fig. 75

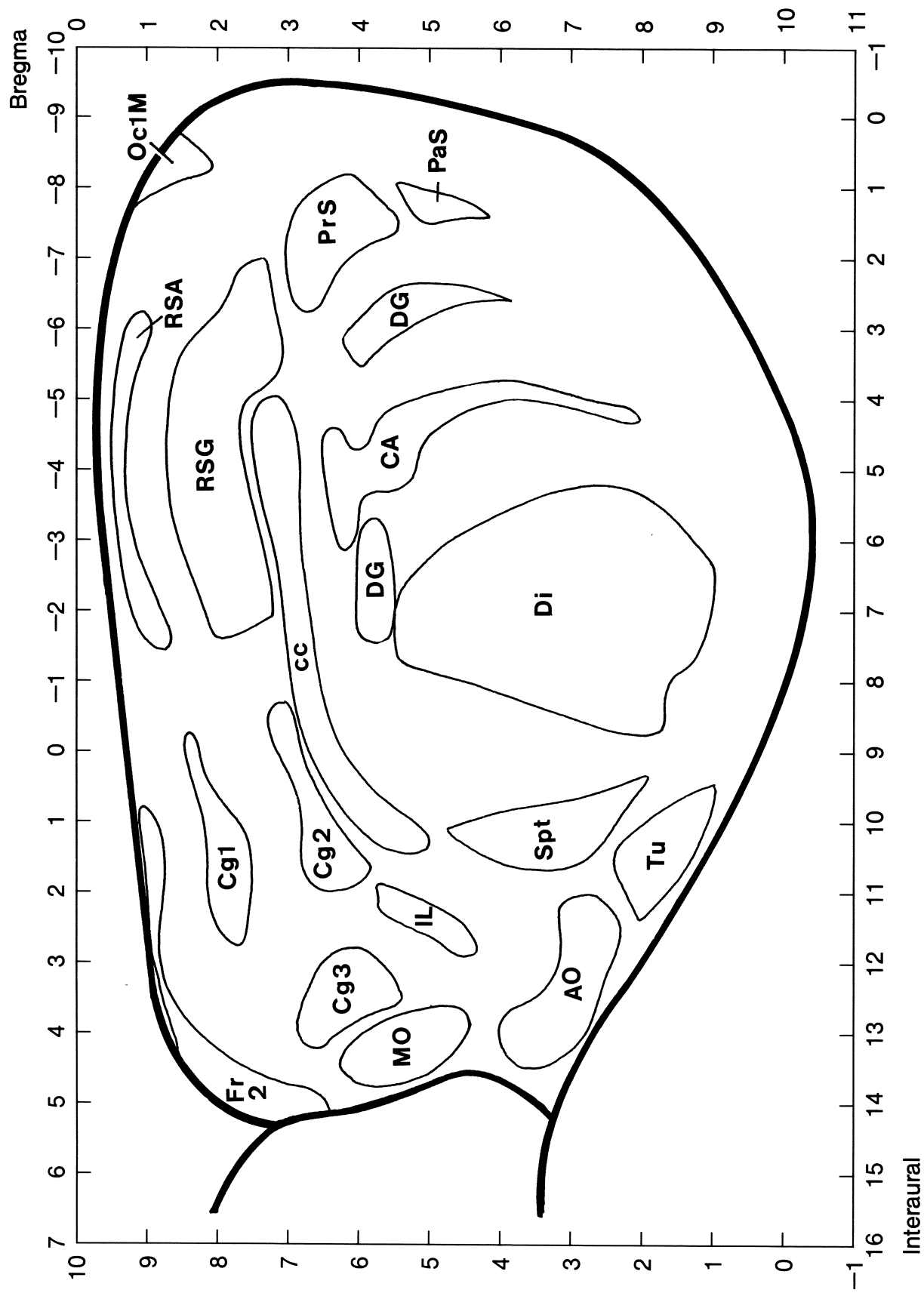


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LIST OF ABBREVIATIONS

Abbreviations used in the figures are listed in alphabetical order; each abbreviation is followed by the full name of the structure

AA	anterior amygdaloid area	Cg3	cingulate cortex, area 3 (medial prefrontal cortex)	Oc1B	occipital cortex, area 1 binocular part (primary visual cortex)
ac	anterior commissure	CGD	central gray, dorsal part	Oc1M	occipital cortex, area 1 monocular part (primary visual cortex)
aca	anterior commissure, anterior part	Cl	claustrum	Oc2L	occipital cortex, area 2 lateral part
Acb	accumbens nucleus	cp	cerebral peduncle	Oc2ML	occipital cortex, area 2 mediolateral part
aci	anterior commissure, intrabulbar part	CPu	caudate putamen (striatum)	Oc2MM	occipital cortex, area 2 mediomedial part
ACo	anterior cortical amygdaloid nucleus	CxA	cortex-amygdala transition zone	opt	optic tract
acp	anterior commissure, posterior part	df	dorsal fornix	ox	optic chiasm
AHi	amygdalohippocampal area	DG	dentate gyrus	Par1	parietal cortex, area 1 (primary somatosensory cortex)
AID	agranular insular cortex, dorsal part (claustrorocortex)	dhc	dorsal hippocampal commissure	Par2	parietal cortex, area 2 (supplementary somatosensory cortex)
AIP	agranular insular cortex, posterior part (claustrorocortex)	Di	diencephalon	PaS	parasubiculum
AIV	agranular insular cortex, ventral part (claustrorocortex)	DPC	dorsal peduncular cortex	pc	posterior commissure
alv	alveus of the hippocampus	En	endopiriform nucleus	Pir	prepiriform cortex ("primary olfactory cortex")
Amg	amygdaloid body	Ent	entorhinal area	PLCo	posterolateral cortical amygdaloid nucleus
AO	anterior olfactory nucleus (retrobulbar region)	Epi	epiphysis cerebri	PMCo	posteromedial cortical amygdaloid nucleus
AOB	accessory olfactory bulb	f	fornix	Pn	pontine nuclei
B	cells of the basal nucleus of Meynert equivalent	fi	fimbria of the hippocampus	PRh	perirhinal area
BL	basolateral amygdaloid nucleus	FL	forelimb area	PrS	presubiculum
BLV	basolateral amygdaloid nucleus, ventral part	fmi	forceps minor	py	pyramidal tract
BM	basomedial amygdaloid nucleus	fmj	forceps major	RSA	agranular retrosplenial cortex
BOl	olfactory bulb	fr	fasciculus retroflexus	RSG	granular retrosplenial cortex
CA	Ammon's horn	Fr1	frontal cortex, area 1 (primary motor cortex)	S	subiculum
CA1	field CA1 of Ammon's horn	Fr2	frontal cortex, area 2	SC	superior colliculus
CA2	field CA2 of Ammon's horn	Fr3	frontal cortex, area 3	sc	splenium of the corpus callosum
CA3	field CA3 of Ammon's horn	FStr	fundus striati	scp	superior cerebellar peduncle
CA4	field CA4 of Ammon's horn (hilus fasciae dentatae)	gcc	genu of the corpus callosum	SFI	septohippocampal nucleus
Cb	cerebellum	GP	globus pallidus	SFO	subformical organ
cc	corpus callosum	Gu	gustatory cortex	SHi	septohippocampal nucleus
Ce	central amygdaloid nucleus	HDB	nucleus of the horizontal limb of the diagonal band (Broca)	SHy	septohypothalamic nucleus
CeL	central amygdaloid nucleus, lateral part	HL	hindlimb area	sm	stria medullaris
CeM	central amygdaloid nucleus, medial part	Hy	hypothalamus	sox	supraoptic decussation
CG	central (periaqueductal) gray	I	intercalated nuclei of the amygdala	Spt	septum
Cg1	cingulate cortex, area 1 (medial prefrontal cortex)	IC	inferior colliculus	st	stria terminalis
Cg2	cingulate cortex, area 2 (medial prefrontal cortex)	ic	internal capsule	str	superior thalamic radiation
		IG	indusium griseum	Te1	temporal cortex, area 1 (primary auditory cortex)
		IL	infralimbic area of the medial frontal cortex	Te2	temporal cortex, area 2
		La	lateral amygdaloid nucleus	Te3	temporal cortex, area 3
		LO	lateral orbital area	tfp	transverse fibers of the pons
		lo	lateral olfactory tract	Th	thalamus
		LOT	nucleus of the lateral olfactory tract	TS	triangular septal nucleus (interstitial nucleus of the ventral hippocampal commissure)
		LOTD	nucleus of the lateral olfactory tract, dorsal part	TT	taenia tecta
		LS	lateral septal nucleus	Tu	olfactory tubercle
		LSD	lateral septal nucleus, dorsal part	VDB	nucleus of the vertical limb of the diagonal band (Broca)
		LSI	lateral septal nucleus, intermediate part	vhc	ventral hippocampal commissure
		LSV	lateral septal nucleus, ventral part	VLO	ventrolateral orbital area
		mcp	middle cerebellar peduncle	VO	ventral orbital area
		Me	medial amygdaloid nucleus	xscp	decussation of the superior cerebellar peduncle
		MG	medial geniculate nucleus		
		ml	medial lemniscus		
		mlf	medial longitudinal fasciculus		
		MO	medial orbital area		
		MP	medial mammillary nucleus, posterior part		
		MS	medial septal nucleus		
		mt	mammillothalamic tract		